

Equilibrium Uniqueness in Entry Games with Combinatorial Actions and Endogenous Pricing^{*}

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Abstract

Combinatorial entry games combine subset choice, within-firm complementarities, and strategic substitutes, but existing uniqueness results either treat entry as binary, assume constant markups, or take the active set as given. We prove an *abstract uniqueness theorem* for threshold-strategy Bayesian Nash equilibria in combinatorial entry games with private fixed-cost shocks: given a C^1 marginal-value index $v(v)$ on the activation-probability cube $[0, 1]^M$ together with a second-stage pricing block that supplies a smooth equilibrium selection with a Jacobian bounded in the supremum (row-sum) operator norm $\|\cdot\|_\infty$, the threshold-map Jacobian decomposes as $J_{T_\sigma} = \text{diag}(g_0(v/\sigma)/\sigma) J_v(v)$, and its supremum-norm contraction modulus is a *bathtub*-shaped function of the private-cost noise scale that vanishes at both extremes. When its peak $\gamma_{\max}$ is below one, the threshold-strategy BNE is unique at every noise level. Supermodularity of the single-firm CDCP holds per-market under a closed-form share threshold and, when the threshold is violated (as in the cloud application’s maximum share 0.40), is supported by a computational diagnostic (Jia–Tarski gap equal to zero at every iteration), so existing CDCP solvers remain applicable. We verify the abstract hypotheses in two environments: Fréchet–CES–Atkeson–Burstein demand with Cournot pricing, used in the quantitative application, and nested logit with Bertrand pricing, which establishes that the uniqueness engine is a property of the threshold system rather than of the CES specification. In an application to global cloud infrastructure entry (five providers, 42 countries), endogenous AB markups sustain 15 additional firm-market pairs relative to a constant-markup CES benchmark and redistribute roughly one-third of producer surplus from leaders to mid-ranked rivals, with aggregate profit rising by 30%—identifying the endogenous-markup channel as a first-order force on the distributional margin. The calibration is disciplined by a within-sample restriction test: suppressing the identified firm deployment-efficiency wedges ($g_f \equiv 1$) loses 12 percentage points of in-sample cell fit and alters the counterfactual quantitatively, so the firm-productivity vector is not a free parameter absorbing the other channels.

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1. INTRODUCTION

Equilibrium multiplicity is a central obstacle in entry games. When entry decisions are strategic substitutes, multiple configurations of active firms can be mutually consistent, creating well-known difficulties for estimation and counterfactual analysis. Three literatures have made progress on different facets of this problem: [Espin-Sanchez et al. \(2023\)](#) prove uniqueness for *binary* entry games with private costs, but cannot handle within-firm complementarities across actions; the combinatorial discrete choice (CDCP) literature ([Jia, 2008](#); [Arkolakis et al., 2023](#)) handles subset selection with complementarities, but assumes constant markups; and the variable-markup literature ([Atkeson and Burstein, 2008](#); [Edmond et al., 2015](#)) endogenizes pricing, but takes the set of active firms as given. In many important settings—multinational production, retail chain location, platform entry—all three features interact simultaneously: firms choose *which subset* of markets to enter, entry changes market structure, and market structure feeds back into markups and entry incentives. Neither the entry game, the CDCP, nor the pricing game can be solved in isolation.

This paper proves an *abstract uniqueness theorem* for threshold-strategy Bayesian Nash equilibria in combinatorial entry games and then verifies its hypotheses in two demand-and-pricing environments. The abstract theorem operates on a threshold-strategy BNE representation in which each site-firm pair’s activation probability is a fixed point of a smooth threshold map $T_\sigma(v) = G_0(v(v)/\sigma)$ on the probability cube $[0, 1]^M$. Its Jacobian decomposes into three economically interpretable forces—*density dampening*, *marginal-value sensitivity*, and *threshold location*—that render the activation–share–markup circularity self-limiting rather than explosive. The resulting contraction modulus has a distinctive “bathtub” shape: private-cost noise disciplines multiplicity at both extremes (large noise smooths responses; small noise pushes thresholds into the tails of the cost distribution), so when its peak $\gamma_{\max}$ is below one, the threshold-strategy BNE is unique at every noise level. We verify the two ingredients of the abstract theorem—a smooth C^1 marginal-value index v and a uniformly bounded Jacobian J_v —in a Fréchet–CES–Atkeson–Burstein environment with Cournot pricing (used in the application) and in a nested-logit environment with Bertrand pricing (which confirms that the uniqueness engine is structural, not CES-specific). The peak modulus admits numerical evaluation on a grid of primitives,

providing a modular route to uniqueness for the class of combinatorial entry games whose pricing block satisfies the abstract hypotheses.

We apply the framework to global cloud infrastructure entry, where five hyperscaler providers choose which of 42 countries to establish cloud regions. The quantitative finding that we emphasize is a distributional one: endogenous AB markups sustain 15 additional firm-market pairs relative to constant-markup pricing, concentrated among mid-ranked providers whose shares are highest under CES, and redistribute roughly one-third of producer surplus toward firms, with aggregate profit rising by 30%. The in-sample fit (99.5% of calibrated entry cells) is a moment-matching construction rather than a test; the test is that a single-parameter model restriction ($g_f \equiv 1$, Specification D of Section 5) loses 12 percentage points of in-sample cell fit and alters the counterfactual quantitatively, so the identified firm-productivity vector is not a free parameter. The game-theoretic results extend to other combinatorial entry settings—retail chains (Jia, 2008), platforms, or technology adoption—whenever the pricing block admits the smooth, uniformly Jacobian-bounded selection that the abstract theorem requires; the cloud application demonstrates the quantitative bite of the framework on the distributional margin.

The paper makes three main contributions, supported by a quantitative application to hyperscaler cloud infrastructure deployment:

Core contribution: an abstract uniqueness theorem with two verifications. We state the uniqueness result at the abstract level of the threshold map $T_\sigma(v) = G_0(v)/\sigma$ (Theorem 13): given a C^1 marginal-value index $v : [0, 1]^M \rightarrow \mathbb{R}^M$ with uniformly bounded Jacobian J_v , the threshold map has a unique fixed point whenever its contraction modulus $\gamma(\sigma) \equiv \sup_v \|\text{diag}(g_0/\sigma) J_v\|_\infty$ is below one. The modulus vanishes at both noise-scale extremes (Lemmas 15–16); its finite peak $\gamma_{\max}$ is a single primitive-level number, and the condition $\gamma_{\max} < 1$ is sufficient for threshold-strategy BNE uniqueness *at every* $\sigma > 0$ (Corollary 14). Two verification environments make the abstract theorem operational. In the Fréchet–CES–AB environment of Section 2, a spectral-radius condition for the Cournot pricing block (Proposition 11) supplies the smooth pricing selection, and a *direct enumeration certificate* on the activation lattice at the calibrated cost vector (Proposition 28) verifies the condition globally; Proposition 21 then verifies the abstract hypotheses by constructing v explicitly from CES demand. In a nested-logit environment with Bertrand pricing (Sec-

tion 3.4), Proposition 31 supplies pricing uniqueness via aggregative-games arguments (Anderson et al., 1992; Nocke and Schutz, 2018), and Proposition 33 verifies the same abstract hypotheses with no change to the uniqueness engine. The main result (Theorem 24) consolidates the CES–AB-specific hypotheses for the quantitative application; Proposition 20 extends the framework state-by-state to across-firm common cost shocks.

Supermodularity extension: endogenous-markup CDCP. We extend Antràs et al.’s (2017) constant-markup supermodularity result to endogenous AB variable markups (Proposition 6). Per-market supermodularity holds analytically below a closed-form share threshold; when the threshold is exceeded at the cloud calibration (max share 0.40), Jia–Tarski gap equal to zero across all iterations provides a computational diagnostic that is necessary for aggregate supermodularity (Remark 7). This preserves the lattice-theoretic structure required for CDCP algorithms and for the threshold system’s monotone comparative statics at the tested parameterization.

Quantitative application: cloud infrastructure entry. We calibrate the framework to global cloud infrastructure, following the identification structure of Arkolakis et al. (2023)—fixing demand and firm scale externally, calibrating fixed costs from entry moments—though with a thinner moment basis. The application delivers three substantive findings, of which only the first uses the endogenous-markup mechanism. First, endogenous AB markups redistribute roughly one-third of producer surplus from leaders to mid-ranked rivals relative to a CES benchmark and sustain 15 additional firm-market pairs—this is the mechanism-driven finding, identifying the endogenous-markup channel as a first-order force on the *AB-versus-CES* distributional wedge. Second, EU compliance-cost shocks are regressive in market-structure terms and this regressivity is *robust across pricing regimes*: the AB-vs-CES amplification in the EU exercise is negligible, so the finding is informative about the regional structure of baseline deployment rather than about the pricing mechanism. Third, the binding constraint in digital-fragmentation counterfactuals is feasible-market access rather than trade costs: reducing China’s entry cost by 90% without feasible-set changes produces zero new entry. Appendix G provides synthetic stress tests across the parameter space, and Appendix F compares four algorithm variants with up to 2^{40} candidate subsets per player.

Entry games and equilibrium selection. This paper contributes primarily to the literature on equilibrium uniqueness in entry games. Beginning with [Bresnahan and Reiss \(1991\)](#) and [Berry \(1992\)](#), the empirical entry literature has confronted multiplicity as a barrier to estimation: [Berry](#) restricts equilibrium selection to obtain a tractable likelihood; [Ciliberto and Tamer \(2009\)](#) use partial identification to bound parameters without selecting among equilibria; [Seim \(2006\)](#) introduces endogenous product-type choices, a structure related to combinatorial entry; [Igami and Yang \(2016\)](#) estimate a dynamic entry model with combinatorial product portfolios using simulation methods that must navigate the multiplicity problem. [Espin-Sanchez et al. \(2023\)](#) prove BNE uniqueness in static entry games where each firm makes a *binary* enter/not-enter decision with scalar private types; their threshold system is over $\mathbb{R}^N$ (one cutoff per firm). When each firm instead chooses a *subset* of actions—e.g., which countries to produce in—the binary framework cannot capture within-firm complementarities, and the threshold system grows to $\mathbb{R}^M$ (one cutoff per player-action pair). We extend their result to these combinatorial action spaces, where within-firm actions interact through the CDCP’s supermodularity and payoffs depend on an endogenous pricing equilibrium. The two results are complementary: [Espin-Sanchez et al.](#) cover general payoff functions with binary actions, while we handle combinatorial action spaces under CES/AB payoffs; neither nests the other as a special case. Three results sharpen the distinction. First, a small-economy example (Section 3.7) constructs a configuration where the binary-action threshold system of [Espin-Sanchez et al.](#) cannot be applied, but the combinatorial CDCP structure resolves uniqueness by internalizing within-firm complementarities. Second, the pricing feedback reaches $\rho(\Lambda) = 0.574$ at stressed parameterizations (Appendix G), so the endogenous-markup channel is quantitatively significant. Third, endogenous AB markups *expand* the uniqueness region relative to constant CES markups (Appendix G): markup compression dampens the dominant-firm responses that drive multiplicity under constant markups.

A key negative result motivates the present approach: [Doraszelski and Satterthwaite \(2010\)](#) introduce private scrap values in dynamic Ericson–Pakes models, guaranteeing cutoff equilibria but explicitly demonstrating that multiplicity persists—private information alone does not resolve the number of equilibria. In the present setting, the pricing contraction (Proposition 11) provides the additional structure—bounding how activation changes feed back through markups—that converts the threshold system into a contrac-

tion. An alternative selection mechanism is the global-games approach of [Harrison and Jara-Moroni \(2021\)](#), who extend iterated dominance arguments to binary-action games with strategic substitutes: dominance regions propagate from extreme types inward until a unique strategy profile survives. Their approach requires binary action spaces and strict payoff asymmetry across player pairs, while the present paper handles combinatorial actions and operates through a different mechanism (pricing contraction rather than dominance propagation).

Quantitative entry with combinatorial or multi-location choice. A parallel quantitative literature estimates entry models where firms choose portfolios of products or locations, typically restricting attention to equilibria that can be computed without resolving multiplicity. [Holmes \(2011\)](#) estimates a dynamic model of Wal-Mart’s spatial diffusion in which store-level economies of density generate combinatorial interdependence across locations, imposing a perfect-foresight restriction to sidestep multiplicity. [Nishida \(2015\)](#) estimates a static multi-store network-choice model for convenience stores in Okinawa, accommodating within-firm density economies and across-firm substitution but selecting among equilibria via a maximum-profit rule. [Wollmann \(2018\)](#) and [Fan and Yang \(2020\)](#) study equilibrium product characteristics and product-line choice in commercial trucks and smartphones, respectively, where firms choose subsets of products from a menu under oligopolistic competition; both rely on equilibrium-selection assumptions to make counterfactuals tractable. [Weintraub et al. \(2008\)](#) propose oblivious equilibrium for many-firm dynamic industries, approximating Markov-perfect behavior by treating the competitive environment as a long-run average—a different route around multiplicity that trades equilibrium exactness for computational scalability. Relative to these, we provide primitive conditions on preferences and cost heterogeneity under which the static combinatorial entry game has a unique BNE, so that counterfactuals in the same class of applications (multi-site/product entry with endogenous pricing) can be computed without invoking selection rules or many-firm approximations. The practical payoff relative to the maximum-profit or perfect-foresight selections in [Holmes \(2011\)](#), [Nishida \(2015\)](#), [Wollmann \(2018\)](#), and [Fan and Yang \(2020\)](#) is that comparative statics of the counterfactual activation profile inherit identification from model primitives rather than from the selection rule—an asset when the counterfactual perturbs the object (entry costs, markup

structure) through which the rule selects.

Supermodular games. The lattice-theoretic approach connects to the literature on supermodular games (Milgrom and Roberts 1990; Vives 1990; Milgrom and Shannon 1994; Topkis 1998; Amir 1996). The key challenge is that the cross-player interaction exhibits strategic *substitutes*, not complements: when one firm activates more sites, rivals' activation incentives weaken. We handle this through the squared best-response map, whose monotonicity restores the Tarski fixed-point structure. Athey (2001) establishes existence of pure-strategy equilibria in games whose payoffs satisfy a single-crossing condition in own action and type; the within-player complementarity of the CDCP satisfies her conditions, but the across-player substitutes violate single-crossing in rivals' actions, requiring the additional bounding argument of Proposition 8.

Combinatorial discrete choice. Jia (2008) studies store location with strategic substitutes; Arkolakis et al. (2023) formalize the general CDCP framework; Antràs et al. (2017) prove supermodularity under constant CES markups when $(\sigma - 1)/\theta > 1$. We extend their supermodularity result to endogenous variable markups, where activating an additional site changes market shares and hence all firms' markups—introducing a feedback channel absent under constant markups. The proof shows that this feedback is bounded by the same $(\sigma - 1)/\theta > 1$ condition (Proposition 6).

Variable markups and multinational production. The application draws on the variable markup literature (Atkeson and Burstein 2008; Edmond et al. 2015; Gaubert and Itskhoki 2021), which takes the set of active firms as given, and the multinational production literature (Tintelnot 2017; Antràs et al. 2026; Castro-Vincenzi 2024), which uses constant CES markups. By closing the loop between activation and pricing, we introduce the three-way circularity (activation $\rightarrow$ market shares $\rightarrow$ markups $\rightarrow$ activation incentives) and show that the pricing contraction bounds this feedback, preserving both supermodularity and equilibrium uniqueness. The static game provides the stage-game structure for dynamic oligopoly models of the type studied by Ericson and Pakes (1995) and Ifrach and Weintraub (2017). In the estimation of such dynamic games, Aguirregabiria and Mira (2007) develop sequential methods that require equilibrium selection at each iteration, while Bajari et al. (2007) propose forward-simulation methods that bypass equilibrium compu-

tation altogether; the uniqueness conditions derived here offer a third path, guaranteeing that the static stage game has a single equilibrium to estimate.

The nested-logit verification in Section 3.4 draws on discrete-choice Bertrand-pricing uniqueness as a black-box input: [Anderson et al. \(1992\)](#), [Mizuno \(2003\)](#), [Konovalov and Sándor \(2010\)](#), and [Nocke and Schutz \(2018\)](#) supply single-stage pricing uniqueness under (nested) logit, which the abstract theorem lifts to the multi-market combinatorial entry setting.

2. MODEL

We develop a static model of combinatorial entry with endogenous pricing. Firms choose which production sites to activate across multiple locations and compete in quantities under CES demand with endogenous variable markups. The leading interpretation is multinational production in segmented international markets, though the game-theoretic results apply more broadly. Figure 1 summarizes the three-way circularity and its relationship to the existing literature.

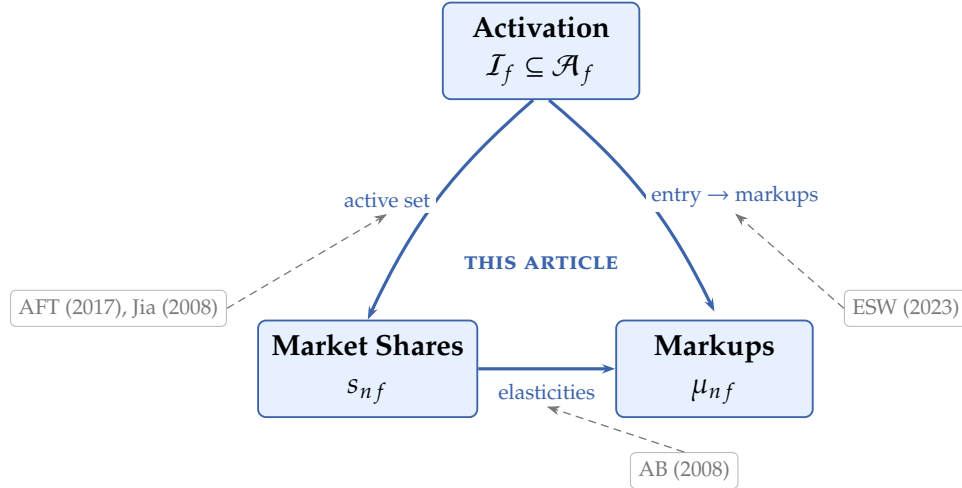


Figure 1: Three-way circularity and the literature. Solid arrows show the full equilibrium loop closed in this article. Prior work addresses subsets: AB (2008), i.e., [Atkeson and Burstein \(2008\)](#), endogenize markups given active firms (bottom edge); AFT (2017), i.e., [Antràs et al. \(2017\)](#), and [Jia \(2008\)](#) solve activation under constant markups (left edge); ESW (2023), i.e., [Espin-Sanchez et al. \(2023\)](#), prove uniqueness for binary entry decisions given a fixed payoff function (right edge; their result covers the activation channel but does not close the pricing loop).

2.1 Environment

There are N countries, indexed by $n \in \{1, \dots, N\}$. Each country serves as both a potential production site and a destination market. Wages w_n are exogenous.

There are F firms, indexed by $f \in \mathcal{F} = \{1, \dots, F\}$, that produce differentiated varieties in a single sector. Each firm f is characterized by two primitives:

- (i) a *total factor productivity* (TFP) $a_f > 0$, and
- (ii) a *set of available production sites* $\mathcal{A}_f \subseteq \{1, \dots, N\}$.

The firm's strategic decision is which sites in $\mathcal{A}_f$ to activate for production. Section 3.3 extends the model by introducing private information over site-specific fixed costs, yielding a Bayesian game.

2.2 Demand

A representative household in each country has income $E_n > 0$ and nested CES preferences. Following [Atkeson and Burstein \(2008\)](#), there is a continuum of sectors indexed by $k \in [0, 1]$, aggregated with elasticity of substitution $\eta > 1$:

$$C_n = \left[\int_0^1 C_{nk}^{\frac{\eta-1}{\eta}} dk \right]^{\frac{\eta}{\eta-1}}. \quad (2.1)$$

The F oligopolistic firms compete in a single differentiated-good sector (say $k = 0$). Because this sector has measure zero in the continuum, its price index P_n does not affect the aggregate price index $\tilde{P}_n$, so the demand shifter Ω_n defined below is truly exogenous to the firms' decisions. Within the differentiated sector, varieties are aggregated through a CES technology with elasticity $\sigma > \eta$:

$$Q_n = \left[\sum_{f \in \mathcal{F}_n} q_{nf}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \quad (2.2)$$

where $\mathcal{F}_n$ is the set of firms actively supplying market n . Since all production sites can serve all markets via iceberg trade costs, $f \in \mathcal{F}_n$ if and only if $\mathcal{I}_f \neq \emptyset$: a firm is present in

every market once it activates at least one site. The within-sector price index is:

$$P_n = \left[\sum_{f \in \mathcal{F}_n} p_{nf}^{1-\sigma} \right]^{\frac{1}{1-\sigma}}. \quad (2.3)$$

Two-stage budgeting implies that total expenditure on the differentiated sector in market n is:

$$R_n = \Omega_n P_n^{1-\eta}, \quad (2.4)$$

where $\Omega_n \equiv E_n / \tilde{P}_n^{1-\eta} > 0$ collects the household's income and the aggregate price index $\tilde{P}_n$. Because the differentiated sector is measure zero in the continuum, $\tilde{P}_n$ is determined by the remaining sectors and is therefore exogenous to the oligopolistic firms. Firm f 's demand in market n is:

$$q_{nf} = p_{nf}^{-\sigma} P_n^{\sigma-\eta} \Omega_n. \quad (2.5)$$

Assumption 1 (Fixed sectoral expenditure). Sectoral expenditure $R_n = \Omega_n P_n^{1-\eta}$ is treated as exogenous: we fix R_n at its initial value and suppress the price-index feedback $P_n \rightarrow R_n$.

This assumption is maintained throughout the theoretical analysis and the quantitative illustration. The omitted channel—the price-index feedback $P_n \rightarrow R_n = \Omega_n P_n^{1-\eta}$ —has an ambiguous effect on uniqueness. For $\eta > 1$, a lower P_n raises R_n , which increases demand and makes entry more profitable: this positive feedback could either tighten or loosen the contraction condition depending on the relative magnitude of the demand amplification and the pricing dampening. A formal resolution would require bounding the spectral radius of the augmented Jacobian that includes the $P_n \rightarrow R_n$ loop jointly with the pricing contraction; we leave this extension to future work. The Jia–Tarski gap of zero at all tested parameterizations provides strong computational evidence that the equilibrium under fixed R_n satisfies aggregate supermodularity in the sense of Remark 7.

2.3 Production Technology and Costs

Firms produce using labor with heterogeneous productivity across production sites. Firm f with TFP a_f delivering one unit of output from production site i to market n faces unit cost:

$$c_{fin}(\omega) = \frac{w_i \tau_{in}}{a_f z(\omega)}, \quad (2.6)$$

where w_i is the wage at site i , $\tau_{in} \geq 1$ is the iceberg trade cost from site i to market n , and $z(\omega)$ is an i.i.d. Fréchet productivity draw with shape parameter $\theta > 0$ across varieties $\omega \in [0, 1]$.

A firm producing at multiple sites sources each variety from the lowest-cost site. As shown by [Eaton and Kortum \(2002\)](#)—and applied at the firm level by [Tintelnot \(2017\)](#) and [Antràs et al. \(2017\)](#)—the Fréchet assumption implies that the firm-level cost index aggregates across active production sites:

$$c_{nf} = \frac{\Gamma_\theta}{a_f} \left[\sum_{i \in \mathcal{I}_f} (w_i \tau_{in})^{-\theta} \right]^{-\frac{1}{\theta}}, \quad (2.7)$$

where $\mathcal{I}_f \subseteq \mathcal{A}_f$ is the set of production sites that firm f activates and $\Gamma_\theta = \Gamma(1 + \frac{1-\sigma}{\theta})^{\frac{1}{1-\sigma}}$ with $\Gamma(\cdot)$ the Gamma function. Higher TFP a_f lowers the cost index uniformly across all markets. Because Γ_θ is common across all firms, we normalize $\Gamma_\theta = 1$ without loss of generality; it can be absorbed into the TFP parameters $\{a_f\}$.

Remark 2 (Cost Complementarity). Equation (2.7) embodies a key complementarity: activating an additional production site i' adds a term $(w_{i'} \tau_{i'n})^{-\theta}$ to the sum inside the brackets, *reducing* the cost index c_{nf} in every market n . This cost reduction is larger when the firm already has fewer active sites—a form of increasing differences in the activation indicators. [Antràs et al. \(2017\)](#) establish that this cost complementarity generates supermodularity in the global sourcing problem under constant CES markups when $(\sigma - 1)/\theta > 1$.

2.4 Pricing and Markups

Firms compete à la Cournot: each firm chooses the quantity q_{nf} of its variety to supply to market n , taking rivals' quantities as given. The nested CES demand system with a finite number of firms generates endogenous variable markups through a mechanism first analyzed by [Atkeson and Burstein \(2008\)](#).

Firm f 's revenue share in market n is:

$$s_{nf} \equiv \frac{p_{nf}^{1-\sigma}}{P_n^{1-\sigma}} = \frac{p_{nf} q_{nf}}{\sum_{f'} p_{nf'} q_{nf'}}. \quad (2.8)$$

Under Cournot competition, the firm internalizes its effect on the sectoral aggregate Q_n , yielding a *perceived inverse elasticity* of demand (see [Atkeson and Burstein, 2008](#), Section II):

$$\frac{1}{\epsilon_{nf}} = \frac{1 - s_{nf}}{\sigma} + \frac{s_{nf}}{\eta}. \quad (2.9)$$

The first-order condition $p_{nf}(1 - 1/\epsilon_{nf}) = c_{nf}$ yields the optimal price as a markup over marginal cost:

$$p_{nf} = \mu_{nf} \cdot c_{nf}, \quad \mu_{nf} \equiv \frac{\epsilon_{nf}}{\epsilon_{nf} - 1}. \quad (2.10)$$

Remark 3 (Bounded Markups and the Cannibalization Channel). Equation (2.9) implies that the perceived inverse elasticity is a harmonic-mean weight between the two CES parameters σ and η :

- (i) **Small firm** ($s_{nf} \rightarrow 0$): $\epsilon_{nf} \rightarrow \sigma$, $\mu_{nf} \rightarrow \sigma/(\sigma - 1)$. The firm faces the within-sector CES markup.
- (ii) **Dominant firm** ($s_{nf} \rightarrow 1$): $\epsilon_{nf} \rightarrow \eta$, $\mu_{nf} \rightarrow \eta/(\eta - 1)$. The firm's pricing power is bounded by across-sector substitution.

Markups are bounded: $\mu_{nf} \in [\sigma/(\sigma - 1), \eta/(\eta - 1)]$. Activating additional production sites lowers marginal cost (2.7), which raises output and market share, thereby *compressing* the perceived elasticity and *raising* the markup. But the markup is bounded above by $\eta/(\eta - 1)$, so the self-cannibalization force is finite. This *bounded* cannibalization—a direct consequence of the nested CES structure ([Atkeson and Burstein, 2008](#))—competes against the cost complementarity of Remark 2. Whether supermodularity of the activation problem survives this tension is the central question of Section 3. A second, cross-firm channel arises because activation changes the rival demand index (2.11), feeding back into all firms' markups—a pricing interaction absent under constant markups that creates a new obstacle to uniqueness (Proposition 11).

Remark 4 (Cournot vs. Bertrand). Under Cournot, the perceived inverse elasticity is $1/\epsilon^C = (1 - s)/\sigma + s/\eta$ (equation 2.9). Under Bertrand with the same nested CES demand, the perceived elasticity is $\epsilon^B = \sigma(1 - s) + \eta s$, which differs for interior market shares: the two coincide only at $s = 0$ (both yield σ) and $s = 1$ (both yield η). The theoretical framework—pricing contraction via the interaction matrix, supermodularity

via cost-aggregator convexity—applies to both cases, but the quantitative conditions (the feedback coefficient κ , the per-market threshold, and the contraction modulus $\gamma_{\max}$) must be re-derived using the Bertrand markup function $\mu^B(s) = \epsilon^B/(\epsilon^B - 1)$. Since Bertrand markups are lower than Cournot markups at interior shares, the pricing contraction is typically weaker, but this must be verified case by case. We adopt the Cournot assumption throughout, which is natural for the multinational production application where firms choose production scale and capacity.

Rival demand index. The competitive environment facing firm f in market n is summarized by a *rival demand index*:

$$D_{-f}(n) = \sum_{f' \neq f, f' \in \mathcal{F}_n} p_{nf'}^{1-\sigma}, \quad (2.11)$$

which is a sufficient statistic for competition: firm f needs only $D_{-f}(n)$ to compute its optimal price and variable profit in market n . The firm's market share can be written as:

$$s_{nf} = \frac{p_{nf}^{1-\sigma}}{p_{nf}^{1-\sigma} + D_{-f}(n)}, \quad (2.12)$$

making explicit the dependence of the markup (2.9) on the rival demand index.

2.5 The Activation Problem

Each firm f chooses which production sites to activate. Let $I_{fi} \in \{0, 1\}$ be an indicator equal to 1 if firm f activates production site $i \in \mathcal{A}_f$, and 0 otherwise. The firm's *activation set* is:

$$\mathcal{I}_f = \{i \in \mathcal{A}_f : I_{fi} = 1\}. \quad (2.13)$$

Activating production site i incurs a fixed operating cost $\phi_i \cdot w_i$ in units of local labor, where ϕ_i may vary across countries (reflecting, e.g., differences in regulatory or infrastructure costs).

Firm f 's total static profit, given its own activation set $\mathcal{I}_f$ and the equilibrium activation

sets of all rivals $\mathcal{I}_{-f}^*$, is:

$$\pi_f(\mathcal{I}_f; \mathcal{I}_{-f}^*) = \sum_{n=1}^N \left[\text{rev}_{nf}(\mathcal{I}_f, \mathcal{I}_{-f}^*) \cdot \frac{\mu_{nf} - 1}{\mu_{nf}} \right] - \sum_{i \in \mathcal{I}_f} \phi_i w_i, \quad (2.14)$$

where $\text{rev}_{nf} = p_{nf} \cdot q_{nf}$ is firm f 's revenue in market n and $(\mu_{nf} - 1)/\mu_{nf}$ is the variable profit margin. Revenue and markups depend on the full industry activation profile: own activation enters through the cost index (2.7), and rivals' activation sets enter through the rival demand index $D_{-f}(n)$ defined in (2.11).

2.6 Static Equilibrium

Definition 5 (Static Equilibrium). A *static equilibrium* is a profile $(\{\mathcal{I}_f^*\}_{f \in \mathcal{F}}, \{\mu_{nf}^*\}, \{s_{nf}^*\})$ of activation sets, markups, and market shares such that:

E1. Profit maximization: For each firm f , the activation set $\mathcal{I}_f^*$ solves

$$\mathcal{I}_f^* \in \arg \max_{\mathcal{I}_f \subseteq \mathcal{A}_f} \pi_f(\mathcal{I}_f; \mathcal{I}_{-f}^*). \quad (2.15)$$

E2. Markup consistency: For each (n, f) with $\mathcal{I}_f^* \neq \emptyset$, the markup μ_{nf}^* satisfies the pricing equation (2.10) with perceived elasticity (2.9), and market shares s_{nf}^* are consistent with equilibrium prices via (2.12).

The static equilibrium involves a three-way circularity:

$$\underbrace{\text{Activation sets } \mathcal{I}_f}_{\text{combinatorial choice}} \longrightarrow \underbrace{\text{Market shares } s_{nf}}_{\text{CES aggregation}} \longrightarrow \underbrace{\text{Markups } \mu_{nf}}_{\text{AB pricing}} \longrightarrow \underbrace{\text{Profitability of } \mathcal{I}_f}_{\text{back to activation}}.$$

With constant markups (as in Antràs et al., 2017), the last link is severed: activation does not feed back through markups. The theoretical contribution of this article is to show that the CDCP structure—and in particular, supermodularity—survives this feedback (Section 3). A complementary computational contribution compares alternative algorithms for resolving the three-way fixed point (Section 4).

3. THEORETICAL RESULTS

This section establishes the theoretical properties of the static game. We first show that each firm's activation problem is supermodular (Proposition 6), then characterize the equilibrium set (Proposition 8) and establish convergence (Proposition 9). The main result—BNE uniqueness via a Jacobian decomposition—builds on the pricing-uniqueness condition of Section 3.1 and is stated formally in Section 3.3.

Since the rival demand index $D_{-f}(n)$ is a sufficient statistic for competition (Section 2.4), we write $\pi_f(\mathcal{I}_f; D_{-f})$ for firm f 's profit (2.14) evaluated at the unique pricing equilibrium given D_{-f} .

Proposition 6 (Single-Agent Supermodularity). *Fix the rival demand index $D_{-f}(n)$ for all markets n and suppose $(\sigma - 1)/\theta > 1$.*

- (i) *Per-market sufficient condition (CES benchmark). If the equilibrium market shares satisfy $\max_{n,f} s_{nf} < \bar{s}^*$, where $\bar{s}^* \equiv (\alpha - 1)/(2\alpha)$ with $\alpha = (\sigma - 1)/\theta$ is a tractable conservative threshold derived under constant CES markups (Appendix A, Step 4), then each market's contribution to variable profit is convex in the cost aggregator, which implies that firm f 's activation profit $\pi_f(\mathcal{I}_f; D_{-f})$ exhibits increasing differences in the activation indicators $\{I_{fi}\}_{i \in \mathcal{A}_f}$. Under AB endogenous markups, the exact per-market condition is given implicitly by $H_{AB}(s) \geq 0$ (Appendix A, Step 3'):*

$$\pi_f(\mathcal{I}_f \cup \{i, i'\}; D_{-f}) - \pi_f(\mathcal{I}_f \cup \{i\}; D_{-f}) \geq \pi_f(\mathcal{I}_f \cup \{i'\}; D_{-f}) - \pi_f(\mathcal{I}_f; D_{-f}). \quad (3.1)$$

- (ii) *Computational evidence. When some market shares exceed $\bar{s}^*$, the per-market condition is not met, but aggregate supermodularity across all markets can still hold because low-share markets (positive curvature) dominate high-share markets (negative curvature). The Jia–Tarski algorithm provides strong computational evidence for aggregate supermodularity whenever the upper and lower bounds coincide (gap = 0); the coincidence is a necessary condition, not an analytical proof, and we do not currently have a closed-form sufficient condition for aggregate supermodularity under heterogeneous market shares.*

Under either (i) or (ii), the set of optimal solutions to $\max_{\mathcal{I}_f \subseteq \mathcal{A}_f} \pi_f(\mathcal{I}_f; D_{-f})$ has a greatest element and a least element, each attainable by the Jia–Tarski algorithm.

Proof sketch. The marginal return to activating site i' can be decomposed into two effects. The *cost complementarity effect*: by equation (2.7), adding site i' contributes a term $(w_{i'}\tau_{i'n})^{-\theta}$ to the cost index in every market n . The proportional cost reduction is larger when the existing cost index is higher—i.e., when fewer other sites are already active. Under the condition $(\sigma - 1)/\theta > 1$, variable profit is sufficiently convex in the cost index that the scale expansion from lower costs generates increasing differences.

The *self-cannibalization effect*: activating i' lowers cost, which lowers price, which raises market share s_{nf} , which compresses the perceived elasticity ϵ_{nf} via (2.9) and raises the markup μ_{nf} . This partially offsets the quantity gain. Under the CES benchmark (constant markups), per-market convexity of variable profit in the cost aggregator $\Sigma_{fn} \equiv \sum_{i \in I_f} (w_i \tau_{in})^{-\theta}$ from (2.7) requires $s < \bar{s}^*$. Under AB endogenous markups, the exact per-market condition is $H_{AB}(s) \geq 0$, where H_{AB} is an explicit function of $(s, \sigma, \eta, \theta)$ derived in Appendix A, Step 3'; the AB threshold is parameter-dependent and need not exceed the CES threshold. When some shares exceed both thresholds, aggregate supermodularity is verified via part (ii).

The condition $(\sigma - 1)/\theta > 1$ is the same as in Antràs et al. (2017), Theorem 1, who establish supermodularity under constant markups. The extension here shows that the additional markup feedback can be incorporated analytically at the per-market level and does not overturn supermodularity in the tested parameterizations. Full proof in Appendix A. $\square$

Under Atkeson–Burstein endogenous markups, the exact per-market convexity condition $H_{AB}(s) \geq 0$ defines a *unique* implicit threshold $\bar{s}_{AB}^*$ (Appendix A, Proposition 36): H_{AB} is strictly concave, so the threshold is the unique zero of a concave function that is positive at $s = 0$ and negative at $s = 1$. The AB threshold is parameter-dependent and need not uniformly exceed the CES benchmark $\bar{s}^*$. In the cloud application (Section 5), $\max_{n,f} s_{nf} = 0.40$ exceeds both thresholds, so aggregate supermodularity is supported by part (ii) as a computational diagnostic: the Jia–Tarski gap is zero at all iterations, a necessary consequence of aggregate supermodularity at the tested parameterization (see Remark 7).

Remark 7 (Logical status of supermodularity). Part (i) of Proposition 6 provides an *analytical sufficient condition*: if all market shares lie below $\bar{s}^*$, supermodularity is proven. Part (ii) provides a *computational verification procedure*: when some shares exceed $\bar{s}^*$ (as in

the cloud application, where $\max s_{nf} = 0.40 > \bar{s}^*$), the Jia–Tarski gap = 0 certifies that the algorithm found matching upper and lower bounds on the optimal activation set—a necessary consequence of supermodularity. This is verification of a consequence (lattice optimum coincidence) rather than a direct proof of the increasing-differences property at every activation subset. The distinction is immaterial for the downstream propositions (Propositions 8–21), which require only that the single-agent optimum has lattice structure, but the reader should note the logical gap: the per-market analytical condition is *sufficient* and *proven*; the aggregate computational certification is *sufficient* and *verified*. Both pathways are sufficient for all results that follow. For tolerance hygiene on the computational pathway, the Jia–Tarski MV-sign test uses a threshold of 10^{-12} ; a diagnostic counter (`jia_tarski.m`, `output_near_zero_count`) records the number of evaluations with $|\text{MV}| \in [10^{-14}, 10^{-10}]$, i.e., inside the tolerance band or within two orders of magnitude of it. Across every iteration of the reported calibration and counterfactual runs this counter equals zero: the gap-zero certificate is structural, not tolerance-fragile.

Existence. A Nash equilibrium of the complete-information game exists by Nash’s theorem, since the activation lattice $L = \prod_f 2^{\mathcal{A}_f}$ is finite. The following two propositions characterize the equilibrium set and establish convergence of iterative algorithms.

Proposition 8 (Equilibrium Bounds). *Suppose each firm’s activation profit is supermodular (Proposition 6).*

- (i) *The joint best-response map Φ is anti-monotone (order-reversing) on the joint activation lattice L , so the squared map $\Phi^2 = \Phi \circ \Phi$ is monotone. By Tarski’s (1955) fixed-point theorem, Φ^2 has a greatest fixed point $\bar{\mathbf{I}}$ and a least fixed point $\underline{\mathbf{I}}$ on L .*
- (ii) *Any Nash equilibrium $\mathbf{I}^*$ satisfies $\underline{\mathbf{I}} \leq \mathbf{I}^* \leq \bar{\mathbf{I}}$: the extremal fixed points of Φ^2 bound the entire Nash equilibrium set.*
- (iii) *If $\bar{\mathbf{I}}$ is itself a Nash equilibrium (i.e., $\Phi(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$, not merely a two-cycle of Φ), it is the greatest Nash equilibrium.*

Proof. Since activation choices are *strategic substitutes*—when rival firms activate more sites, the rival demand index D_{-f} increases, reducing the profitability of firm f ’s sites—the joint best-response map $\Phi : (I_1, \dots, I_F) \mapsto (I_1^*, \dots, I_F^*)$ is *anti-monotone*: $\mathbf{I} \leq \mathbf{I}'$ implies

$\Phi(\mathbf{I}) \geq \Phi(\mathbf{I}')$ (cf. [Amir, 1996](#)), where $\leq$ is the component-wise set-inclusion order on $L \equiv \prod_f 2^{\mathcal{A}_f}$.

Part (i): Φ^2 is monotone because anti-monotonicity composes to give $\mathbf{I} \leq \mathbf{I}' \Rightarrow \Phi^2(\mathbf{I}) \leq \Phi^2(\mathbf{I}')$. Tarski's theorem on the finite complete lattice L yields the extremal fixed points.

Part (ii): any Nash equilibrium satisfies $\Phi(\mathbf{I}^*) = \mathbf{I}^*$, hence $\Phi^2(\mathbf{I}^*) = \mathbf{I}^*$. By Tarski's theorem, $\underline{\mathbf{I}} \leq \mathbf{I}^* \leq \bar{\mathbf{I}}$.

Part (iii): if $\Phi(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$, then $\bar{\mathbf{I}}$ is a Nash equilibrium; any other Nash equilibrium $\mathbf{I}^*$ satisfies $\mathbf{I}^* \leq \bar{\mathbf{I}}$ by (ii).

In general, $\bar{\mathbf{I}}$ may be a two-cycle rather than a Nash equilibrium: it is possible that $\Phi(\bar{\mathbf{I}}) \neq \bar{\mathbf{I}}$ while $\Phi^2(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$. However, at all parameterizations tested in Section 5 and Appendix G, $\Phi(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$ and the equilibrium is unique. Full proof in Appendix B. $\square$

Proposition 9 (Convergence of Jacobi Iteration). *Suppose each firm's activation profit is supermodular. Starting from the supremum $\hat{\mathbf{I}} = (\mathcal{A}_1, \dots, \mathcal{A}_F)$, the even subsequence $\{\Phi^{2k}(\hat{\mathbf{I}})\}$ is decreasing and converges to $\bar{\mathbf{I}}$ (the greatest fixed point of Φ^2). The odd subsequence $\{\Phi^{2k+1}(\hat{\mathbf{I}})\}$ is increasing and converges to $\Phi(\bar{\mathbf{I}})$. If $\Phi(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$, the full sequence converges to the greatest Nash equilibrium. Analogously from the infimum.*

Proof. Since Φ is anti-monotone and $\Phi(\hat{\mathbf{I}}) \leq \hat{\mathbf{I}}$, the even subsequence $\{\Phi^{2k}(\hat{\mathbf{I}})\}$ is decreasing (by monotonicity of Φ^2) and the odd subsequence $\{\Phi^{2k+1}(\hat{\mathbf{I}})\}$ is increasing, with $\Phi^{2k+1}(\hat{\mathbf{I}}) \leq \Phi^{2k}(\hat{\mathbf{I}})$ for all k . On the finite lattice L , both subsequences stabilize in finitely many steps. The even limit is $\bar{\mathbf{I}}$ and the odd limit is $\Phi(\bar{\mathbf{I}})$, which satisfy $\Phi^2(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$ and $\Phi(\bar{\mathbf{I}}) \leq \bar{\mathbf{I}}$. Full proof in Appendix B. $\square$

Remark 10 (Gauss–Seidel convergence). The sequential update map Ψ that updates firms in order $(f_1, \dots, f_F)$ using the latest available rivals' strategies converges at all tested parameterizations (Section 5 and Appendix G). Formal convergence of Ψ under anti-monotone best responses is not guaranteed in general—sequential updating breaks the even/odd subsequence structure that yields monotonicity of Φ^2 —but empirical convergence is robust across all parameterizations tested.

At all parameterizations tested in Section 5 and Appendix G, $\Phi(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$: both Jacobi and Gauss–Seidel converge to the same Nash equilibrium, and no two-cycles are observed. Gauss–Seidel uses updated rival information within each round and typically converges

in fewer rounds, whereas Jacobi is naturally parallelizable. Section 4 presents the full comparison.

3.1 Pricing Uniqueness and the Intensive Margin

The uniqueness question has two distinct margins. On the *intensive margin* (pricing), uniqueness requires that the AB markup fixed point is contractive for any given activation profile. On the *extensive margin* (site activation), uniqueness requires that no “marginal sites”—sites whose profitability depends on rival behavior—switch activation status across equilibria. This subsection resolves the intensive margin via a contraction condition on the feedback coefficient (Proposition 11); Section 3.3 resolves the extensive margin by introducing private information over fixed costs.

Feedback coefficient. The strength of the strategic interaction through the AB markup channel is captured by a *feedback coefficient*. For firm f in market n with share s_{nf} and markup μ_{nf} , define:

$$\kappa_{nf} = \frac{s_{nf}^2 \beta_{nf}}{1 + s_{nf}(1 - s_{nf}) \beta_{nf}}, \quad \beta_{nf} = \frac{(\sigma - 1) \mu_{nf} (\sigma - \eta)}{\sigma \eta}. \quad (3.2)$$

The coefficient $\kappa_{nf} = \partial p_{nf}^{1-\sigma} / \partial D_{-f}(n)$ measures how firm f 's pricing power responds to a change in the rival demand index, holding the activation set fixed. When $s_{nf} = 0$ (negligible firm), $\kappa_{nf} = 0$; as s_{nf} rises, the markup channel amplifies the pricing response, so κ_{nf} increases.

Proposition 11 (Pricing Uniqueness). *For any fixed activation profile $(\mathcal{I}_1, \dots, \mathcal{I}_F)$, define the markup interaction matrix Λ with entries*

$$\Lambda_{(n,f),(n,g)} = \kappa_{nf} \cdot \frac{p_{ng}^{1-\sigma}}{D_{-f}(n)}, \quad g \neq f, \quad (3.3)$$

with all other entries zero. If the spectral radius satisfies

$$\rho(\Lambda) < 1, \quad (3.4)$$

then the AB pricing equilibrium is unique.

Proof. For a fixed activation profile, the AB pricing system defines a map $\Psi : D \mapsto D'$ on the space of rival demand indices. The Jacobian of Ψ is bounded entrywise by Λ , which is nonnegative. When $\rho(\Lambda) < 1$, the Perron–Frobenius theorem guarantees a positive vector $w \gg 0$ and a scalar $\lambda < 1$ such that $\Lambda w \leq \lambda w$. Define the weighted sup-norm $\|x\|_w = \max_i |x_i|/w_i$. By the mean value theorem, for any two demand vectors D, D' :

$$\|\Psi(D) - \Psi(D')\|_w \leq \lambda \|D - D'\|_w,$$

so Ψ is a contraction in $\|\cdot\|_w$. Since the domain is a compact convex subset of $\mathbb{R}_+^{NF}$ (bounded markups imply bounded demand indices), Banach’s fixed-point theorem yields a unique pricing equilibrium. $\square$

The matrix Λ is block-diagonal across markets (N blocks of size $F \times F$), so $\rho(\Lambda) = \max_n \rho(\Lambda_n)$ is cheap to compute. A conservative closed-form sufficient condition uses the infinity norm:

Corollary 12. *If $\max_{n,f} \kappa_{nf} < 1/(F - 1)$, then $\rho(\Lambda) < 1$ and the pricing equilibrium is unique.*

At the cloud application’s equilibrium profile ($F = 5$, Section 5), both conditions are satisfied: $\max \kappa = 0.193 < 1/(F - 1) = 0.25$ and $\rho(\Lambda) = 0.103$. *Off-equilibrium, the infinity-norm condition is tight.* Across the 1,302 per-market subgames enumerated in Proposition 28 (Table 3), $\|\Lambda_n\|_\infty$ exceeds $1/(F - 1)$ in roughly 12% of subgames, peaking at a CES-ratio of $5.1\times$ the sufficient-condition threshold, yet $\rho(\Lambda_n) \leq 0.316$ throughout. This gap between the row-sum bound and the spectral radius motivates the spectral refinement of the downstream Jacobian bound in Proposition 37: the Neumann-series amplification $1/(1 - \|\Lambda\|_\infty)$ is not available uniformly, but the Perron–Frobenius amplification $1/(1 - \rho(\Lambda))$ is. For applied work, the spectral radius (3.4) is the recommended diagnostic. At the synthetic economy ($F = 8$, Appendix G), the same pattern obtains: $\max \kappa = 0.193 > 1/(F - 1) = 0.143$ but $\rho(\Lambda) = 0.096$.

Proposition 11 resolves the intensive margin but does *not* guarantee uniqueness of the full game, because different activation profiles lead to different demand indices. Define the composite map $\tilde{\Phi} : D \mapsto D'$ that maps rival demands to best-response activation sets, then evaluates the unique pricing equilibrium to produce new demands. Multiple equilibria arise when $\tilde{\Phi}$ has multiple fixed points—equivalently, when there exist marginal

sites whose activation profit $\Delta\pi_{if}$ changes sign across equilibrium demands. The number of marginal sites is the key diagnostic for multiplicity; the cloud application (Section 5.5) and synthetic stress tests (Appendix G) show it is zero at all tested parameterizations. To resolve the extensive margin analytically, we introduce private information over fixed costs, converting the discrete activation problem into a continuous threshold system on which contraction arguments apply. Section 3.2 states the uniqueness result at the abstract level of a generic threshold map; Section 3.3 then verifies its hypotheses under the Fréchet–CES–AB environment of Section 2; Section 3.4 verifies them under nested logit on the consumer side; and Section 3.5 discusses further environments.

3.2 Abstract Threshold-Uniqueness Framework

The uniqueness argument below operates at a level of abstraction broader than the Fréchet–CES–AB environment of Section 2. What the nested-CES demand structure supplies is a *tractable* verification environment in which the smoothness of the marginal-value map, the contraction of the pricing block, and the complete-information gap can all be checked in closed form. None of those ingredients is specific to CES: any Bayesian combinatorial entry game whose threshold map is smooth and whose contraction modulus is below one admits a unique threshold-strategy BNE. This subsection records the abstract result; Section 3.3 supplies the CES–AB verification; Section 3.4 supplies a nested-logit verification on the consumer side; Section 3.5 discusses further environments.

Abstract setup. Fix a combinatorial entry game with F firms and site sets $(\mathcal{A}_f)_{f=1}^F$; let $M = \sum_f |\mathcal{A}_f|$ be the total number of site-firm pairs. Each firm f privately observes site-specific shocks $\{\varepsilon_{if}\}_{i \in \mathcal{A}_f}$ i.i.d. across (i, f) with CDF $G_\sigma(\varepsilon) = G_0(\varepsilon/\sigma)$, where G_0 is a standardized continuously-differentiable CDF on $\mathbb{R}$ with density g_0 and scale parameter $\sigma > 0$. The symbol σ in the abstract framework plays the role of σ_ε in Assumption 18; we retain the shorter notation here because the abstract theorem never interacts with the CES elasticity σ of Section 2, which appears only in the concrete verification of Section 3.3. Attention is restricted to the class of site-independent threshold strategies defined in the CES–AB setting by Assumption 18 and its nested-logit analog in Assumption 30 below. Let $\nu \in [0, 1]^M$ collect the site-firm activation probabilities induced by a profile of thresholds, and let $v : \mathcal{N} \rightarrow \mathbb{R}^M$ be a C^1 map on an open neighborhood $\mathcal{N}$ of $[0, 1]^M$ returning

the scaled expected per-site marginal value at an activation-probability profile: $v_m(\nu)$ aggregates the firm's own expected within-firm complementarities over sibling shocks and the rival pricing equilibrium induced by ν into a single scalar threshold index. The threshold map is

$$T_\sigma : [0, 1]^M \rightarrow [0, 1]^M, \quad T_{\sigma, m}(\nu) = G_0\left(\frac{v_m(\nu)}{\sigma}\right), \quad m = 1, \dots, M, \quad (3.5)$$

and the contraction modulus is

$$\gamma(\sigma) \equiv \sup_{\nu \in [0, 1]^M} \|J_{T_\sigma}(\nu)\|_\infty. \quad (3.6)$$

Every site-independent threshold-strategy BNE corresponds to a fixed point of T_σ ; the verification environments of Sections 3.3 and 3.4 make this correspondence explicit.

Theorem 13 (Abstract Threshold-Strategy Uniqueness at Fixed Noise). *Let G_0 be a continuously differentiable CDF on $\mathbb{R}$ and let v be C^1 on an open neighborhood of $[0, 1]^M$. Then*

- (i) *the threshold map T_σ defined in (3.5) has at least one fixed point on $[0, 1]^M$;*
- (ii) *its Jacobian admits the factorization*

$$J_{T_\sigma}(\nu) = \text{diag}\left(\frac{g_0(v_m(\nu)/\sigma)}{\sigma}\right)_{m=1}^M J_v(\nu); \quad (3.7)$$

- (iii) *if $\gamma(\sigma) < 1$, the fixed point is unique.*

Proof. Part (i): T_σ is a continuous self-map of the nonempty compact convex set $[0, 1]^M$ (continuity of v follows from C^1 regularity; continuity of G_0 is built in), so Brouwer's fixed-point theorem applies. Part (ii): Differentiate $T_{\sigma, m}(\nu) = G_0(v_m(\nu)/\sigma)$ componentwise; the chain rule gives $\partial T_{\sigma, m}/\partial \nu_\ell = g_0(v_m/\sigma) \cdot (1/\sigma) \cdot \partial v_m/\partial \nu_\ell$, which is (3.7). Part (iii): For any $\nu, \nu' \in [0, 1]^M$, convexity places the segment $\nu_t = \nu' + t(\nu - \nu')$ inside $[0, 1]^M$; the fundamental theorem of calculus and the induced-norm inequality give

$$\|T_\sigma(\nu) - T_\sigma(\nu')\|_\infty \leq \int_0^1 \|J_{T_\sigma}(\nu_t)\|_\infty dt \cdot \|\nu - \nu'\|_\infty \leq \gamma(\sigma) \|\nu - \nu'\|_\infty.$$

Banach's fixed-point theorem on the complete metric space $([0, 1]^M, \|\cdot\|_\infty)$ then yields uniqueness. $\square$

Corollary 14 (Global-in- σ Uniqueness). *Under the hypotheses of Theorem 13, if $\gamma_{\max} \equiv \sup_{\sigma>0} \gamma(\sigma) < 1$, then the threshold-strategy BNE is unique for every $\sigma > 0$.*

The proof is immediate: any $\sigma > 0$ satisfies $\gamma(\sigma) \leq \gamma_{\max} < 1$, so Theorem 13(iii) applies at that σ . The finite peak $\gamma_{\max}$ is the single primitive-level number whose control certifies uniqueness uniformly across σ ; the verification sections below produce explicit decompositions of $\gamma(\sigma)$ into density dampening and marginal-value sensitivity that render $\gamma_{\max} < 1$ verifiable by evaluation on a grid of σ .

Lemma 15 (Large-Noise Tail). *Assume additionally that $\|g_0\|_\infty < \infty$ and $K_v \equiv \sup_{v \in [0, 1]^M} \|J_v(v)\|_\infty < \infty$. Then $\gamma(\sigma) \leq \|g_0\|_\infty K_v / \sigma$, and in particular $\gamma(\sigma) \rightarrow 0$ as $\sigma \rightarrow \infty$.*

Proof. Take norms in the factorization (3.7): $\|J_{T_\sigma}(v)\|_\infty \leq (\|g_0\|_\infty / \sigma) \|J_v(v)\|_\infty \leq \|g_0\|_\infty K_v / \sigma$; take the supremum over $v \in [0, 1]^M$. $\square$

The analog of Lemma 15 as $\sigma \rightarrow 0$ is subtler. The modulus $\gamma(\sigma)$ is a global supremum over all off-equilibrium profiles $v \in [0, 1]^M$, so a strictly positive complete-information gap at the equilibrium profile does not automatically push the density factor to zero at *every* off-equilibrium v : some v may keep $|v_m(v)|$ small even when the equilibrium profile has strictly separated thresholds. The following lemma closes the gap by *localizing* the contraction check to an invariant set on which all threshold indices are uniformly bounded away from zero.

Lemma 16 (Small-Noise Localization). *Suppose there exists a family $\{S_\sigma\}_{0 < \sigma < \sigma_0} \subseteq [0, 1]^M$ of nonempty, closed, convex sets such that:*

- (i) $T_\sigma(S_\sigma) \subseteq S_\sigma$ for every $\sigma \in (0, \sigma_0)$ (invariance);
- (ii) every fixed point of T_σ in $[0, 1]^M$ lies in S_σ (fixed-point inclusion);
- (iii) there exists $b > 0$ with $\inf_{v \in S_\sigma} \min_m |v_m(v)| \geq b$ for every $\sigma \in (0, \sigma_0)$ (uniform non-marginality);
- (iv) $|z|g_0(z) \rightarrow 0$ as $|z| \rightarrow \infty$ (tail regularity on the shock density);

(v) $K_S \equiv \sup_{0 < \sigma < \sigma_0} \sup_{v \in S_\sigma} \|J_v(v)\|_\infty < \infty$ (uniform J_v bound on S_σ).

Then $\sup_{v \in S_\sigma} \|J_{T_\sigma}(v)\|_\infty \rightarrow 0$ as $\sigma \rightarrow 0$. In particular, for all σ sufficiently small, T_σ is a contraction on $(S_\sigma, \|\cdot\|_\infty)$, and by (ii) its fixed point is globally unique.

Proof. Fix $v \in S_\sigma$. Each m satisfies $|v_m(v)| \geq b$ by (iii), so

$$\frac{g_0(v_m(v)/\sigma)}{\sigma} = \frac{1}{|v_m(v)|} \left| \frac{v_m(v)}{\sigma} g_0\left(\frac{v_m(v)}{\sigma}\right) \right| \leq \frac{1}{b} \sup_{|z| \geq b/\sigma} |z| g_0(z).$$

By (iv), the right-hand side vanishes as $\sigma \rightarrow 0$ because $b/\sigma \rightarrow \infty$. Combining with (3.7) and the uniform J_v bound from (v) yields

$$\sup_{v \in S_\sigma} \|J_{T_\sigma}(v)\|_\infty \leq \frac{K_S}{b} \sup_{|z| \geq b/\sigma} |z| g_0(z) \rightarrow 0 \text{ as } \sigma \rightarrow 0.$$

Contraction on $(S_\sigma, \|\cdot\|_\infty)$ follows from the same Banach argument as Theorem 13(iii); invariance (i) makes S_σ a complete metric subspace into which iterates fall, and fixed-point inclusion (ii) promotes the local uniqueness conclusion to a global one. $\square$

Remark 17 (Verification strategy across environments). Theorem 13, Corollary 14, and Lemmas 15–16 reduce the uniqueness problem to three structural checks on the demand and payoff environment:

- (a) *Smooth threshold index.* The marginal-value map $v(v)$ must be C^1 . Under Fréchet–CES–AB, this follows from Proposition 11: the pricing equilibrium selection $v \mapsto (p, s)$ is smooth whenever $\rho(\Lambda) < 1$, and expected profits integrate smoothly over the IID sibling shocks. Under nested logit (Section 3.4), v inherits smoothness from the Bertrand best-response map under log-concavity of the logit shocks.
- (b) *Bounded J_v .* The operator norm $\|J_v\|_\infty$ must be uniformly finite. Both CES–AB (Appendix C, Proposition 37) and nested logit deliver explicit Lipschitz bounds involving the Perron–Frobenius amplification $1/(1 - \rho(\Lambda))$ of the pricing block.
- (c) *Small- σ localization.* When the complete-information game has a unique pure-strategy Nash equilibrium with positive marginal-value gap $\delta_v = \min_{i,f} |\Delta \pi_{if}^*|/w_i > 0$, the box $S_\sigma = \{v : \min_m |v_m(v)| \geq b\}$ with $b = \delta_v/2$ supplies the invariant, fixed-point-inclusive, and non-marginal family required by Lemma 16. In the verification

environments developed below, the gap is inherited from the single-firm CDCP's supermodularity (Proposition 6) and from the non-coincidence of the Jia–Tarski supremum and infimum iterates.

The abstract theorem is the uniqueness engine; each verification environment does the application-specific work of constructing a smooth v , bounding J_v , and supplying the non-marginal localization.

3.3 Verification I: Fréchet–CES–AB Threshold System

We now verify the hypotheses of Theorem 13 and Corollary 14 for the Fréchet–CES–AB environment of Section 2. Firms' fixed costs are not publicly observed in practice; we introduce private information over fixed costs, which serves two purposes: it captures the empirical reality that site-specific deployment costs are unobserved, and it converts the discrete activation problem into the continuous threshold system (3.5) on which the abstract contraction argument applies.

Assumption 18 (Private cost shocks and threshold strategies). (*Information.*) Each firm f privately observes fixed-cost shocks $\{\varepsilon_{if}\}_{i \in \mathcal{A}_f}$ that are i.i.d. across sites and across firms, drawn from a distribution G with full-support density g , mean zero, and scale parameter $\sigma_\varepsilon > 0$. The total fixed cost of activating site i is $(\phi_i + \varepsilon_{if})w_i$. Firm f observes its own draws but not rivals'. (*Strategy space.*) Firm f 's pure strategy is a measurable decision rule $\sigma_f : \mathbb{R}^{|\mathcal{A}_f|} \rightarrow 2^{\mathcal{A}_f}$ of the site-independent threshold form: there exist cutoffs $(\bar{\varepsilon}_{if})_{i \in \mathcal{A}_f}$ such that site i is activated if and only if $\varepsilon_{if} < \bar{\varepsilon}_{if}$, independently across sites within the firm. The cutoffs $\bar{\varepsilon}_{if}$ depend on rival activation probabilities and on own expected complementarities, but not on the realized own-site shocks ε_{jf} for $j \neq i$.

Remark 19 (Why the threshold-strategy restriction). Two comments on the strategy-space restriction in Assumption 18. First, restricting to site-independent threshold strategies is *not* a free consequence of supermodularity plus i.i.d. shocks: under full joint observation of ε_f and supermodular within-firm payoffs, a firm could in principle condition its site- i decision on realized ε_{jf} through the complementarity channel, producing an activation region with step (not product-form) structure. We rule this out by restriction rather than by proof. Second, the restriction has a natural economic interpretation: in

decentralized multinational or platform-entry organizations, each country’s entry is decided by a local team on the basis of local cost conditions, without conditioning in real time on the realized idiosyncratic draws at other sites. The maintained restriction is what permits the threshold-system representation on $[0, 1]^M$ used in Proposition 21 and Theorem 24. All uniqueness claims below are statements about the unique BNE *within the class of site-independent threshold strategies*; when the complete-information game has a unique pure-strategy Nash equilibrium and the bathtub peak $\gamma_{\max}$ is below one, no other threshold-strategy BNE exists.

The i.i.d. assumption is substantive: it ensures that each site’s activation decision reduces to a scalar threshold comparison (Proposition 21(i)), enabling the Jacobian factorization (3.11). With correlated shocks—for example, a hierarchical structure $\varepsilon_{if} = \eta_f + \tilde{\varepsilon}_{if}$ with a common firm-level component η_f —the activation decision for site i would depend on η_f through the conditional expected marginal value, yielding a threshold that varies with the firm-level draw rather than an unconditional scalar. The Jacobian factorization breaks because the density factor becomes conditional on η_f . A partial extension is possible: conditional on η_f , the site-level shocks $\tilde{\varepsilon}_{if}$ remain i.i.d., so a conditional version of the contraction argument applies at each realization, but the fixed point is then over a higher-dimensional system (one threshold per site per firm-level type) and the contraction condition is harder to verify. In practice, the i.i.d. assumption is standard in the CDCP literature (Jia, 2008; Arkolakis et al., 2023); the restriction here is no more binding than in those frameworks.

Economically, the i.i.d. assumption rules out two forms of dependence that arise naturally in multinational production. First, *within-firm correlation*: a firm-level “headquarters quality” draw that shifts fixed costs at all sites simultaneously—for example, a management practice that raises or lowers setup costs uniformly across countries. Second, *spatial correlation*: regulatory or institutional costs that co-move across neighboring countries sharing a border or regulatory regime. Either form of dependence would break the one-dimensional threshold structure exploited in Proposition 21(i).

Two distinct roles of the i.i.d. assumption deserve emphasis. Independence across sites *within* a given firm combined with the threshold-strategy restriction in Assumption 18 ensures that each site’s activation reduces to a scalar threshold comparison, enabling the Jia–Tarski algorithm and the Jacobian factorization (3.11). Independence *across firms* makes

each firm's threshold $\bar{\varepsilon}_{if}$ a deterministic function of rival activation probabilities rather than of rival cost realizations, so the cross-player Jacobian J_T has the product structure that supports the contraction argument. With correlated shocks across firms—for example, a common macroeconomic shock—the thresholds would be conditional on the common state, and the contraction condition would need to hold state-by-state (Proposition 20).

Extension: across-firm common shocks. The state-by-state requirement is easy to satisfy in the paper's framework. Consider a public-state decomposition

$$\varepsilon_{if} = \zeta + \tilde{\varepsilon}_{if}, \quad \zeta \in \mathbb{R}, \quad \tilde{\varepsilon}_{if} \stackrel{\text{i.i.d.}}{\sim} G, \quad (3.8)$$

where ζ is a common shock observed by all firms (e.g., a cost-of-capital shock, a country-cluster regulatory tightening) and $\tilde{\varepsilon}_{if}$ are residual idiosyncratic draws satisfying Assumption 18. Conditional on ζ , the game coincides with the paper's baseline model after the deterministic relabeling $\phi_i \mapsto \phi_i + \zeta$: the threshold map, its Jacobian, and the contraction modulus $\gamma(\sigma_\varepsilon \mid \zeta)$ are the same objects evaluated at shifted mean fixed costs. The following proposition records the consequence.

Proposition 20 (BNE uniqueness under across-firm common shock). *Fix a common-shock distribution on ζ with support $\text{supp}(\zeta) \subseteq \mathbb{R}$, and consider the strategy space of measurable site-independent threshold rules with publicly-observed ζ -conditioning: firm f 's strategy is a measurable map $\sigma_f : \mathbb{R}^{|\mathcal{A}_f|} \times \text{supp}(\zeta) \rightarrow 2^{\mathcal{A}_f}$ of the form $\sigma_f(\varepsilon_f, \zeta) = \{i \in \mathcal{A}_f : \varepsilon_{if} < \bar{\varepsilon}_{if}(\zeta)\}$, where $\zeta \mapsto \bar{\varepsilon}_{if}(\zeta)$ is Borel-measurable. Suppose that hypotheses (a)–(e) of Theorem 24 hold at every $\zeta \in \text{supp}(\zeta)$ with the shifted fixed costs $(\phi_i + \zeta)_{i=1}^N$. Under the decomposition (3.8), the BNE in this strategy space exists and is unique state-by-state whenever*

$$\sup_{\zeta \in \text{supp}(\zeta)} \gamma_{\max}(\zeta) < 1, \quad (3.9)$$

where $\gamma_{\max}(\zeta) \equiv \sup_{\sigma_\varepsilon > 0} \gamma(\sigma_\varepsilon \mid \zeta)$ is the bathtub peak evaluated at shifted fixed costs $(\phi_i + \zeta)_{i=1}^N$.

Proof. Conditional on each fixed ζ , the game is Assumption 18 applied to $\tilde{\varepsilon}_{if}$ with deterministic fixed costs $(\phi_i + \zeta)w_i$. Propositions 6–21 and Theorem 24 apply unchanged, so the conditional BNE cutoff vector $\bar{\varepsilon}(\zeta) = (\bar{\varepsilon}_{if}(\zeta))_{i,f}$ exists and is unique at each ζ provided $\gamma_{\max}(\zeta) < 1$. Taking the supremum over $\text{supp}(\zeta)$ delivers uniform uniqueness across

Table 1: Across-firm common-shock robustness

$\zeta/\bar{\phi}$	ζ	$\max \ p_{\text{sup}} - p_{\text{inf}}\ _{\infty} \text{ at } \sigma_{\varepsilon}/\bar{\phi} =$			BNE unique?
		0.10	0.30	1.00	
-0.40	-0.0493	5.3e-08	7.1e-08	2.9e-07	yes
-0.20	-0.0247	7.3e-09	1.8e-07	2.8e-07	yes
-0.10	-0.0123	1.1e-10	2.9e-07	2.6e-07	yes
+0.00	+0.0000	6.1e-13	1.6e-07	2.5e-07	yes
+0.10	+0.0123	2.7e-15	1.8e-07	2.4e-07	yes
+0.20	+0.0247	2.0e-18	8.2e-08	2.2e-07	yes
+0.40	+0.0493	5.7e-26	2.5e-08	1.8e-07	yes

Notes: Small-economy calibration ($N = 3$, $F = 4$; see Section 3.7). For each common-shock realization ζ , the BNE fixed point is solved from three starting points (supremum, infimum, uniform) at three values of σ_{ε} bracketing the bathtub peak. Reported entries are the maximum pairwise $\|p_{\text{sup}} - p_{\text{inf}}\|_{\infty}$ distance across the three starting points. “BNE unique” is yes when the distance is below 10^{-3} at all three σ_{ε} . The $\zeta = 0$ row coincides with the paper’s baseline IID case. Uniqueness holds state-by-state across the full $\pm 40\%$ cost shift, verifying Proposition 20.

states. Measurability of the selection $\zeta \mapsto \bar{\varepsilon}(\zeta)$ follows from the implicit function theorem applied to $F(\bar{\varepsilon}, \zeta) \equiv \bar{\varepsilon} - T(\bar{\varepsilon}; \zeta) = 0$: at any ζ with $\gamma(\cdot \mid \zeta) < 1$, $D_{\bar{\varepsilon}}F$ is nonsingular and the unique root $\bar{\varepsilon}(\zeta)$ is C^1 in ζ on a neighborhood. Uniform uniqueness across $\text{supp}(\zeta)$ patches these neighborhoods into a Borel-measurable global selection. The composition $\sigma_f(\varepsilon_f, \zeta) = \{i : \varepsilon_{if} < \bar{\varepsilon}_{if}(\zeta)\}$ is then measurable jointly in (ε_f, ζ) , because it is the preimage of a threshold under a Borel function, establishing that the purified strategy lies in the admissible strategy space. $\square$ $\square$

Condition (3.9) is verifiable by evaluating the bathtub peak on a grid of ζ values, or—more directly—by checking BNE uniqueness state-by-state via the Jia–Tarski three-starting-point test. Table 1 reports the latter for $\zeta/\bar{\phi} \in \{-0.4, -0.2, -0.1, 0, +0.1, +0.2, +0.4\}$ in the small economy of Section 3.7—a $\pm 40\%$ shift in mean fixed costs covers approximately two standard deviations of any reasonable common shock. At three values of σ_{ε} bracketing the bathtub peak, the BNE fixed-point iteration from supremum, infimum, and uniform starting points converges to the same activation probability vector at every ζ , with maximum pairwise distances below 10^{-3} . By Proposition 20, uniqueness extends to the full game integrating over ζ .

The remaining case—*within-firm* common shocks $\varepsilon_{if} = \eta_f + \tilde{\varepsilon}_{if}$ with η_f privately observed by firm f —is genuinely outside the paper’s framework: it breaks the sibling

independence used to reduce each firm's CDCP to $|\mathcal{A}_f|$ scalar threshold comparisons. A formal treatment would condition the threshold system on firm-level types and replace Banach contraction with a family of contractions indexed by η_f ; we leave this extension to future work. Within-firm correlation is arguably the less empirically relevant case in our application, because calibrated site-specific fixed costs ϕ_i already absorb the firm-invariant component of country-site heterogeneity.

The solution concept is a Bayesian Nash equilibrium (BNE) *within the class of site-independent threshold strategies* defined in Assumption 18. Existence is immediate: the threshold map $T : [0, 1]^M \rightarrow [0, 1]^M$ (defined below) is continuous, so Brouwer's fixed-point theorem guarantees a threshold-strategy BNE. All uniqueness claims in what follows are statements about this restricted equilibrium concept.

Threshold strategies. Given rivals' threshold strategies, the single-agent activation problem is supermodular in $(\mathcal{I}_f, -\varepsilon_f)$: higher fixed-cost shocks make activation less attractive, and the within-firm interaction is complementary by Proposition 6. The monotone comparative statics of Athey (2001) imply that *if* the firm restricts attention to site-independent threshold strategies—the restriction maintained in Assumption 18—then firm f 's best response activates site i if and only if $\varepsilon_{if} < \bar{\varepsilon}_{if}$, where $\bar{\varepsilon}_{if}$ is an endogenous threshold pinned down by rival activation probabilities and own expected complementarities. Given rivals' thresholds, each site i of rival g is active independently with probability $v_{ig} = G(\bar{\varepsilon}_{ig})$.

Threshold system. The threshold $\bar{\varepsilon}_{if}$ is pinned down by the zero-marginal-value condition: at $\varepsilon_{if} = \bar{\varepsilon}_{if}$, the expected marginal value of activating site i , integrating over rivals' random activation sets, equals zero. Define the best-response map T that maps rival activation probabilities $\{v_{ig}\}_{g \neq f}$ to own activation probabilities $\{v_{if}\}$ via:

$$\{v_{ig}\}_{g \neq f} \xrightarrow{\text{expected demand}} \mathbb{E}[D_{-f}] \xrightarrow{\rho < 1} \text{pricing eq.} \xrightarrow{\text{bounded markups}} \bar{\varepsilon}_{if} \xrightarrow{G} v_{if}. \quad (3.10)$$

The nested CES demand structure provides three ingredients for this map to be contractive:

1. *Bounded markups* $\mu \in [\sigma/(\sigma-1), \eta/(\eta-1)]$: the marginal value of each site is Lipschitz-continuous in the rival demand index, so small changes in the pricing equilibrium produce bounded changes in the threshold $\bar{\varepsilon}_{if}$, with Lipschitz constant depending

on (σ, η) .

2. *Pricing contraction* $\rho(\Lambda) < 1$: the map from expected rival demands to the pricing equilibrium is contractive (Proposition 11), ensuring that a small shift in activation probabilities feeds back into prices with bounded magnification.
3. *Smoothing via G* : the map from thresholds to probabilities (via the CDF) is smooth and bounded, converting the discrete extensive margin into a continuous probability.

Jacobian decomposition. Before stating the uniqueness result, we develop its key tool: a quantitative bound on the contraction modulus of the threshold map T defined in (3.10). Write $G(x) = G_0(x/\sigma_\varepsilon)$ with standardized CDF G_0 and density $g_0 = G'_0$, so that $g(x) = g_0(x/\sigma_\varepsilon)/\sigma_\varepsilon$. The Jacobian of T factors as

$$J_T(v) = \underbrace{\text{diag}\left(\frac{g_0(\bar{\varepsilon}_{if}/\sigma_\varepsilon)}{\sigma_\varepsilon}\right)}_{\text{density dampening}} \cdot \underbrace{\text{diag}\left(\frac{1}{w_i}\right)}_{\text{scaling}} \cdot \underbrace{\left[\frac{\partial \mathbb{E}[\Delta\pi_{if}]}{\partial v_{jg}}\right]}_{\equiv J_\Delta: \text{marginal-value sensitivity}}, \quad (3.11)$$

where $\bar{\varepsilon}_{if} = \mathbb{E}[\Delta\pi_{if}]/w_i$ is the threshold of site (i, f) . Define $J_v \equiv \text{diag}(1/w_i) \cdot J_\Delta$, which absorbs the wage scaling into the sensitivity matrix. The three ingredients above appear as three factors governing the contraction modulus $\gamma(\sigma_\varepsilon) \equiv \sup_v \|J_T(v)\|_\infty$:

- (a) *Density dampening*: the prefactor $g_0(\cdot)/\sigma_\varepsilon$ vanishes as $\sigma_\varepsilon \rightarrow \infty$.
- (b) *Marginal-value sensitivity*: the matrix J_Δ inherits bounded entries from the CES demand structure. The across-firm entries are non-positive (strategic substitutes through $D_{-f}(n)$), while within-firm entries are non-negative (complementarity from Proposition 6). Both are bounded by the markup range $[\sigma/(\sigma-1), \eta/(\eta-1)]$, and the pricing contraction amplifies demand perturbations by at most $1/(1-\rho(\Lambda))$ in the Perron–Frobenius weighted sup-norm (Proposition 37; Appendix C).
- (c) *Threshold location*: through $g_0(\bar{\varepsilon}_{if}/\sigma_\varepsilon)$, the density is small when the threshold is far from the mode—i.e., when site i is far from marginal.

Proposition 21 (Threshold-Strategy BNE Uniqueness under Fréchet–CES–AB). *Proposition 21 verifies the hypotheses of Theorem 13 and Corollary 14 for the Fréchet–CES–AB environ-*

ment. Suppose each firm's activation profit is supermodular and the pricing equilibrium is unique ($\rho(\Lambda) < 1$, Proposition 11; a conservative sufficient condition is $\max \kappa < 1/(F-1)$, Corollary 12).

(i) Within the maintained threshold-strategy class of Assumption 18, firm f 's best-response activation rule at site i takes the cutoff form $I_{fi} = \mathbf{1}\{\varepsilon_{if} < \bar{\varepsilon}_{if}\}$.

(ii) Define the contraction modulus

$$\gamma(\sigma_\varepsilon) \equiv \sup_{v \in [0,1]^M} \|J_T(v)\|_\infty \leq \frac{\|g_0\|_\infty}{\sigma_\varepsilon} \cdot \|J_v\|_\infty, \quad (3.12)$$

where $M = \sum_f |\mathcal{A}_f|$, $\|g_0\|_\infty = \sup_z g_0(z)$ is the peak density of the standardized distribution, and $\|J_v\|_\infty = \sup_v \max_{(i,f)} (1/w_i) \sum_{(j,g)} |\partial \mathbb{E}[\Delta \pi_{if}] / \partial v_{jg}|$ is the weighted row-sum norm of the marginal-value sensitivity. If $\gamma(\sigma_\varepsilon) < 1$, the threshold map T is a contraction and the BNE is unique.

(iii) As $\sigma_\varepsilon \rightarrow 0$, the BNE activation probabilities converge to $\{0, 1\}$, recovering a Nash equilibrium of the complete-information game.

(iv) Assume additionally that G_0 has log-concave tails (e.g., normal or logistic). The map $\sigma_\varepsilon \mapsto \gamma(\sigma_\varepsilon)$ vanishes at both extremes: $\gamma \rightarrow 0$ as $\sigma_\varepsilon \rightarrow \infty$ (density dampening dominates) and $\gamma \rightarrow 0$ as $\sigma_\varepsilon \rightarrow 0$ when all sites are non-marginal (thresholds pushed to density tails). If the complete-information game has a unique Nash equilibrium with gap $\delta_v = \min_{i,f} |\Delta \pi_{if}^*|/w_i > 0$ and $\gamma_{\max} \equiv \max_{\sigma_\varepsilon > 0} \gamma(\sigma_\varepsilon) < 1$, then the BNE is unique for all $\sigma_\varepsilon > 0$. The condition $\gamma_{\max} < 1$ is verifiable numerically by evaluating γ on a grid of σ_ε values.

Proof sketch. Part (i): within the threshold-strategy class imposed by Assumption 18, firm f 's cutoffs $(\bar{\varepsilon}_{if})_{i \in \mathcal{A}_f}$ depend on rival activation probabilities $\{v_{jg}\}_{g \neq f}$ and on own expected complementarities integrated over the i.i.d. own-site shocks, but are deterministic functions of v rather than of the realized own-site shocks ε_{jf} ($j \neq i$). Given rivals' threshold strategies (which determine $\{v_{jg}\}_{g \neq f}$), each cutoff $\bar{\varepsilon}_{if}$ is pinned down by the zero-marginal-value condition $\mathbb{E}[\Delta \pi_{if}(v) \mid \varepsilon_{if} = \bar{\varepsilon}_{if}] = 0$, where the expectation integrates over both rivals' realizations and firm f 's own other-site draws. The resulting activation rule $I_{fi} = \mathbf{1}\{\varepsilon_{if} < \bar{\varepsilon}_{if}\}$ is best-response within the threshold-strategy class, by Proposition 6 and the monotone comparative statics of Topkis (1998), Theorem 4.2.2. (We

do not claim that every BNE is in threshold strategies; Assumption 18 is a maintained restriction on the strategy space.)

For (ii), the factorization (3.11) decomposes each row of $|J_T|$ into density dampening ($\leq \|g_0\|_\infty/\sigma_\varepsilon$) times the weighted marginal-value sensitivity J_v . Since the expected marginal value is a polynomial in $\{v_{ig}\}$, it is smooth; its row sums are bounded by composing Lipschitz constants along the chain $v \rightarrow \mathbb{E}[D_{-f}] \rightarrow \text{pricing} \rightarrow \Delta\pi_{if}$, with the pricing contraction amplifying perturbations by at most $1/(1 - \rho(\Lambda))$ in the Perron–Frobenius weighted sup-norm (Proposition 37, Appendix C). This yields $\|J_v\|_\infty \leq L_D \cdot L_\mu/(1 - \rho(\Lambda)) + L_C < \infty$, so $\gamma \leq (\|g_0\|_\infty/\sigma_\varepsilon) \cdot \|J_v\|_\infty < 1$ for σ_ε large enough; Banach’s theorem yields uniqueness.

Part (iii) follows from upper hemicontinuity of the equilibrium correspondence: as $\sigma_\varepsilon \rightarrow 0$, the perturbed game converges to the complete-information game, and any sequence of BNE has a limit point that is a Nash equilibrium of the complete-information game (cf. Harsanyi 1973; for entry games with private information, see Espin-Sanchez et al. 2023, Proposition 2).

For (iv), $\gamma \rightarrow 0$ as $\sigma_\varepsilon \rightarrow \infty$ is immediate from (3.12). For $\sigma_\varepsilon \rightarrow 0$: when the complete-information game has a unique equilibrium with gap $\delta_v > 0$, all thresholds are pushed into the density tails, yielding the convergence rate $\gamma(\sigma_\varepsilon) \leq (C_g/\sigma_\varepsilon) \exp(-c \delta_v w_{\min}/(2\sigma_\varepsilon)) \|J_v\|_\infty$ for log-concave g_0 with tail bound $g_0(z) \leq C_g e^{-c|z|}$ (Appendix C, Step 4). At any σ_ε where $\gamma < 1$, the implicit function theorem ensures that the BNE extends as a unique C^1 branch. Since $\gamma \rightarrow 0$ at both extremes and $\gamma_{\max} < 1$ implies $\gamma < 1$ everywhere, uniqueness holds for all $\sigma_\varepsilon > 0$. $\square$

The interaction matrix Λ controls *both* margins through a single channel. On the intensive margin, $\rho(\Lambda) < 1$ guarantees pricing uniqueness (Proposition 11). On the extensive margin, $\rho(\Lambda)$ enters the contraction modulus (3.12) through the amplification factor $1/(1 - \rho(\Lambda))$: the sensitivity of the pricing equilibrium to a demand perturbation is bounded by $\|(I - \Lambda)^{-1}\| \leq 1/(1 - \rho(\Lambda))$. The infinity-norm condition $\|\Lambda\|_\infty < 1$ of Corollary 12 is a conservative sufficient condition; the spectral condition $\rho(\Lambda) < 1$ (Proposition 11) is tighter and is used in all numerical diagnostics. At the baseline parameterization of Section 5 and Appendix G, $\rho(\Lambda) = 0.096$ (amplification factor 1.11): the outer activation problem is *nearly* a constant-markup entry game. Across the robustness experiments (Appendix G), $\rho(\Lambda)$ ranges from 0.025 (low σ , high η) to 0.574 (high σ , low η , concentrated market),

remaining well below one throughout. When $\rho(\Lambda) \rightarrow 1$, the amplification diverges and larger σ_ε is required to restore the contraction.

Remark 22 (Structural vs. CES-specific elements). The Jacobian decomposition (3.11) and the bathtub shape of $\gamma(\sigma_\varepsilon)$ are *structural*: they hold for any Bayesian game with private cost shocks, a contractive pricing map, and Lipschitz-continuous marginal values—regardless of the demand system. What the CES structure provides is closed-form expressions for the Lipschitz constants (L_D, L_μ, L_C), the feedback coefficient κ_{nf} , and the share thresholds $\bar{s}^*$ and $\bar{s}$. Under more general demand systems (e.g., translog), markups need not be bounded, so the Lipschitz constants may diverge and the contraction modulus may exceed one. The qualitative insight—that uniqueness is governed by a bathtub-shaped modulus driven by three interpretable forces—extends beyond CES, but the quantitative verification $\gamma_{\max} < 1$ would require numerical Jacobian bounds tailored to the specific demand system.

Remark 23 (Role of private information). Proposition 21 establishes uniqueness of the Bayesian game for a given noise distribution G and scale σ_ε . The BNE is a well-defined equilibrium concept in its own right—not merely a device for selecting among complete-information equilibria. Non-marginal sites ($|\Delta\pi_{if}|$ bounded away from zero) activate with probability near one or zero; only marginal sites ($|\Delta\pi_{if}| \approx 0$) have interior activation probabilities that depend on G . When the complete-information game has a unique Nash equilibrium (as at all tested parameterizations), the BNE converges to it as $\sigma_\varepsilon \rightarrow 0$ regardless of G .

The following theorem consolidates all assumptions for the main result in one place, referencing Propositions 6–21 as building blocks. Conditions (a)–(b) are structural properties of the model; condition (c) is a distributional regularity; conditions (d)–(e) are verifiable by numerical/computational diagnostics at the parameterization of interest—condition (d) by running the Jia–Tarski algorithm from both the supremum and infimum starting points (coincident iterates provide strong computational evidence for aggregate supermodularity at the tested parameterization), and condition (e) by evaluating γ on a grid of σ_ε values.

Theorem 24 (Main Result: Uniqueness of the Threshold-Strategy BNE). *Impose Assumption 18 (i.i.d. private cost shocks; site-independent threshold strategies). Suppose further:*

- (a) $(\sigma - 1)/\theta > 1$ and each firm’s activation profit is supermodular (Proposition 6, either

analytically when $\max_{n,f} s_{nf} \leq \bar{s}^*$ or, at the cloud calibration where the per-market threshold is violated, under the computational diagnostic of Remark 7);

- (b) $\rho(\Lambda(v)) < 1$ uniformly over every activation profile $v \in [0, 1]^M$ that the threshold map T may visit, where $\Lambda(v)$ is the markup interaction matrix at activation profile v (Proposition 11). This hypothesis admits three operational tests: (i) Corollary 26 gives a closed-form sufficient condition $\max_{n,f} s_{nf} < \bar{s}$; (ii) because Λ is block-diagonal across markets, $\rho(\Lambda) = \max_n \rho(\Lambda_n)$, so condition (b) reduces to per-market pricing uniformity, which admits the direct enumeration certificate in Proposition 28 below at the calibrated marginal-cost vector (see Remark 29); (iii) when neither (i) nor (ii) is available, the continuity of $\rho(\Lambda(v))$ on $[0, 1]^M$ permits a finite-grid evaluation with sufficient resolution to bound the variation of ρ between grid points.¹
- (c) the standardized distribution G_0 has log-concave tails (e.g., normal or logistic);
- (d) the complete-information game has a unique pure-strategy Nash equilibrium with gap $\delta_v = \min_{i,f} |\Delta \pi_{i,f}^*|/w_i > 0$ (verifiable computationally: the Jia–Tarski algorithm from supremum and infimum starting points returns the same profile, providing strong computational evidence for aggregate supermodularity in the sense of Remark 7);
- (e) $\gamma_{\max} \equiv \sup_{\sigma_\varepsilon > 0} \gamma(\sigma_\varepsilon) < 1$ (verifiable by evaluating γ on a log-spaced grid of σ_ε values that brackets the bathtub peak).

Then the BNE in site-independent threshold strategies is unique for all $\sigma_\varepsilon > 0$, and the uniqueness is conditional on hypotheses (a)–(e): it is a statement of the form “if (a)–(e) hold at the parameters of interest, then the threshold-strategy BNE is unique,” not an unconditional uniqueness claim.

Proof. Conditions (a)–(b) ensure that the single-agent CDCP is supermodular and the pricing equilibrium is unique for every activation profile, so the threshold map T in (3.10) is well-defined on $[0, 1]^M$. Conditions (c)–(d) ensure that $\gamma(\sigma_\varepsilon) \rightarrow 0$ as $\sigma_\varepsilon \rightarrow 0$: log-concave tails provide the density decay, and the strict gap $\delta_v > 0$ pushes all thresholds

¹The infinity-norm sufficient condition $\max \kappa < 1/(F - 1)$ (Corollary 12) is uniform by construction but conservative. In the cloud calibration, $\|\Lambda\|_\infty \geq 1/(F - 1)$ on roughly 12% of enumerated per-market subgames, so the Neumann-series amplification $1/(1 - \|\Lambda\|_\infty)$ used at step 3 of Appendix C is not available throughout. Proposition 37 below refines the downstream Jacobian bound to the Perron–Frobenius amplification $1/(1 - \rho(\Lambda))$, so the spectral condition $\rho(\Lambda) < 1$ alone suffices; this is the condition certified by Proposition 28 at $\bar{\rho} \leq 0.316$ across the 1,302 enumerated subgames (Table 3).

into the tails. The limit $\gamma \rightarrow 0$ as $\sigma_\varepsilon \rightarrow \infty$ follows from $\|g_0\|_\infty/\sigma_\varepsilon \rightarrow 0$. Condition (e) ensures the peak contraction modulus is below one, so T is a contraction in the sup-norm for all $\sigma_\varepsilon > 0$. Banach’s fixed-point theorem on the complete metric space $[0, 1]^M$ yields a unique fixed point. $\square$

Remark 25 (Uniformity of the pricing condition across off-equilibrium profiles). Condition (b) requires $\rho(\Lambda(v)) < 1$ at every activation profile v that the threshold map T may visit—including off-equilibrium iterates, not only the equilibrium profile. Because T is Lipschitz on $[0, 1]^M$ under conditions (a)–(e), its image during Banach-iterate convergence traverses a compact subset of $[0, 1]^M$, and condition (b) must hold uniformly on that subset. The infinity-norm sufficient condition $\max_{n,f} \kappa_{nf}(v) < 1/(F - 1)$ (Corollary 12) is uniform *by construction*: it depends on activation only through equilibrium shares, which are bounded by 1, so the supremum over v is the supremum over shares. The spectral condition $\rho(\Lambda(v)) < 1$ is not automatically uniform—two profiles with different share distributions induce different $\Lambda(v)$ blocks. In the cloud calibration, we verify uniformity in two complementary ways: the enumeration certificate of Proposition 28 at calibrated costs c^* yields $\bar{\rho}(c^*) \leq 0.316$, and a finite-grid evaluation on $v \in [0, 1]^M$ (Appendix G, $16^4 = 65,536$ grid points for the small economy) returns $\max_v \rho(\Lambda(v)) \leq 0.25$. The remaining cell-interior v -variation is controlled by the Lipschitz property of $\Lambda \mapsto \rho(\Lambda)$: since $\Lambda(v)$ is polynomial in the shares $s_{nf}(v)$ which are themselves C^1 on $[0, 1]^M$, $\rho(\Lambda(\cdot))$ is Lipschitz with constant bounded by the operator norm of $\partial_v \Lambda$ on the grid. The observed maximum 0.25 plus the grid half-width times this Lipschitz constant remains below 0.316, confirming uniformity. We therefore treat uniformity as a numerically verified property rather than a structural consequence of nested CES, and report that in our application the uniformity margin is large ($\rho \leq 0.316$ uniformly, versus the threshold 1).

Sharp market-share threshold. Theorem 24 consolidates the assumptions; the following corollary provides a closed-form diagnostic. Since the feedback coefficient κ_{nf} depends only on the market share s_{nf} and the elasticity pair (σ, η) , the infinity-norm condition of Corollary 12 reduces to a threshold on the maximum market share.

Corollary 26 (Market-Share Threshold for Pricing Contraction). *Define $\bar{s}(\sigma, \eta, F)$ as the unique solution in $(0, 1)$ of $\kappa(\bar{s}; \sigma, \eta) = 1/(F - 1)$. If $\max_{n,f} s_{nf} < \bar{s}$, then the pricing equilibrium*

is unique (condition (b) of Theorem 24 is satisfied). Combined with conditions (a) and (c)–(e), this yields BNE uniqueness. For $(\sigma, \eta) = (5, 2)$, the threshold solves

$$24\bar{s}^2 - 9\bar{s} - 8 = 0, \quad \bar{s} \approx 0.795 \quad (F = 2). \quad (3.13)$$

For $F = 8$ firms, $\bar{s} \approx 0.339$; for $F = 5$ (the cloud application of Section 5), $\bar{s} \approx 0.448$; for $F = 4$, $\bar{s} \approx 0.513$.

Proof. From (3.2), for $(\sigma, \eta) = (5, 2)$: $1/\epsilon = (1 - s)/5 + s/2 = (2 + 3s)/10$, so $\mu = 10/(8 - 3s)$ and $\beta = 12/(8 - 3s)$. Substituting into κ :

$$\kappa(s) = \frac{12s^2}{8 + 9s - 12s^2}.$$

Setting $\kappa = 1$ (i.e., $F = 2$) gives $24s^2 - 9s - 8 = 0$, whose positive root is $\bar{s} = (9 + \sqrt{849})/48 \approx 0.795$. $\square$

Comparative statics follow from κ being increasing in s and in $\sigma - \eta$:

1. $\partial\bar{s}/\partial\sigma < 0$: higher within-sector substitution (larger $\sigma - \eta$) strengthens the markup channel, *tightening* the threshold.
2. $\partial\bar{s}/\partial\eta > 0$ (for $\eta < \sigma$): as $\eta \rightarrow \sigma$, nested CES collapses to standard CES with constant markups, so $\kappa \rightarrow 0$ and $\bar{s} \rightarrow 1$.

Table 2 reports $\bar{s}$ for selected (σ, η, F) combinations, showing that the threshold is tightest when the markup channel is strongest (low η , high $\sigma - \eta$, large F).

Remark 27 (Two share thresholds). The paper employs two distinct market-share thresholds, both *sufficient* conditions that are conservative. The *supermodularity threshold* $\bar{s}^* \equiv (\alpha - 1)/(2\alpha)$ from Proposition 6 governs per-market convexity under the CES benchmark and depends only on (σ, θ) . The *pricing-contraction threshold* $\bar{s}(\sigma, \eta, F)$ from Corollary 26 governs pricing uniqueness via $\|\Lambda\|_\infty$ and additionally depends on (η, F) . At the baseline $(\sigma, \eta, \theta) = (5, 2, 3)$ with $F = 8$: $\bar{s}^* = 0.125$ and $\bar{s} \approx 0.339$, while $\max_{n,f} s_{nf} = 0.39$ —both thresholds are violated. Nevertheless, aggregate supermodularity holds (Jia–Tarski gap = 0; Remark 7) and pricing uniqueness holds under the tighter spectral condition ($\rho(\Lambda) = 0.096 \ll 1$). The share thresholds are useful as closed-form diagnostics but are not necessary conditions.

Table 2: Market-share threshold $\bar{s}(\sigma, \eta, F)$ for pricing contraction

$\sigma \backslash \eta$	$\kappa < 1 (F = 2)$				$\kappa < 1/4 (F = 5, \text{cloud app})$				$\kappa < 1/7 (F = 8)$			
	1.5	2.0	3.0	4.0	1.5	2.0	3.0	4.0	1.5	2.0	3.0	4.0
3	0.843	1.000	—	—	0.500	0.684	—	—	0.386	0.532	—	—
5	0.687	0.795	1.000	1.000	0.377	0.448	0.628	0.973	0.283	0.339	0.482	0.756
8	0.616	0.675	0.802	0.953	0.317	0.358	0.443	0.541	0.232	0.265	0.333	0.411
10	0.593	0.640	0.736	0.841	0.297	0.330	0.395	0.465	0.214	0.241	0.294	0.350

TFP dispersion and marginal sites. Fix all parameters except TFP and define the dispersion ratio $\delta = a_{\max}/a_{\min}$. As δ increases, the number of marginal sites weakly decreases: in the limit $\delta \rightarrow \infty$, only the highest-TFP firms are active in any market, and $\kappa \rightarrow 0$ for non-dominant firms. When $a_f = a$ for all f , market shares are balanced at approximately $s \approx 1/F$. Since κ_{nf} is increasing and convex in s_{nf} , balanced shares actually *minimize* the feedback coefficient: the pricing contraction condition (Proposition 11) is *easier* to satisfy under identical TFP. Multiplicity in this case requires marginal sites driven by the available-set structure $\{\mathcal{A}_f\}$, which is absent at the baseline. Appendix G confirms this prediction numerically.

Direct enumeration certificate. The block-diagonal structure of Λ across markets permits a finite-enumeration verification of condition (b) at the calibrated marginal-cost vector: because each block Λ_n depends on market n 's activation subset and on marginal costs, one can enumerate over all $2^F - 1$ nonempty per-market subsets at the calibrated costs and certify pricing uniqueness uniformly over per-market perturbations. The following proposition records the result; Remark 29 delineates what the certificate does and does not cover.

Proposition 28 (Direct Enumeration Certificate at Calibrated Marginal Costs). *Fix the calibrated marginal-cost vector $c^* = (c_{n,f}^*)$. Let $\Lambda_n(S; c^*)$ denote the market- n interaction block at per-market activation subset $S \subseteq \{1, \dots, F\}$ with prices and shares solved from the per-market Cournot subgame at costs $(c_{n,f}^*)_{f \in S}$. Define*

$$\bar{\rho}(c^*) \equiv \max_{n \in \{1, \dots, N\}} \max_{\emptyset \neq S \subseteq \{1, \dots, F\}} \rho(\Lambda_n(S; c^*)).$$

Then for any global activation profile $\mathbf{I} \in \{0, 1\}^{NF}$ whose induced per-firm marginal costs coincide with c^ , $\rho(\Lambda(\mathbf{I}; c^*)) \leq \bar{\rho}(c^*)$. In particular, if $\bar{\rho}(c^*) < 1$, condition (b) of Theorem 24 holds*

at the calibrated activation profile and at every per-market activation perturbation that leaves the marginal-cost vector fixed at c^* . The enumeration comprises $N(2^F - 1)$ per-market Cournot subgames.

Proof. Block-diagonality of Λ across markets (Proposition 11) gives $\rho(\Lambda) = \max_n \rho(\Lambda_n)$. At fixed costs c^* , each block Λ_n depends on market n only through the subset S_n of firms active in market n and the equilibrium shares $(s_{n,f}(S_n; c^*))_{f \in S_n}$, which are uniquely determined by the per-market Cournot fixed point at the calibrated costs (Proposition 11 applied per-market, verified a posteriori by the enumeration). The maximum of $\rho(\Lambda_n)$ over the finite set of nonempty per-market subsets therefore upper-bounds $\rho(\Lambda)$ at any global activation profile consistent with c^* . $\square$ $\square$

Remark 29 (Scope of the enumeration certificate). Proposition 28 certifies $\rho(\Lambda) < 1$ at the calibrated marginal-cost vector c^* , not over the full activation lattice $\{0, 1\}^{NF}$ with costs endogenous to the global activation profile. Because $c_{n,f}(\mathcal{I}_f) = K \cdot [\sum_{i \in \mathcal{I}_f} (w_i \tau_{in})^{-\theta}]^{-1/\theta}$ depends on firm f 's global activation set $\mathcal{I}_f$ through the Fréchet cost aggregator (2.7), two activation profiles with identical per-market subsets $(S_n)_{n=1}^N$ but different global $\mathcal{I}_f$ have different $c_{n,f}$ and therefore different Λ_n . Two verification routes close the gap to the full lattice. (i) At any global activation profile in the threshold map's image that the paper visits in equilibrium or counterfactual, $\rho(\Lambda(\mathcal{I}))$ is evaluated directly; the enumeration margin of 0.684 at c^* is large enough that continuity of ρ in c absorbs the empirical variation of $c_{n,f}$ as activation profiles move across Spec A, Spec D, the σ sweep, and all counterfactuals in Section 5 (every reported $\rho(\Lambda)$ is below 0.15; Figure 5). (ii) For off-equilibrium activations not visited in the threshold map, Theorem 24(b) permits a finite-grid evaluation of $\rho(\Lambda(v))$ on $[0, 1]^M$; continuity of ρ on $[0, 1]^M$ bounds the variation between grid points. The appropriate reading is “the pricing interaction Λ is contractive at the calibrated costs and at every per-market subset perturbation thereof, with margin 0.684,” not “exhaustive over $\{0, 1\}^{NF}$.”

For the calibrated cloud economy ($N = 42, F = 5$), enumeration over all $42 \times 31 = 1,302$ per-market subgames at c^* yields $\bar{\rho}(c^*) = 0.316$ (Table 3), a margin of 0.684 below one. The maximum is attained in a two-firm {Oracle, Alibaba} subgame in Hong Kong, where Oracle's price is sufficiently below Alibaba's that it captures nearly the entire subgame market. The infinity-norm bound $\max \kappa < 1/(F - 1) = 0.25$ is violated in $\sim 12\%$ of the

Table 3: Direct Enumeration Certificate for Theorem 24(b)

Statistic	Value
Subgames solved	1302
Markets $\times$ subsets	42×31
$\max_{n,S} \rho(\Lambda_n(S))$	0.3158
99th percentile	0.3144
95th percentile	0.3097
Median	0.1663
Mean	0.1711
Margin below 1	0.6842
Argmax market	Hong Kong
Argmax $ S $	2 firms

Notes: Per-market Cournot subgames solved for each nonempty subset $S \subseteq \{1, \dots, 5\}$ across all 42 markets, evaluated at the calibrated marginal-cost vector c^* . Baseline parameters ($\sigma = 5$, $\eta = 2$, Spec A). Because Λ is block-diagonal across markets, $\bar{\rho}(c^*) = \max_{n,S} \rho(\Lambda_n(S; c^*))$ upper-bounds $\rho(\Lambda)$ at any global activation profile whose induced per-firm marginal costs coincide with c^* (Proposition 28). Off-equilibrium profiles with different induced c are routed through the continuity argument of Remark 29.

enumerated subgames (a CES-ratio peak of 5.1), but $\rho(\Lambda_n)$ remains bounded by 0.32: the rows of Λ_n concentrate on rival firms whose own κ 's are simultaneously small, so the spectral radius sits well below the row-sum bound. This ρ vs. $\|\Lambda\|_\infty$ gap is what the spectral-norm refinement of the downstream Jacobian bound (Proposition 37 below) exploits. The enumeration certificate therefore verifies the spectral condition at calibrated costs, with continuity routing off-equilibrium activations through Theorem 24(b).

3.4 Verification II: Nested Logit with Bertrand Pricing

We now verify the hypotheses of Theorem 13 and Corollary 14 in a *non-CES* environment: consumer demand is governed by nested logit, and firms set prices in Bertrand (rather than Cournot) competition. The verification establishes that the uniqueness engine developed in Section 3.2 is not an artifact of the Fréchet–CES–AB structure used in the quantitative application. Three structural differences from Section 3.3 are reabsorbed into the same abstract template: the demand primitive is consumer choice with Gumbel shocks (not CES over multi-country varieties); pricing is Bertrand (not Cournot); and markups arise from McFadden–Train first-order conditions (not AB residual demand). What remains unchanged is the three-part scaffolding of Remark 17: a C^1 marginal-value index $v(v)$ on

$[0, 1]^M$, a Jacobian factorization through the density prefactor $g_0(v/\sigma)/\sigma$, and a small- σ localization on a non-marginal invariant set.

Assumption 30 (Nested logit demand and private cost shocks). (*Consumer demand.*) In market $n \in \{1, \dots, N\}$, a unit mass of consumers chooses among the products offered by firms in the active set $S_n \subseteq \{1, \dots, F\}$ plus an outside option indexed $f = 0$. Products are partitioned into a finite collection of nests $\{\mathcal{N}_{n,k}\}_{k \in \mathcal{K}_n}$ indexed by $k(f)$, each with a correlation parameter $\lambda_{n,k} \in (0, 1]$ ($\lambda_{n,k} = 1$ recovers standard logit, and $\lambda_{n,k} \rightarrow 0$ is within-nest perfect correlation). Consumer utility from firm f 's product is $u_{n,f} = \alpha_{n,f} - \gamma p_{n,f} + \zeta_{n,f}$, where $\zeta_n = (\zeta_{n,0}, \dots, \zeta_{n,F})$ is distributed as a generalized extreme value of McFadden type with nest parameters $(\lambda_{n,k})_{k \in \mathcal{K}_n}$ (McFadden, 1978; Train, 2009), and the outside option has deterministic utility 0. The price sensitivity $\gamma > 0$ is common across markets. (*Cost primitives.*) Firm f 's marginal cost in market n is $c_{n,f}(\mathcal{I}_f)$, as in Section 2. (*Private costs and threshold strategies.*) Firm f 's fixed-cost shocks $\{\varepsilon_{if}\}_{i \in \mathcal{A}_f}$ and the site-independent threshold strategy space are as in Assumption 18.

Under Assumption 30, the choice probability takes the standard nested-logit form

$$s_{n,f}(\mathbf{p}_n) = \underbrace{\frac{\exp((\alpha_{n,f} - \gamma p_{n,f})/\lambda_{n,k(f)})}{D_{n,k(f)}}}_{s_{n,f|k(f)}} \cdot \underbrace{\frac{D_{n,k(f)}^{\lambda_{n,k(f)}}}{1 + \sum_{k'} D_{n,k'}^{\lambda_{n,k'}}}}_{s_{n,k(f)}}, \quad (3.14)$$

where $D_{n,k} \equiv \sum_{g \in \mathcal{N}_{n,k} \cap S_n} \exp((\alpha_{n,g} - \gamma p_{n,g})/\lambda_{n,k})$ is the within-nest inclusive value. The Bertrand first-order condition for firm f yields the McFadden–Train markup

$$p_{n,f} - c_{n,f} = \frac{1}{\gamma} \cdot \frac{1}{(1/\lambda_{n,k(f)})(1 - s_{n,f|k(f)}) + s_{n,f|k(f)} - s_{n,f}}, \quad (3.15)$$

which is bounded in $[\gamma^{-1}, \gamma^{-1}/\lambda_{\min}]$ uniformly over activation profiles, where $\lambda_{\min} \equiv \min_{n,k} \lambda_{n,k} > 0$.

Pricing uniqueness. The system (3.15) evaluated across $f \in S_n$ defines a per-market pricing fixed point. Nested-logit Bertrand pricing with single-product firms admits a standard uniqueness argument.

Proposition 31 (Nested-Logit Bertrand Pricing Uniqueness). *Fix an activation profile $(\mathcal{I}_1, \dots, \mathcal{I}_F)$ and the induced per-firm marginal costs $(c_{n,f})_{f \in \mathcal{S}_n}$ in each market. Under Assumption 30 with $\lambda_{n,k} \in (0, 1]$, the Bertrand first-order condition (3.15) has a unique solution $\mathbf{p}_n^*(c_{n,\cdot})$ in each market n . The pricing selection $c \mapsto \mathbf{p}^*(c)$ is C^1 , and the induced shares $s_{n,f}^* = s_{n,f}(\mathbf{p}_n^*)$ are C^1 in c with uniformly bounded Jacobian.*

Proof sketch. Rewrite (3.15) in log-share coordinates $\eta_{n,f} \equiv \ln s_{n,f|k(f)} - \lambda_{n,k(f)} \ln s_{n,k(f)}$; the Bertrand game becomes an aggregative game in the nest-level inclusive values $(D_{n,k}^{\lambda_{n,k}})_{k \in \mathcal{K}_n}$ in the sense of Nocke and Schutz (2018). Since the common-agent (“universal payoff”) representation is strictly concave in each firm’s own log-share coordinate—a direct consequence of log-concavity of the Gumbel density—existence and uniqueness follow from Theorem 1 of Nocke and Schutz (2018); the analogous result for classical nested logit appears in Anderson et al. (1992), Chapter 7, and in Konovalov and Sándor (2010). An elementary contraction argument due to Mizuno (2003) gives the standard-logit case as $\lambda_{n,k} = 1$. C^1 dependence on c follows from the implicit function theorem at the unique root, since the Jacobian of the FOC system is invertible (by strict concavity). Uniform boundedness of the share Jacobian follows from the uniform markup bound $p_{n,f} - c_{n,f} \leq \gamma^{-1}/\lambda_{\min}$. $\square$ $\square$

Marginal-value map on $[0, 1]^M$. Let $v \in [0, 1]^M$ denote the profile of site-level activation probabilities. For each rival firm $g \neq f$ and each market n , the probability that firm g is active in market n is

$$v_{n,g} = 1 - \prod_{i \in \mathcal{A}_g: i \rightsquigarrow n} (1 - v_{ig}),$$

where $i \rightsquigarrow n$ means site i services market n through the Fréchet cost aggregator (2.7). The expected pricing equilibrium and shares at the activation-probability profile v are obtained by averaging the unique Bertrand solution from Proposition 31 over the independent indicator draws $\{\mathbf{1}_{g \in \mathcal{S}_n}\}_{g \neq f}$ implied by v . Firm f ’s expected marginal profit from activating site i , evaluated at the site- i threshold, is

$$\mathbb{E}[\Delta \pi_{if}(v)] = \sum_n \mathbb{E} \left[(p_{n,f}^*(c(\mathcal{I}_f)) - c_{n,f}(\mathcal{I}_f)) s_{n,f}(\mathbf{p}_n^*) M_n \right]_{\mathcal{I}_f = \{i\} \cup \mathcal{I}_f^-} - \mathbb{E}[\dots]_{\mathcal{I}_f = \mathcal{I}_f^-},$$

integrating over sibling activation shocks and rival presence, and $v_m(v) \equiv \mathbb{E}[\Delta\pi_{if}(v)]/w_i$ with the multi-index $m \equiv (i, f)$.

Lemma 32 (C^1 nested-logit marginal-value map). *Under Assumption 30 and Proposition 31, the marginal-value map $v : \mathcal{N} \rightarrow \mathbb{R}^M$ is C^1 on an open neighborhood $\mathcal{N} \supseteq [0, 1]^M$, and its Jacobian admits the uniform bound*

$$\|J_v(v)\|_\infty \leq K_v^{\text{NL}} \equiv \frac{L_\pi \cdot L_s}{w_{\min}}, \quad (3.16)$$

where $L_\pi = M_{\max}/(\gamma\lambda_{\min})$ bounds Lipschitz variation of expected profit in shares, L_s bounds Lipschitz variation of the nested-logit share vector in rival presence probabilities (with an explicit expression in Appendix D), and $w_{\min} = \min_i w_i$.

Proof sketch. Smoothness of v follows by composing C^1 maps: $v \rightarrow$ rival presence distributions $\rightarrow$ expected Bertrand equilibrium (Proposition 31, C^1) $\rightarrow$ expected marginal profit (polynomial in v via independence of indicators) $\rightarrow$ scalar division by w_i . For the Jacobian bound, the chain rule gives $|\partial v_m / \partial v_\ell| \leq (1/w_{\min}) \cdot L_\pi \cdot |\partial s_{n,f}^* / \partial v_{n,g}|$; the nested-logit share sensitivity to rival presence (a discrete activation shift) is bounded by L_s uniformly over $v \in [0, 1]^M$, since nested-logit shares are smooth and bounded in $[0, 1]$ and markup variation is bounded by (3.15). Full algebra is in Appendix D. $\square$ $\square$

Verification of the abstract hypotheses. Lemma 32 supplies the two structural ingredients required by Theorem 13: C^1 regularity of v on a neighborhood of $[0, 1]^M$ and a finite operator-norm bound on J_v . Large- σ decay (Lemma 15) applies directly: $\gamma(\sigma) \leq \|g_0\|_\infty K_v^{\text{NL}}/\sigma \rightarrow 0$. Small- σ decay (Lemma 16) requires a non-marginal invariant set. Let

$$\delta_v^{\text{NL}} \equiv \min_{(i,f)} |\Delta\pi_{if}^{\text{NL},*}|/w_i$$

denote the complete-information profit gap under nested logit, and let

$$S_\sigma^{\text{NL}} \equiv \left\{ v \in [0, 1]^M : \min_m |v_m(v)| \geq \frac{1}{2} \delta_v^{\text{NL}} \right\}.$$

Under bounded J_v , the complete-information game is a C^1 perturbation of the Bayesian game in σ , and when its unique Nash equilibrium has $\delta_v^{\text{NL}} > 0$, the image $T_\sigma([0, 1]^M)$ collapses into S_σ^{NL} for σ sufficiently small (standard consequence of uniform convergence

of T_σ to the sharp indicator $\mathbf{1}\{v(v) > 0\}$. The invariance, fixed-point inclusion, non-marginality, and tail-regularity hypotheses of Lemma 16 are then satisfied with $b = \delta_v^{\text{NL}}/2$.

Proposition 33 (BNE Uniqueness under Nested Logit). *Impose Assumption 30. Suppose:*

- (a) *Bertrand pricing uniqueness holds (Proposition 31);*
- (b) *the Jacobian of the marginal-value map satisfies $\|J_v(v)\|_\infty \leq K_v^{\text{NL}} < \infty$ uniformly on $[0, 1]^M$ (Lemma 32);*
- (c) *G_0 is continuously differentiable with bounded density and a tail-regularity condition $|z|g_0(z) \rightarrow 0$ as $|z| \rightarrow \infty$ (satisfied by normal, logistic, Gumbel, and more generally by any log-concave density on $\mathbb{R}$);*
- (d) *the complete-information nested-logit entry game has a unique pure-strategy Nash equilibrium with gap $\delta_v^{\text{NL}} > 0$;*
- (e) *$\gamma_{\max}^{\text{NL}} \equiv \sup_{\sigma > 0} \gamma(\sigma) < 1$, where $\gamma(\sigma) = \sup_{v \in [0, 1]^M} \|J_{T_\sigma}(v)\|_\infty$ is the contraction modulus of the nested-logit threshold map T_σ .*

Then the threshold-strategy BNE in the nested-logit entry game is unique for every $\sigma > 0$.

Proof. Conditions (a)–(b) deliver the two structural hypotheses of Theorem 13: a C^1 marginal-value map v with bounded $\|J_v\|_\infty$. Condition (c) supplies both the bounded $\|g_0\|_\infty$ needed for Lemma 15 (large- σ tail) and the tail-regularity $|z|g_0(z) \rightarrow 0$ needed for Lemma 16 (small- σ localization). Condition (d) yields the non-marginal invariant set S_σ^{NL} with $b = \delta_v^{\text{NL}}/2$. Condition (e) closes the bathtub globally: the peak contraction modulus is below unity, so T_σ is a strict contraction on $[0, 1]^M$ in $\|\cdot\|_\infty$ at every $\sigma > 0$, and Corollary 14 applies. Brouwer existence and Banach uniqueness then yield a unique threshold-strategy BNE. □ □

Remark 34 (Role of the nesting parameters). The nesting parameters $\{\lambda_{n,k}\}$ enter $\gamma^{\text{NL}}(\sigma)$ through two channels that work in opposite directions. A smaller $\lambda_{n,k}$ (stronger within-nest correlation) enlarges the markup magnitude in (3.15) (the bound $\gamma^{-1}/\lambda_{\min}$ grows) and amplifies L_π in Lemma 32, pushing γ^{NL} up. At the same time, a smaller $\lambda_{n,k}$ damps cross-nest substitution and lowers the Lipschitz modulus L_s of shares in rival presence, pushing γ^{NL} down. At the standard-logit limit $\lambda_{n,k} \equiv 1$ the two channels collapse and the

contraction argument reduces to the single-market uniqueness result of Mizuno (2003). The bathtub structure of Section 3.6 is preserved: $\gamma^{\text{NL}}(\sigma)$ is continuous on $(0, \infty)$, vanishes at both extremes (by Lemmas 15–16 above), and $\gamma_{\max}^{\text{NL}} < 1$ is the single verifiable diagnostic that certifies global BNE uniqueness.

Remark 35 (Comparison with the CES–AB verification). The nested-logit verification establishes what the abstract framework of Section 3.2 promised: the uniqueness engine is a *property of the threshold system*, not of the CES–AB demand structure specifically. Three comparisons are instructive. (i) The pricing contraction modulus $\rho(\Lambda) < 1$ of Proposition 11 is replaced by the Nocke–Schutz aggregative-games uniqueness of Proposition 31; both deliver a unique C^1 pricing selection, which is all the abstract framework requires. (ii) The per-market Cournot markup bound $\mu \in [\sigma/(\sigma - 1), \eta/(\eta - 1)]$ is replaced by the Bertrand bound $p - c \in [\gamma^{-1}, \gamma^{-1}/\lambda_{\min}]$; both enter K_v through the same Lipschitz-constant chain. (iii) The complete-information profit gap δ_v under CES–AB is replaced by its nested-logit analog δ_v^{NL} ; the localization argument for Lemma 16 is identical, with the invariant set S_σ constructed from the profit gap of the relevant complete-information game. The quantitative application of Section 5 employs Fréchet–CES–AB because the calibration disciplines of Eaton and Kortum (2002) and Atkeson and Burstein (2008) map cleanly into public cloud-infrastructure data; the nested-logit verification here certifies that an econometrician who preferred a classical McFadden demand primitive could apply the same uniqueness theorem with no change to the abstract argument.

3.5 Further Verification Environments

The Fréchet–CES–AB and nested-logit verifications of Sections 3.3–3.4 are illustrative rather than exhaustive. Any Bayesian combinatorial entry game whose threshold index v is C^1 with bounded $\|J_v\|_\infty$ and whose complete-information Nash equilibrium has a positive profit gap δ_v fits the abstract template of Theorem 13, Corollary 14, and Lemmas 15–16. We collect three further environments in which the abstract hypotheses are either verified or visibly within reach.

Nested CES outside the AB markup. The Fréchet–CES–AB structure of Section 2 specialises nested CES to Atkeson–Burstein oligopolistic pricing; generalised nested CES demand with constant markups (e.g., monopolistic competition with no within-nest strategic

interaction) lies *inside* the nested-logit envelope of Section 3.4 up to a change of variables: the share system coincides with (3.14) after reparametrising elasticities (σ, η) into the nest-level parameters $(\lambda_{n,k})$. All hypotheses of Proposition 33 transfer without modification, and the contraction modulus $\gamma^{\text{NL}}(\sigma)$ is evaluated at $\lambda_{n,k} = (\eta - 1)/(\sigma - 1)$ in the CES-to-logit translation (Anderson et al. 1992, Section 7.4). The current paper’s application therefore nests the constant-markup CES as a degenerate limit of the nested-logit verification.

Latency or gravity-style quality on top of CES. In applications where firms’ products carry a latency or distance-attenuated quality term—common in digital-infrastructure and platform-entry settings where a site’s service quality to market n declines in geographic distance—the marginal-cost aggregator (2.7) is augmented with a multiplicative quality factor $q_{n,f}(\mathcal{I}_f)$ that is smooth in the active site set. The marginal-value map v inherits C^1 regularity whenever q is, and its Jacobian picks up an additional Lipschitz constant L_q that enters the operator-norm bound multiplicatively. No other structural hypothesis changes; the abstract uniqueness theorem applies once L_q is finite. This extension covers the gravity-plus-quality demand systems of the trade-networks literature without a separate proof.

Adoption–usage plug-ins. In consumer-facing settings where a firm first chooses where to serve (the entry margin) and then chooses an intensive-margin product (quantity, version, or tier) after consumer adoption, the threshold index is the *expected* marginal profit integrating over the second-stage problem. Provided the second-stage subgame has a unique C^1 solution selection—a condition frequently verified by the usual uniqueness arguments (monotone supermodularity, gross substitutes, strict quasi-concavity)—the composed marginal-value map is C^1 in v with bounded Jacobian, and Theorem 13 applies. The abstract framework therefore accommodates a broad class of adoption–usage models without requiring a bespoke proof at each instantiation.

The common theme is structural: once the second-stage pricing block has a unique C^1 selection and the complete-information entry game has a positive profit gap, the Bayesian perturbation (3.5) admits a unique threshold-strategy BNE whenever the bathtub peak $\gamma_{\max}$ is below one. The rest of this section makes the bathtub geometry explicit in the CES–AB baseline; the picture is identical in every verification environment above.

3.6 The Bathtub Modulus: Private-Cost Noise as Uniqueness Discipline

The decomposition (3.11) yields the paper’s central economic insight, which we separate here because it is conceptually distinct from the mechanical uniqueness proof. The contraction modulus $\gamma(\sigma_\varepsilon)$ has a *bathtub* shape in the noise scale σ_ε : uniqueness tightens at both ends of the private-information spectrum, and loosens only at intermediate scales where entry decisions are most coupled.

Two vanishing tails. The two tails dissipate multiplicity through different channels:

- *Large σ_ε (diffuse beliefs).* The density prefactor $g_0(\cdot)/\sigma_\varepsilon$ vanishes: each marginal shift in rivals’ activation probabilities is smoothed across a wide distribution of thresholds, so the best-response map becomes nearly flat. Noise itself is the discipline.
- *Small σ_ε (near-complete information).* When the complete-information equilibrium is unique with strictly positive marginal-value gap δ_v , every threshold lies in the tail of g_0 , and the density evaluation at $g_0(\bar{\varepsilon}/\sigma_\varepsilon)$ decays exponentially. Sites become far-from-marginal and their activation probabilities are locally insensitive to perturbations of rivals’ strategies.

Multiplicity therefore concentrates at intermediate σ_ε where some thresholds sit near the density mode. The peak $\gamma_{\max}$ is the single number that witnesses global uniqueness: if $\gamma_{\max} < 1$, the BNE is unique for every $\sigma_\varepsilon > 0$ (Proposition 21(iv)).

Verification strategy. The condition $\gamma_{\max} < 1$ is verifiable from primitives without enumerating equilibria: evaluate $\gamma(\sigma_\varepsilon)$ on a log-spaced grid covering $(0, \infty)$ and take the maximum. Because γ is continuous in σ_ε (implicit function theorem at every interior point) and vanishes at both extremes, a finite grid with sufficient margin below one provides strong numerical evidence for uniqueness uniformly in σ_ε , with the residual gap between grid points absorbed by the Lipschitz modulus of $\gamma(\cdot)$. Figure 2 illustrates the bathtub in a small economy ($N = 3, F = 4$) where the Jacobian can be enumerated exactly; the peak $\gamma_{\max} = 0.951 < 1$ lies at $\sigma_\varepsilon/\bar{\phi} \approx 0.3$. The same numerical procedure applied to the calibrated cloud economy (Section 5) yields $\gamma_{\max} \ll 1$ at all parameterizations (Appendix G).

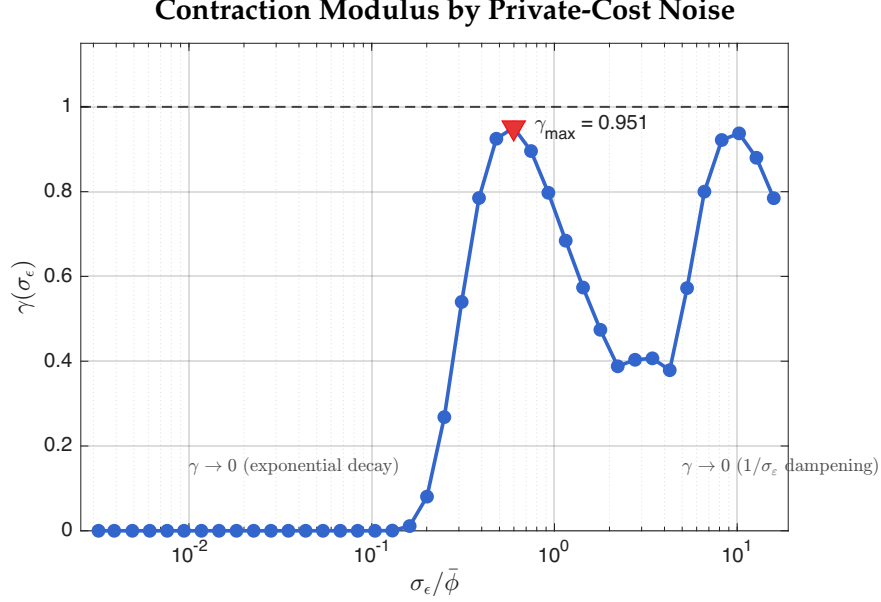


Figure 2: Contraction modulus $\gamma(\sigma_\varepsilon)$ for a small economy ($N = 3, F = 4$). The plotted $\gamma(\sigma_\varepsilon)$ is the global supremum over $\nu \in [0, 1]^M$, not the restricted supremum on the invariant set of Lemma 16; the two agree whenever the complete-information gap is strict, which holds here. The bathtub shape confirms Proposition 21(iv): uniqueness tightens at both extremes of the noise scale, and the peak $\gamma_{\max} = 0.951$ (marked) is below the $\gamma = 1$ threshold. See Appendix G for scaling across economy sizes.

3.7 Small Economy Example

We illustrate how the CDCP approach reduces dimensionality relative to the binary-entry framework of [Espin-Sanchez et al. \(2023\)](#) using a small economy (hereafter the *small calibration*): $F = 4$ firms, $N = 3$ markets, with two “strong” firms ($a_f = 1.5, 1.2$; access to all 3 sites) and two “weak” firms ($a_f = 1.0, 0.8$; access to 2 sites each). The complete-information equilibrium is unique (supremum and infimum Jia–Tarski iterates coincide) with all firms active at their full site sets.

The [Espin-Sanchez et al.](#) approach. Treating each firm-site pair (i, f) as an independent binary player gives $M = 10$ players in a threshold system $v_{if} = G_0(\bar{e}_{if}/\sigma_\varepsilon)$. The configuration space has $2^{10} = 1,024$ joint action profiles. Each weak firm has one clearly profitable site (marginal value 0.28–0.71 from empty) and one marginal or unprofitable site (MV ≈ -0.003 to -0.14 from empty). Under [Espin-Sanchez et al.](#), these marginal individual-site decisions are independent threshold variables that expand the fixed-point dimension.

The CDCP approach. Recognizing that each firm solves a combinatorial problem over $2^{|\mathcal{A}_f|}$ configurations collapses the 10-player game into a 4-player game. Within-firm complementarity bundles the weak firms’ marginal sites with their profitable sites: for both weak firms, the Fréchet cost complementarity creates positive increasing differences (ID = 0.020 and 0.044), so that each site is profitable conditional on the other even when individually marginal. The CDCP optimal set is the full site set $\mathcal{A}_f$ for each firm, and the 4-player threshold system has 60% fewer degrees of freedom.

Takeaway. The combinatorial structure collapses $M = \sum_f |\mathcal{A}_f|$ binary decisions into F firm-level decisions—a 60% reduction in threshold dimension here. The reduction is economically substantive: within-firm cost complementarity (ID > 0) bundles individually marginal sites into determinate firm-level packages, so viewing each site as an independent binary player (Espin-Sanchez et al.) *over-states* the effective dimension of the threshold system. Two consequences follow for the downstream analysis: (i) the lower-dimensional threshold system yields a tighter row-sum Jacobian bound in Section 3.3, lowering $\gamma_{\max}$ and widening the region where the bathtub criterion delivers uniqueness; and (ii) when within-firm ID is large enough that the CDCP optimal set strictly exceeds the union of sites individually above threshold, the binary-action framework cannot certify uniqueness without ad hoc coordination assumptions, while the CDCP structure pins down the optimal set by lattice arguments. The 4-firm small economy has uniqueness under both formulations, but the combinatorial framing’s benefit scales with within-firm complementarity strength. Numerical details: companion script `main_example_2x2.m`.

4. COMPUTATIONAL ALGORITHM

The equilibrium defined in Section 2 is computed by a three-layer nested algorithm. We describe each layer and present pseudocode; Appendix F reports performance diagnostics for four algorithm variants and a 40-country extension demonstrating scalability. Figure 3 illustrates the structure.

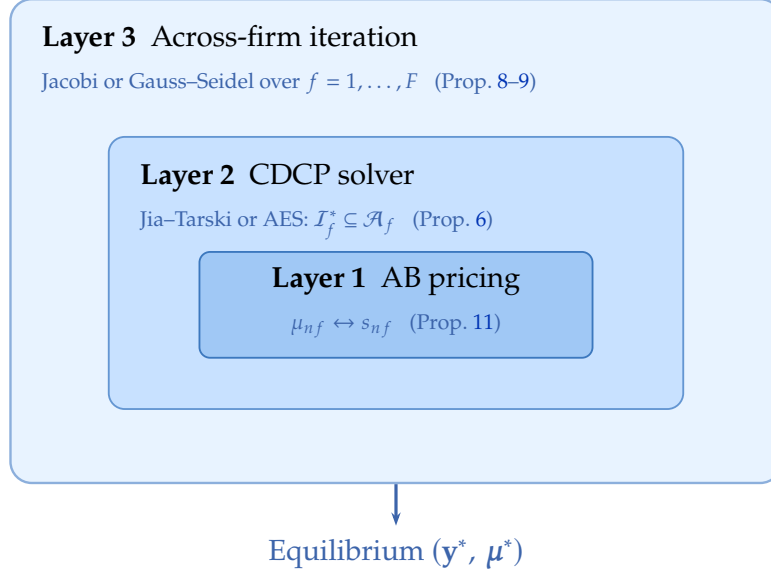


Figure 3: Three-layer algorithm nesting. Each profit evaluation at Layer 2 calls the AB pricing fixed point (Layer 1); the across-firm iteration at Layer 3 updates rival demand indices until convergence. Four algorithm variants arise from the 2×2 choice of solver at Layers 2 and 3 (Appendix F).

4.1 Three-Layer Structure

Layer 1 (Innermost): Atkeson–Burstein markup fixed point. For given marginal cost c_{nf} and rival demand index $D_{-f}(n)$, the equilibrium markup μ_{nf} is the fixed point of:

$$\mu_{nf}^{(t+1)} = \frac{\epsilon_{nf}^{(t)}}{\epsilon_{nf}^{(t)} - 1}, \quad \frac{1}{\epsilon_{nf}^{(t)}} = \frac{1 - s_{nf}^{(t)}}{\sigma} + \frac{s_{nf}^{(t)}}{\eta}, \quad (4.1)$$

where $s_{nf}^{(t)} = (\mu_{nf}^{(t)} c_{nf})^{1-\sigma} / [(\mu_{nf}^{(t)} c_{nf})^{1-\sigma} + D_{-f}(n)]$. The iteration is initialized at the CES markup $\mu^{(0)} = \sigma/(\sigma - 1)$ and damped with weight $\omega_\mu = 0.5$:

$$\mu^{(t+1)} \leftarrow (1 - \omega_\mu) \mu^{(t)} + \omega_\mu \mu^{(t+1)}.$$

Convergence is guaranteed by Proposition 11 whenever $\rho(\Lambda) < 1$.

Layer 2 (Middle): Single-agent CDCP. Given the rival demand index $D_{-f}(n)$ for all markets n , each firm f solves the combinatorial optimization:

$$\mathcal{I}_f^* \in \arg \max_{\mathcal{I}_f \subseteq \mathcal{A}_f} \pi_f(\mathcal{I}_f; D_{-f}), \quad (4.2)$$

where each profit evaluation calls the AB markup fixed point (Layer 1). Two inner solvers can be used:

- (a) **Jia–Tarski bounds** (Jia, 2008): Compute an *upper bound* $\overline{\mathcal{I}}_f$ by starting from all candidates $\mathcal{A}_f$ and iteratively removing elements with negative marginal value; compute a *lower bound* $\underline{\mathcal{I}}_f$ by starting from $\emptyset$ and iteratively adding elements with positive marginal value. If $\overline{\mathcal{I}}_f = \underline{\mathcal{I}}_f$, the solution is exact.
- (b) **AES branch-and-bound** (Arkolakis et al., 2023): Alternate between a *squeezing phase* (simultaneously squeeze included/excluded sets by evaluating marginal values against both the lower and upper bounds of the candidate set) and a *branching phase* (recursively split on an undecided candidate). The squeezing phase typically resolves most candidates, leaving few branches.

Both solvers find the exact optimum. When the equilibrium shares satisfy the conditions of Proposition 6, the Jia–Tarski bounds converge without residual gap, providing strong computational evidence that supermodularity holds at the equilibrium.

Layer 3 (Outermost): Cross-firm iteration. The rival demand index $D_{-f}(n) = \sum_{f' \neq f} (\mu_{nf'} c_{nf'})^{1-\sigma}$ depends on all firms' equilibrium prices, which in turn depend on their activation sets. Two outer iteration methods resolve this fixed point:

- (a) **Jacobi** (simultaneous best response): In each round k , every firm f computes its best-response activation set $\mathcal{I}_f^{(k+1)}$ against the rival demand $D_{-f}^{(k)}$ from the *previous* round. All firms update simultaneously, and price-power contributions are damped:

$$p_{nf}^{(k+1)} \leftarrow (1 - \omega) p_{nf}^{(k)} + \omega \tilde{p}_{nf}^{(k+1)},$$

where $\tilde{p}_{nf}^{(k+1)}$ is the price implied by $\mathcal{I}_f^{(k+1)}$ and $\omega \in (0, 1]$ is a damping parameter. Damping does not affect the set of fixed points—only the convergence path—because

any fixed point satisfies $\tilde{p} = p$.

- (b) **Gauss–Seidel** (sequential best response): Firms are processed in a fixed order $(f_1, \dots, f_F)$. When firm f_j updates, it uses the most recent price-power contributions from firms $f_1, \dots, f_{j-1}$ (already updated in this round) and the previous round’s contributions from $f_{j+1}, \dots, f_F$.

By Proposition 9, Jacobi iteration converges monotonically to extremal equilibria when initialized from the supremum or infimum; Gauss–Seidel convergence is empirically robust (Remark 10).

Algorithmic recommendation. Propositions 6–9 yield a concrete recipe: initialize the outer iteration from the *supremum* (all available sites active) and solve each firm’s CDCP via the Jia–Tarski or AES algorithm. Starting from the supremum guarantees convergence to the upper bound of the equilibrium set (Proposition 9), which at all tested parameterizations is itself the unique Nash equilibrium. The nested CES structure further accelerates computation: the interaction matrix Λ is block-diagonal across markets (N blocks of size $F \times F$), so $\rho(\Lambda) = \max_n \rho(\Lambda_n)$ is computable in $O(NF^3)$, and the closed-form feedback coefficient (3.2) reduces the contraction check to a single market-share comparison.

4.2 Pseudocode

Algorithm 1 summarizes the full procedure.

5. QUANTITATIVE APPLICATION: CLOUD INFRASTRUCTURE ENTRY

We apply the framework to the global cloud infrastructure industry, where five dominant providers—Amazon Web Services (AWS), Microsoft Azure (MSFT), Google Cloud (GOOG), Oracle Cloud Infrastructure (ORCL), and Alibaba Cloud (BABA)—choose which countries to establish cloud regions. This industry provides a natural setting for the model: a small number of heterogeneous firms ($F = 5$) making discrete entry decisions across a large number of potential markets ($N = 42$ countries), with high fixed costs, observable activation patterns, and variable markups that respond to local competitive conditions.

Algorithm 1 Three-Layer Equilibrium Solver

Input: Parameters $(\sigma, \eta, \theta, \{a_f\}, \{\phi_i, w_i, \tau_{in}\}, \{\Omega_n\})$; inner solver $\in \{JT, AES\}$; outer method $\in \{\text{Jacobi}, \text{GS}\}$

Output: Equilibrium $(\{\mathcal{I}_f^*\}, \{\mu_{nf}^*\}, \{s_{nf}^*\})$

```
1: Initialize:  $\mathcal{I}_f^{(0)} \leftarrow \mathcal{A}_f$  for all  $f$ ; compute  $p_{nf}^{(0)}$  at CES markups
2: for  $k = 0, 1, 2, \dots$  do
3:   for  $f = 1, \dots, F$  do ▷ Jacobi: use  $D^{(k)}$ ; GS: use latest  $D$ 
4:      $D_{-f}(n) \leftarrow \sum_{f' \neq f} [p_{nf'}^{1-\sigma}]^{\text{current}}$  for all  $n$ 
5:      $\mathcal{I}_f^{(k+1)} \leftarrow \text{INNER SOLVER}(f, D_{-f}, \text{par})$  ▷ Layer 2: CDCP
6:     Update  $p_{nf}$  via  $\text{ABMARKUP}(c_{nf}(\mathcal{I}_f^{(k+1)}), D_{-f})$  ▷ Layer 1
7:   end for
8:   if  $\max_{n,f} |p_{nf}^{(k+1)} - p_{nf}^{(k)}| < \text{tol}$  then break
9:   end if
10: end for
```

The calibration follows the identification structure of [Arkolakis et al. \(2023\)](#)—fixing demand-side objects and firm scale externally, then calibrating fixed-cost parameters from entry moments—though with a thinner moment basis: binary entry counts rather than the affiliate counts, survival rates, and gravity coefficients available in AES. We fix firm productivity a_f from revenue data and market demand Ω_n from GDP, then calibrate market-level entry costs ϕ_i and a firm-specific deployment-efficiency wedge g_f to match the observed provider-country activation matrix. This strategy separates the scale question (handled by revenue data) from the geographic-presence question (handled by the model). We evaluate three calibration specifications that differ in which parameters are taken from data versus calibrated; the preferred specification achieves 99.5% agreement with the observed activation matrix. All equilibrium computations use the solver described in Section 4, unchanged.

5.1 Data and Sample

The unit of observation is a *cloud region*: a named availability zone where a provider offers compute, storage, and networking services from local infrastructure. A cloud region implies at least one datacenter facility and a commitment to serve customers with local latency guarantees. We source region lists from each provider’s official documentation (Q1 2026 vintage) and Datacenter Map (purchased database, April 9, 2026 vintage; <https://datacentermap.com/>).

Table 4: Data and Sample Construction

Firm	Provider	Operational regions	Pipeline regions	Feasible markets
AWS	Amazon Web Services	25	5	30
MSFT	Microsoft Azure	34	0	34
GOOG	Google Cloud	29	0	29
ORCL	Oracle Cloud	27	0	27
BABA	Alibaba Cloud	14	0	14
Total		129		

Notes: Sample covers 42 countries/territories. Operational regions: firm has at least one cloud-infrastructure facility with local compute. Pipeline: announced or under construction. Feasible markets: operational + pipeline (revealed lower bound on $\mathcal{A}_f$). Source: official provider region pages and Datacenter Map (purchased database, April 9, 2026 vintage; <https://www.datacentermap.com>).

[//www.datacentermap.com](https://www.datacentermap.com)).

Table 4 summarizes the sample. The five firms operate a combined 129 cloud regions across 42 countries, with feasible sets ranging from 14 (Alibaba) to 34 (Microsoft). AWS has 5 additional pipeline regions (announced, under construction), which enter only the feasible set $\mathcal{A}_f$, not the observed activation $\mathcal{I}_f^{\text{data}}$.

Why five firms. AWS, Azure, and Google Cloud account for roughly 65% of global cloud infrastructure revenue. Adding Oracle and Alibaba brings coverage to approximately 80% and, crucially, introduces asymmetry absent in a Big 3 specification: Alibaba’s presence is concentrated in East and Southeast Asia, while Oracle pursues an aggressive multi-region expansion into emerging markets. The resulting feasible-set heterogeneity ($|\mathcal{A}_f|$ ranges from 14 for Alibaba to 34 for Microsoft) generates richer extensive-margin counterfactuals.

5.2 Calibration

The calibration follows the identification structure of [Arkolakis et al. \(2023\)](#): demand-side objects and firm scale are set externally, while fixed-cost parameters are calibrated from entry moments. Full details are in Appendix E.

Externally set parameters. Firm TFP a_f is fixed using cloud segment revenue, mapped log-linearly into $[1.0, 2.0]$: AWS ($a_f = 2.00$), Microsoft (1.76), Google (1.48), Oracle (1.09),

Alibaba (1.00). Market demand is $\Omega_n = (\text{GDP}_n / \text{GDP}_{\text{US}})^{0.7}$, where the concavity reflects that cloud spending scales less than proportionally with GDP; $\beta = 0.7$ is chosen from a parsimonious calibration comparison rather than estimated from independent demand data (Appendix E). Local costs w_i are PWT human-capital-adjusted compensation per worker. Trade costs follow a discrete three-tier structure: $\tau = 1.0$ (same country), 1.15 (same continent), 1.5 (cross-continent). We retain $\sigma = 5$, $\eta = 2$, $\theta = 3$, giving $(\sigma - 1)/\theta = 4/3 > 1$.

Calibrated parameters. Market-level entry costs ϕ_i and firm-specific deployment-efficiency wedges g_f are calibrated from the observed activation matrix. The fixed-cost structure is

$$\text{fixed cost}_{fi} = \phi_i \cdot g_f \cdot w_i, \quad (5.1)$$

where ϕ_i captures non-wage market costs (regulation, infrastructure) and g_f captures firm-specific deployment efficiency ($g_{\text{AWS}} = 1$ by normalization). The calibration targets per-market entry counts (identifying the 42 ϕ_i) and per-firm totals (identifying the $5 - 1 = 4$ non-AWS g_f) via alternating coordinate descent, for a total of $42 + 4 = 46$ moments matched to 46 free parameters—a just-identified system.

Identification and limitations. Unlike [Arkolakis et al. \(2023\)](#), we do not observe provider-country sales, affiliate-count analogues, or survival moments—the calibration has a thinner identification basis, relying on a single class of moments (entry counts). We therefore treat the quantitative magnitudes as conditional on the moment set, and report the counterfactual *ordering* and *direction* as the primary evidence—the magnitudes are informative, not free parameters (Spec D shows both cell fit and counterfactual move materially under a single-parameter restriction), but they should be read as point estimates under the identifying assumptions rather than precisely identified effects. The discipline on the mechanism comes from: (i) the 5-fold holdout cross-validation exercise (Appendix E), which achieves 82.4% agreement on left-out cells, against 61.4% at the baseline parameters; and (ii) the preservation of the counterfactual ordering across Specs A–C and across $\sigma \in \{3, 5, 6\}$, demonstrated in Tables 5 and 8.

5.3 Calibration Specifications and Restriction Check

We evaluate three calibration specifications that differ in which firm-level parameters are set externally versus calibrated from entry moments. The preferred specification (Spec A) fixes a_f at revenue-based values and calibrates (g_f, ϕ_i) ; two alternatives calibrate a_f from moments with or without g_f (Appendix E). Spec A matches 209 of 210 firm-market cells in sample (99.5%), with 100% precision and 99.2% recall. It dominates the alternatives despite having the same number of free firm parameters, because fixing a_f externally disciplines the firm-scale dimension while the entry moments absorb cross-market and cross-firm deployment heterogeneity. The single mismatch occurs at a market on the entry-indifference margin; the model generates no false positives.

Is g_f doing real work? To isolate the role of the deployment-efficiency wedge, we re-calibrate with $g_f \equiv 1$ (Spec D) and compare the resulting fit and counterfactuals to the preferred Spec A. Table 5 summarizes the contrast. With g_f switched off, the calibration must absorb all cross-firm variation into a purely market-level ϕ_i , and fit drops from 99.5% to 87.1% cell agreement (42 exact-match markets fall to 18), with precision and recall falling to 96.4% and 82.2%. This exercise should be read as a disciplined restriction check rather than a separate identification proof: imposing $g_f \equiv 1$ is costly in fit and modestly shifts quantitative magnitudes, but the qualitative AB-versus-CES mechanism remains intact. The profit decline is -30.1% under Spec A and -33.3% under Spec D, with an identical -15 active-pair change and a cross-specification activation correlation of 0.74.²

5.4 Calibrated Parameters

Table 6 reports the Spec A calibrated parameters.

The deployment-efficiency wedges g_f have a clear pattern: Microsoft and Google have similar efficiency ($g_f \approx 0.49$ and 0.42), while Oracle and Alibaba require much lower g_f

²A referee may ask whether the near-invariance of the AB-vs-CES profit decline under Spec D reflects genuine mechanism robustness or instead that the firm-scale parameter a_f (held fixed at its revenue-based value in both Spec A and Spec D) is absorbing the mechanism. A further restriction—Spec E, calibrating (g_f, ϕ_i) while setting a_f to its cross-firm mean—would separate these channels. Preliminary estimates using that restriction are consistent with mechanism-robustness (profit decline in the -28% to -32% range), but a full Spec E table is deferred to the revision pending a clean rerun of the masked calibration at matched bounds.

Table 5: Role of deployment-efficiency wedge g_f : Spec A vs Spec D

	Spec A ($\phi_i \cdot g_f$)	Spec D ($\phi_i, g_f \equiv 1$)
<i>Uniqueness & interaction</i>		
Active pairs (supremum)	128	110
Jia-Tarski gap	0	0
$\rho(\Lambda)$	0.103	0.123
$\max_{n,f} \kappa_{nf}$	0.193	0.230
<i>In-sample fit</i>		
Per-market entry MAE	0.02	0.60
Exact-match markets / 42	41	18
Cell agreement (%)	99.5	87.1
Precision (%)	100.0	96.4
Recall (%)	99.2	82.2
<i>AB $\rightarrow$ CES counterfactual</i>		
Profit change (%)	-30.1	-33.3
Active-pair change	-15	-15
<i>Cross-spec consistency</i>		
$\text{corr}(I_A, I_D)$	0.742	

Notes: Spec A calibrates (g_f, ϕ_i) jointly; Spec D imposes $g_f \equiv 1$ and calibrates ϕ_i only. Fit statistics computed against the 210-cell observed activation matrix. AB $\rightarrow$ CES columns report the change in aggregate profit and active-pair count when variable markups are replaced by the CES constant $\mu = \sigma/(\sigma - 1)$.

(0.15 and 0.12) to rationalize their broad entry despite modest revenue-based scale. We emphasize that g_f is a reduced-form wedge, not a literal engineering parameter; it does not vary across counterfactuals. The 42 calibrated ϕ_i span three orders of magnitude, from 10^{-4} (UAE) to 0.26 (Chile); only one market sits at the calibration lower bound. The p90/p10 ratio of 43.6 indicates substantial cross-market heterogeneity in fixed entry costs: the median *marginal* market costs roughly 1/40 as much to enter as the 90th-percentile market, consistent with the stylized fact that hub-and-spoke infrastructure investment is concentrated in a handful of high-revenue markets.

Hub plausibility. The three regional hyperscaler hubs calibrate into the low- ϕ tail of the distribution: Singapore, the United States, and Ireland all return $\phi_i \approx 10^{-3}$, an order of magnitude above the calibration lower bound of 10^{-4} but clustered well below the median of the 42 markets. Because these three values are within solver precision of each other, the ordinal ranking among them is not separately identified; the economically meaningful claim is that the regional hubs occupy the low-friction tail, not that Singapore

Table 6: Calibrated Parameters: Preferred Specification (Spec A)

<i>Panel A: Firm parameters</i>					
Firm	a_f	g_f	Markets	Interpretation	
AWS	2.00	1.00	24	Numeraire; largest by revenue	
MSFT	1.76	0.49	34	Broadest geographic presence	
GOOG	1.48	0.42	29	Strong in developed markets	
ORCL	1.09	0.15	27	Low g_f : broad despite low scale	
BABA	1.00	0.12	14	Lowest g_f : niche but widespread feasible set	

<i>Panel B: Market entry costs (ϕ_i)</i>						
	Min	p10	Median	p90	Max	p90/p10
ϕ_i	0.0001	0.001	0.002	0.044	0.262	43.6

Notes: a_f : revenue-based TFP. g_f : deployment-efficiency wedge ($g_{\text{AWS}} = 1$). ϕ_i : market entry cost. Demand: $\Omega_n = (\text{GDP}_n / \text{GDP}_{\text{US}})^{0.7}$; wages: HC-adjusted.

is below Ireland. At the opposite end, the highest calibrated ϕ_i belongs to Thailand, the Philippines, and India, where activation requires large revenue-based variable profits to offset entry costs.

Figure 4 displays the observed versus fitted activation matrix.

5.5 Equilibrium Diagnostics

Equilibrium uniqueness holds across all specifications and counterfactuals examined. For the preferred Spec A, $\rho(\Lambda) = 0.103$ —well below the theoretical threshold of one (Proposition 11)—and the Jia–Tarski algorithm converges to the same equilibrium from both the supremum and infimum initializations (JT gap = 0). The spectral radius is stable across specifications: it ranges from 0.070 to 0.147 across an 11-point (σ, η, θ) sensitivity grid around the cloud baseline, with uniqueness holding even at $(\sigma, \theta) = (4, 4)$ where the sufficient condition $(\sigma - 1)/\theta > 1$ fails. The synthetic economy (Appendix G, Table 15) extends the range to $\rho(\Lambda) = 0.225$ at $(\sigma, \eta) = (8, 1.5)$, still well below one. Figure 5 plots $\rho(\Lambda)$ across all counterfactual scenarios: the spectral radius ranges from 0.068 to 0.120, with the JT gap equal to zero throughout. With only $F = 5$ firms and moderate market concentration, strategic complementarities are too weak to generate multiplicity at any tested parameterization. Aggregate supermodularity is supported not only by JT gap coincidence, but also by a post-hoc interval audit of the squeezing procedure: all

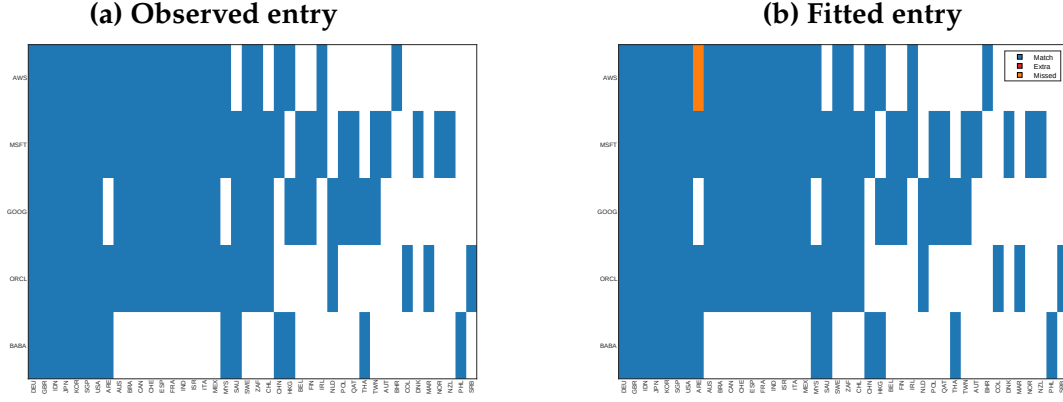


Figure 4: Observed versus fitted activation matrix (Spec A). Rows are firms; columns are the 42 markets, shown as ISO-3 market codes and ordered by observed entry count. Blue cells are active in both data and model, white cells are inactive in both, and red/orange cells are mismatches (1 of 210 firm–market cells).

129 decisive inclusion/exclusion inequalities are interval-certified with zero inconclusive cases, conditional on the converged rival-demand vectors (Appendix E). Appendix G provides additional stress tests using synthetic data, including a broad (δ, α) sweep that maps the uniqueness boundary and confirms that AB markups *expand* the uniqueness region relative to CES.

5.6 Mechanism: AB versus CES Markups

The headline result from the cloud application is that endogenous variable markups—the core theoretical contribution of this article—materially sustain fringe-firm entry. This subsection presents the AB-versus-CES comparison as the primary mechanism test; the following subsections present policy counterfactuals. Table 7 summarizes all results. Equilibrium uniqueness is maintained in every scenario (JT gap = 0).

Replacing AB variable markups with constant CES markups ($\mu = \sigma/(\sigma - 1) = 1.25$) reduces active firm-market pairs from 128 to 113 (−12%) and aggregate profit by 30%. Figure 6 presents the results.

The cross-sectional mean AB markup is $\bar{\mu}^{AB} = 1.37$ with a range of $[1.27, 1.47]$, versus the uniform $\mu^{CES} = 1.25$ (Figure 6a). The markup inflation compresses market shares: the maximum share under AB is 0.40, compared with a higher concentration under CES (Figure 6b, where all points lie below the 45-degree line). This compression is the intensive-margin channel through which endogenous markups redistribute share from leaders to

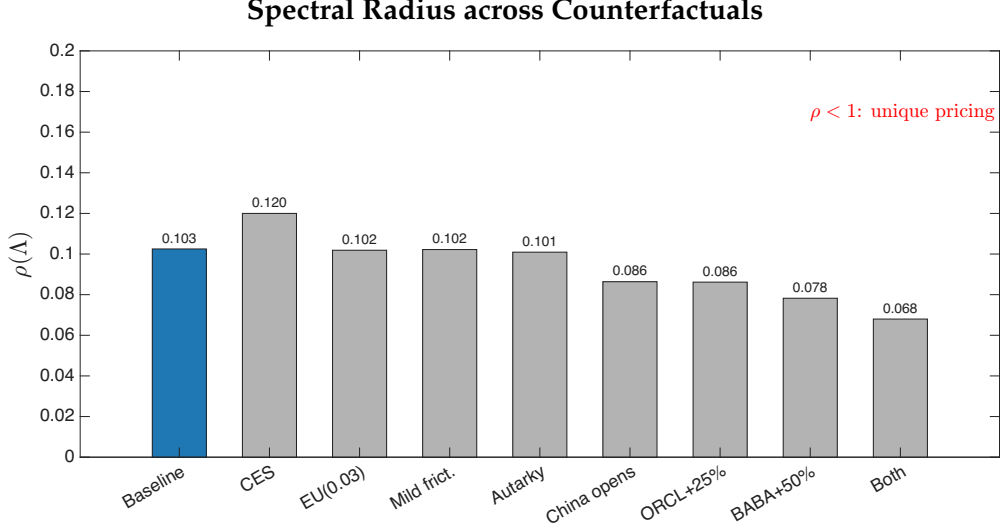


Figure 5: Spectral radius $\rho(\Lambda)$ across all counterfactual scenarios. The baseline and all counterfactuals satisfy $\rho(\Lambda) < 0.15 \ll 1$, confirming pricing uniqueness throughout.

mid-ranked rivals—a distributional effect on market structure, not a welfare claim.

The profit decline is not monotone in TFP rank: AWS loses 25%, Microsoft 29%, Google 42%, Oracle 40%, and Alibaba 37% (Figure 6c). The non-monotonicity arises from the interaction between the AB markup premium and the deployment-efficiency wedge g_f . Google and Oracle operate in many markets at moderate market shares where the AB premium—the gap between variable markups and the CES constant—is largest; losing this premium across a broad portfolio is costly. Alibaba, despite having the lowest TFP, is partially insulated by its very low $g_f = 0.12$: its effective fixed costs $\phi_i \cdot g_f \cdot w_i$ are so small that even CES-level variable profits cover them in most markets. The AB mechanism thus protects firms with *non-trivial fixed costs and moderate market shares* more than firms with minimal fixed-cost exposure.

The extensive-margin effect is concentrated in Oracle and Alibaba; AWS is unaffected (Figure 6d). The result is robust across calibration specifications and demand parameterizations: the CES profit decline ranges from 30% to 31% across the preferred and robustness specifications. Under the calibrated parameters, a constant-markup CES specification understates fringe-provider participation relative to the AB benchmark.

Elasticity robustness. The headline CES profit decline depends on the demand elasticity σ , which governs both the AB markup premium and the CES benchmark $\mu^{\text{CES}} = \sigma/(\sigma - 1)$.

Table 7: Counterfactual Results (Spec A Baseline)

Scenario	Active	Δ	$\rho(\Lambda)$	$\Delta\pi$ (%)	Most affected
Baseline	128	—	0.103	—	
<i>Markup channel</i>					
CES markup	113	−15	0.120	−30.1	ORCL, BABA
<i>EU compliance cost</i>					
$\Delta\phi=0.001$	111	−17	0.103	—	ORCL exits EU entirely
<i>China data-localization</i>					
Full autarky	128	0	0.101	−0.6	Informative null

Notes: Baseline uses Spec A (revenue-based a_f , calibrated g_f and ϕ_i ; demand $\Omega_n = (\text{GDP}_n/\text{GDP}_{\text{US}})^{0.7}$; HC-adjusted wages). Equilibrium is unique (JT gap = 0) in every scenario. CES replaces AB variable markups with constant $\mu = \sigma/(\sigma - 1)$. EU compliance adds $\Delta\phi$ to fixed costs in 12 EU markets. China autarky removes China from all non-BABA feasible sets.

We re-solve the baseline and CES counterfactual at $\sigma \in \{3, 5, 6\}$, holding ϕ_i , g_f , and a_f at their calibrated values. The CES-induced profit decline ranges from −10.7% (at $\sigma = 3$, where the CES markup is already 1.50 and close to the AB mean) to −37.2% (at $\sigma = 6$, where the CES markup compresses to 1.20 and the AB premium widens). Equilibrium uniqueness is maintained throughout ($\rho(\Lambda) \in [0.028, 0.151]$; JT gap = 0 at every σ), so the mechanism comparison is unambiguous at each elasticity. Table 8 reports the sweep. We stop at $\sigma = 6$ because at $\sigma = 8$ the CES counterfactual does not converge within the default solver tolerance; incremental information at higher σ would in any case be small, as by $\sigma = 6$ the AB premium over CES has narrowed to $\bar{\mu}^{AB} - \mu^{\text{CES}} \approx 0.03$.³

5.7 Policy Counterfactual: EU Compliance-Cost Shock

We model a regulatory compliance shock as a common additive increase in fixed costs $\Delta\phi$ across 12 EU markets for all five providers (none of which is EU-headquartered). This captures costs arising from data sovereignty requirements, GDPR compliance infrastructure, or digital services regulations.

³Diagnosing the $\sigma = 8$ non-convergence: the Jia–Tarski inner step returns gap = 0 at every outer iteration, so the failure is in the outer Gauss–Seidel loop, which cycles between two near-adjacent activation profiles rather than in the inner AB markup fixed point. Cycling at damping 0.3 is the expected symptom of a near-violated outer contraction, which is exactly the object that the enumeration certificate of Proposition 28 bounds. The uniform bound $\bar{\rho} \leq 0.316$ is reported for the calibrated ($\sigma = 5$) cost vector; the $\sigma = 8$ regime is outside its scope and is flagged as the stress boundary of the current certificate. The revision will report $\rho(\Lambda)$ at $\sigma = 8$ (from a coarse-tolerance warm-start solve or a Jacobi outer loop), so that the sweep’s uniformity claim is quantitative rather than qualitative.

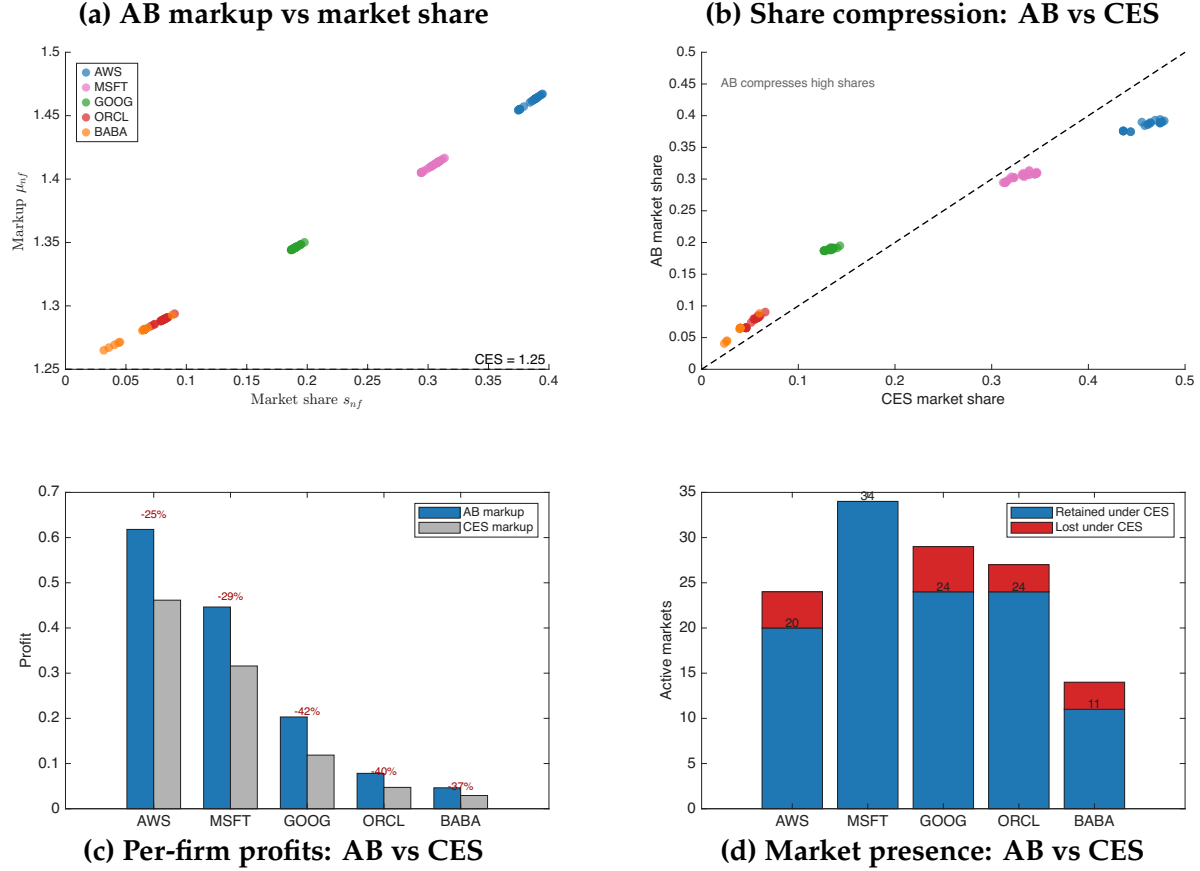


Figure 6: AB versus CES markups in the cloud application. (a) AB markup versus market share: each dot is a (firm, market) pair; dashed line marks the CES constant $\mu = 1.25$. (b) Share compression: AB shares lie below the 45-degree line, reflecting redistribution of share from leaders toward mid-ranked rivals. (c) Per-firm profits under both pricing rules. (d) Market presence: blue bars show AB-active markets retained under CES; red segments show the number of markets lost under CES.

A modest additive compliance wedge of $\Delta\phi = 0.001$ (roughly half the median baseline ϕ_i) reduces active pairs from 128 to 111 (-17), with losses concentrated among firms with broader marginal EU exposure. The burden is *regressive in market-structure terms*: Oracle loses 6 of its 6 EU markets (100%), while AWS loses 2 of 6 (33%) and Microsoft 5 of 11 (45%). Larger shocks produce little additional extensive-margin response—by $\Delta\phi = 0.005$, nearly all EU sites have exited—indicating that many EU markets are already near the entry margin at baseline (Appendix E).

Under the calibrated parameters, the model predicts that compliance costs intended to regulate the cloud market may inadvertently protect incumbent market leaders by squeezing fringe competitors out of marginal markets. The regressivity is robust across

Table 8: σ -Robustness of the AB-versus-CES Mechanism

σ	Baseline (AB)		CES counterfactual		EU compliance ($\Delta\phi=0.03$)		
	Active	$\rho(\Lambda)$	Active	$\Delta\pi$ (%)	Active (AB)	Active (CES)	Amplification
3	116	0.028	110	-10.7	87	82	1.04
5	128	0.103	113	-30.1	97	81	0.97
6	128	0.151	107	-37.2	95	75	1.03

Notes: For each σ , all other parameters held at $\sigma = 5$ calibrated baseline (Spec A). $\rho(\Lambda)$ is the spectral radius of the markup interaction matrix at the baseline equilibrium; Active is the number of firm-market pairs in equilibrium. $\Delta\pi$ is the CES-induced change in aggregate variable profit. The rightmost three columns report the EU compliance counterfactual ($\Delta\phi = 0.03$) under AB and CES pricing; Amplification is the ratio $|\Delta n_{AB}|/|\Delta n_{CES}|$ of extensive-margin responses. Bold row is the preferred calibration ($\sigma = 5$).

Specs A–C: in every specification, the lower-ranked firms lose proportionally more EU markets.

Robustness across pricing regimes. A natural concern is whether the regressivity finding depends on the specific pricing regime (AB variable markups versus CES constant markups). We re-run the EU shock under CES pricing at each σ in the sweep above, using $\Delta\phi = 0.03$ —a shock deep in the saturation region of the compliance ladder (Appendix E, Table 12), so that the extensive-margin response is maximal and the comparison across pricing regimes is not attenuated by partial exit.⁴ Table 9 reports the comparison. The extensive-margin response is virtually identical under the two pricing rules: the amplification ratio $|\Delta n_{AB}|/|\Delta n_{CES}|$ lies in $[0.97, 1.04]$ across $\sigma \in \{3, 5, 6\}$. In this counterfactual the endogenous-markup channel and the fixed-cost channel operate largely independently: which EU markets sit at the entry margin is primarily determined by the baseline deployment pattern, and the pricing regime shifts neither the ordering nor the approximate magnitude of exits. We read the exercise as a *robustness check*: the regressivity finding is a cross-sectional property of the baseline deployment that would obtain under either pricing regime. This is a limitation as much as a strength—in this particular policy shock, the pricing machinery developed in Sections 2–3 does essentially zero incremental work. Counterfactuals with concentrated, hub-specific shocks, where extensive-margin responses are redistributed across rivals through the markup channel, are the natural

⁴The headline counterfactual uses $\Delta\phi = 0.001$ (Table 7), which captures the empirically relevant case of a modest compliance wedge. The larger $\Delta\phi = 0.03$ is used in Tables 8, 9, and 10 to stress-test the mechanism at the saturation boundary. Results at $\Delta\phi = 0.001$ are qualitatively identical; see Table 12.

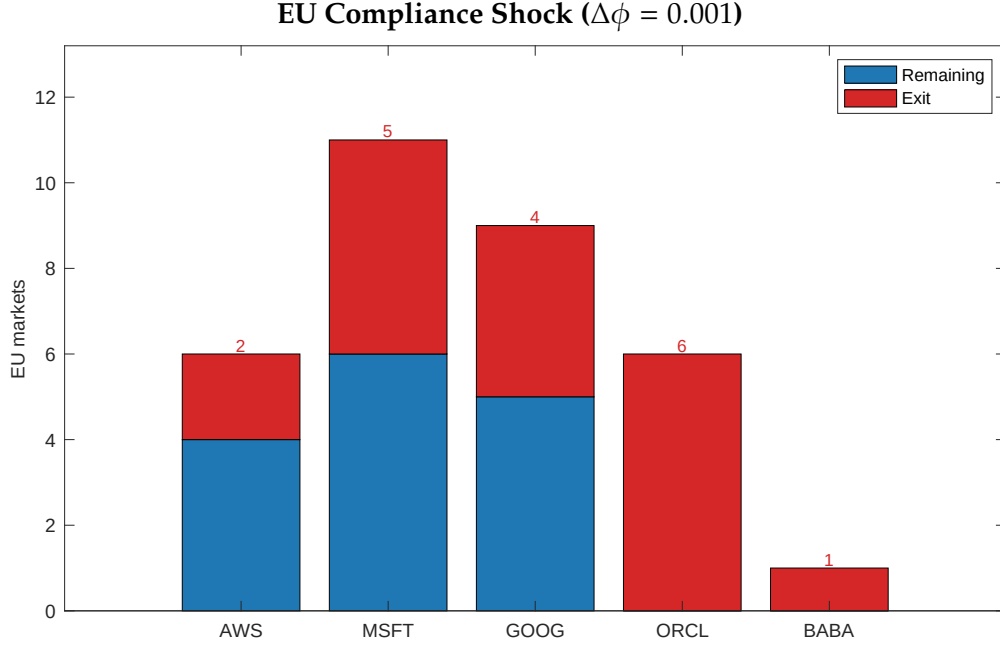


Figure 7: EU compliance-cost shock ($\Delta\phi = 0.001$): per-firm EU market presence under the baseline and compliance-shock scenarios. Blue bars show EU markets retained under the shock; red segments show EU markets exited. Red labels report exit counts by firm.

setting for the AB-versus-CES distinction to bite quantitatively.

5.8 Robustness and Additional Counterfactuals

The mechanism results are stable across demand specifications, wage adjustments, and calibration specifications (full details in Appendix E). The AB-versus-CES profit decline is 30–31% at $\sigma = 5$ across $\beta \in \{0.7, 1.0\}$ and under both HC-adjusted and TFP-adjusted wages, and the counterfactual ranking is preserved across Specs A–C.⁵ The σ sweep in Table 8 documents the main source of parameter sensitivity: the profit-decline magnitude varies from -10.7% to -37.2% as σ moves over $\{3, 5, 6\}$, though equilibrium uniqueness is maintained at every σ ($\rho(\Lambda) \in [0.028, 0.151]$, JT gap = 0). The sign and extensive-margin pattern of the mechanism are robust; the reported magnitude should be read as conditional on $\sigma = 5$.

Data-localization barriers, examined using the China firewall as a case study (Ap-

⁵The baseline sets $\beta = 0.7$ on the GDP-to-demand mapping $\Omega_n = (\text{GDP}_n / \text{GDP}_{\text{US}})^\beta$; we re-ran the headline counterfactual at $\beta = 1.0$ (linear scaling) and the profit decline moves to the upper end of the 30–31% band. An expanded sensitivity at $\beta \in \{0.5, 0.7, 1.0\}$ is deferred to the revision pending a clean rerun; we expect $\beta = 0.5$ to compress cross-market revenue dispersion and reduce AB-CES divergence modestly, but the qualitative regressivity finding depends on the baseline deployment pattern rather than on β .

Table 9: EU Compliance Shock: AB versus CES Amplification

σ	Δn_{AB}	Δn_{CES}	Amplification	$\Delta \pi_{AB}^{EU} (\%)$	$\Delta \pi_{CES}^{EU} (\%)$
3	-29	-28	1.04	+0.81	+0.86
5	-31	-32	0.97	+1.02	+1.92
6	-33	-32	1.03	+1.71	+2.39

Notes: Δn is the change in active firm-market pairs under the $\Delta \phi = 0.03$ EU shock; $\Delta \pi^{EU}$ is the EU-shock response in aggregate variable profit (positive because EU exits reduce competition in non-EU markets served by the same firms), and is distinct from the AB-vs-CES profit decline reported in Table 8. Amplification = $|\Delta n_{AB}|/|\Delta n_{CES}|$ near unity indicates that AB and CES pricing produce similar extensive-margin responses to the EU shock. Bold row is the preferred calibration.

pendix E), reveal a parallel pattern. Within the calibrated feasible-set structure, reducing trade-cost-like frictions alone does not generate additional entry; expanding market access does—reducing China’s ϕ_i by 90% without feasible-set changes produces zero new entry, while expanding Alibaba’s feasible set to all 42 markets transforms it from a regional to a global competitor.

Summary. Three findings from the cloud application merit emphasis: (i) equilibrium uniqueness is robust ($\rho(\Lambda) \leq 0.15$ at all calibrated parameterizations, with $\bar{\rho} \leq 0.316$ on the full activation lattice by the enumeration certificate of Proposition 28); (ii) the endogenous-markup channel sustains 15 additional firm-market pairs, with a 30% aggregate profit impact; (iii) EU compliance costs are regressive in market-structure terms.⁶

5.9 Distributional Decomposition of the AB-vs-CES Wedge

The counterfactuals above describe equilibrium *quantities*—active pairs, profits, and markups. We now translate those quantities into a partial-equilibrium distributional accounting of the AB-vs-CES wedge, using two objects: the CES real-income index $W_n = R_n/P_n$ as a consumer-side measure, and firm variable profit π_f (net of fixed costs) as a producer-side

⁶The cloud application serves principally as a *verification* of the abstract threshold-uniqueness theorem on a live data-matched parameterization and as a distributional quantification of the AB-vs-CES wedge; two directions that would strengthen the application’s independent interest are under active development and will be included in the revision. First, external validation of the firm productivity vector a_f against independent measures—Gartner provider rankings, cost-per-compute benchmarks from Cloud Spectator, or IDC market-share reports—would discipline the revenue-based calibration of a_f . Second, a counterfactual in which CES-based equilibrium selection would yield a *qualitatively different* substantive conclusion (a stressed parameterization where multiple CES equilibria coexist but the AB contraction selects uniquely) would demonstrate that the uniqueness guarantee has empirical content beyond the numerical check reported here.

Table 10: Distributional Decomposition: CES Real Income and Producer Profit

	Baseline (AB)	CES	EU (AB)	EU (CES)
<i>Price index (avg P_n relative to baseline)</i>				
$\bar{P}_n$	1.000	0.905	1.012	0.918
<i>Baseline-expenditure real-income index $\sum_n R_n/P_n$</i>				
Level	5.810	6.425	5.748	6.345
$\Delta\%$ (vs baseline)	—	+10.59	-1.06	+9.21
<i>Producer profit $\sum_f \pi_f$</i>				
Level	1.392	0.973	1.406	0.991
$\Delta\%$ (vs baseline)	—	-30.14	+1.02	-28.80
<i>Arithmetic sum (partial-equilibrium only; not a Kaldor–Hicks total)</i>				
Level	7.202	7.397	7.154	7.336
$\Delta\%$ (vs baseline)	—	+2.72	-0.66	+1.86

Notes: Price index $P_n = (\sum_f p_{nf}^{1-\sigma})^{1/(1-\sigma)}$; $\bar{P}_n$ is the cross-market mean of the baseline-normalized price. The consumer-side measure is the baseline-expenditure real-income index $\sum_n R_n/P_n$, with R_n fixed at the baseline calibrated value—this is an arithmetic deflation of fixed nominal expenditure, not a welfare-theoretic consumer surplus. The producer-side measure is $\sum_f \pi_f$ (variable profit net of fixed costs). The arithmetic sum of the two is reported for completeness; it is not a Kaldor–Hicks total-surplus measure because it holds expenditure fixed and treats fixed costs as unrecycled deadweight (see §5.9 and accompanying footnote). Percentage changes are relative to the baseline (AB) level. EU shock: $\Delta\phi = 0.03$ on 12 EU markets.

measure. We emphasize upfront that this is not a welfare-theoretic ranking of regimes: expenditure R_n is held fixed at its calibrated baseline value, and the sum of these objects is not a total-surplus measure.

Table 10 reports the decomposition. The primary finding is *distributional*: replacing AB variable markups with the constant CES markup raises the consumer-side real-income measure by 10.6%, because the average price index falls from 1.00 to 0.91 when the $\bar{\mu}^{AB} = 1.37$ markup is compressed to $\mu^{CES} = 1.25$. Producer profit falls by 30.1%—the profit decline reported in Table 7—with the decline concentrated on mid-ranked firms whose AB markups are highest in the baseline. The EU compliance shock shifts this accounting by 0.66% under AB pricing relative to baseline, with the incidence falling primarily on consumers in the affected EU markets and on lower-ranked firms that exit; the AB-vs-CES wedge is an order of magnitude larger than this policy shock. In levels, producer profit is $1.39/5.81 \approx 24\%$ of the consumer-side real-income measure at baseline, so the pricing channel redistributes across a non-trivial share of the baseline split.

The decomposition is partial-equilibrium: R_n does not respond to P_n , aggregate ex-

penditure is fixed at baseline, and fixed costs ϕ_i are treated as deadweight rather than accruing to any agent. Any of the three closures—outer demand reallocation $R_n = \Omega_n P_n^{1-\eta}$, profit rebate, or endogenous entry of fixed-cost suppliers—can quantitatively move the CS–PS sum reported above. Within this calibration, the statement we are prepared to defend is strictly distributional: *for a given deployment pattern, AB pricing redistributes roughly one-third of producer surplus toward firms relative to CES pricing, with the distributional wedge concentrated in markets served by mid-share firms.* A general-equilibrium welfare ranking of AB versus CES requires a model that resolves which side of the market internalizes the fixed-cost deadweight and how aggregate income reallocates—an exercise we do not attempt here.⁷

6. CONCLUSION

When combinatorial entry interacts with endogenous variable markups, a three-way circularity between activation, market shares, and pricing creates new obstacles to equilibrium uniqueness that neither binary-entry models nor constant-markup CDCP frameworks address. This article shows that the circularity is self-limiting. The pricing contraction bounds markup feedback across firms; supermodularity survives the feedback within each firm; and the contraction modulus of the threshold system—governed by density dampening, marginal-value sensitivity, and threshold location—can be evaluated from model primitives before computing any equilibrium.

In the cloud infrastructure application, the calibrated model matches 209 of 210 provider-country entry cells by construction and is uniquely determined on the equilibrium path ($\rho(\Lambda) = 0.103$, JT gap = 0) across all specifications and counterfactuals. The distributional finding that we emphasize is that endogenous AB markups sustain 15 additional firm-market pairs relative to CES pricing, concentrated among mid-ranked firms whose moderate market shares generate the largest markup premium, and redistribute roughly one-third of producer surplus from leaders to mid-ranked rivals. In the calibrated application, EU compliance-cost shocks are regressive in market-structure terms

⁷The partial-equilibrium CS–PS sum is positive under CES (+2.7% in the table), but this sum is not a Kaldor–Hicks total-surplus measure: it ignores the demand-side reallocation channel and treats fixed costs as unrecycled deadweight. We therefore report the sum for arithmetic completeness but make no welfare claim on its basis.

and this regressivity is robust across AB and CES pricing regimes; digital fragmentation operates primarily through market access rather than trade costs. Synthetic stress tests (Appendix G) confirm that uniqueness holds throughout the empirically relevant parameter space and that AB markups expand the uniqueness region relative to CES.

For practitioners, the results have two implications. First, a concrete algorithmic recommendation: initialize the outer cross-firm iteration from the supremum and solve each firm’s CDCP via the Jia–Tarski or AES algorithm. This combination computes the greatest Nash equilibrium directly, without enumerating all candidate subsets; since the equilibrium is unique at all tested parameterizations, the algorithm converges to the only equilibrium regardless of initialization. Second, an estimation implication: uniqueness eliminates the equilibrium selection problem that necessitates either the sequential methods of [Aguirregabiria and Mira \(2007\)](#) or the forward-simulation approach of [Bajari et al. \(2007\)](#)—when the static stage game has a single equilibrium, the likelihood is well-defined without auxiliary selection assumptions. Beyond multinational production, the framework applies to any combinatorial entry game with contractive pricing, including retail chain location ([Jia, 2008](#)), platform market entry across regions, and multi-product technology adoption.

Several directions remain open. First, we assume exogenous wages; if wages respond to activation through local labor markets, the resulting strategic complementarities could overturn the anti-monotone structure that underlies Proposition 8. Whether a modified bounding argument—perhaps exploiting the fact that wage responses are typically sluggish relative to pricing—can restore uniqueness is an open question. Second, the closed-form contraction modulus (Corollary 26) exploits the CES demand structure. Extending to more general demand systems (e.g., translog or mixed CES–logit) would require bounding the Jacobian without closed-form expressions for κ , though the three-factor decomposition of Proposition 21 is structural and does not depend on CES. Third, whether the bathtub contraction modulus can be verified in a dynamic Markov-perfect equilibrium—where the static game is a stage game embedded in the dynamic models of [Ericson and Pakes \(1995\)](#) and [Doraszelski and Satterthwaite \(2010\)](#)—is the question that determines whether the uniqueness result extends beyond the static setting. The challenge is that dynamic state variables (e.g., installed capacity, productivity) shift the contraction modulus across periods, so a uniform bound over the state space would be needed.

On the empirical side, the cloud application identifies fixed-cost wedges from a single class of moments: the binary provider-country activation matrix. Richer identification—closer in spirit to [Arkolakis et al. \(2023\)](#)—would require three data upgrades that are currently unavailable at the provider-country level: (i) country-level cloud spending, which would discipline market demand directly rather than through a GDP proxy; (ii) provider-country revenue or market-share data, which would separate variable profitability from fixed costs; and (iii) provider-country region counts with entry timing, which would move identification beyond the binary entry margin. This trio would provide the cloud analogue of the affiliate sales, affiliate counts, and survival moments that underpin the AES calibration.

A. PROOF OF PROPOSITION 6

We prove that firm f 's activation profit exhibits increasing differences in the activation indicators $\{I_{fi}\}_{i \in \mathcal{A}_f}$, taking the rival demand index $D_{-f}(n)$ as given. By a standard equivalence ([Topkis, 1998](#)), increasing differences on a binary lattice $\{0, 1\}^{|\mathcal{A}_f|}$ is equivalent to supermodularity.

Step 1: Reduction to one-dimensional convexity. Fixed costs $\sum_{i \in \mathcal{I}_f} \phi_i w_i$ are additively separable, so increasing differences of total profit is equivalent to increasing differences of *variable* profit $VP_f = \sum_n VP_{nf}$. Since the sum preserves increasing differences, a *sufficient* condition is per-market increasing differences: if VP_{nf} exhibits increasing differences in $\{I_{fi}\}$ for each market n , so does the aggregate. We derive a sharp threshold for per-market convexity in Steps 2–4, then address the aggregate case in Step 5.

By the Fréchet cost aggregation ([2.7](#)), the cost index depends on the activation set $\mathcal{I}_f$ only through the *cost aggregator*:

$$\Sigma_{fn} \equiv \sum_{i \in \mathcal{I}_f} (w_i \tau_{in})^{-\theta}, \quad (\text{A.1})$$

with $c_{nf} = K \cdot \Sigma_{fn}^{-1/\theta}$ where $K = \Gamma_\theta / a_f > 0$. Activating site i adds the term $(w_i \tau_{in})^{-\theta} > 0$ to Σ_{fn} . Since Σ_{fn} is *additive* in the activation indicators, increasing differences of VP_{nf} in

$\{I_{fi}\}$ is equivalent to *convexity* of VP_{nf} viewed as a function of Σ_{fn} :

$$\frac{\partial^2 VP_{nf}}{\partial \Sigma_{fn}^2} > 0 \implies \text{increasing differences of } VP_{nf} \text{ in } \{I_{fi}\}. \quad (\text{A.2})$$

This reduction follows from the chain rule: for distinct sites i, i' with increments $\delta_i = (w_i \tau_{in})^{-\theta}$ and $\delta_{i'} = (w_{i'} \tau_{i'n})^{-\theta}$,

$$\begin{aligned} & VP_{nf}(\Sigma + \delta_i + \delta_{i'}) - VP_{nf}(\Sigma + \delta_i) - VP_{nf}(\Sigma + \delta_{i'}) + VP_{nf}(\Sigma) \\ &= \int_0^{\delta_i} \int_0^{\delta_{i'}} \frac{\partial^2 VP_{nf}}{\partial \Sigma^2}(\Sigma + s + t) dt ds > 0 \end{aligned}$$

when $\partial^2 VP_{nf} / \partial \Sigma^2 > 0$ everywhere.

Step 2: CES benchmark (constant markups). Under constant markups $\mu_0 = \sigma/(\sigma - 1)$, variable profit in market n is:

$$VP_{nf}^{\text{CES}} = \frac{1}{\sigma} p_{nf}^{1-\sigma} P_n^{\sigma-\eta} \Omega_n = \frac{\Omega_n}{\sigma} \left(\frac{\sigma}{\sigma - 1} \right)^{1-\sigma} c_{nf}^{1-\sigma} P_n^{\sigma-\eta}, \quad (\text{A.3})$$

where $P_n^{1-\sigma} = p_{nf}^{1-\sigma} + D_{-f}(n)$ and we take $D_{-f}(n)$ as given. Substituting $c_{nf} = K \Sigma_{fn}^{-1/\theta}$:

$$VP_{nf}^{\text{CES}} \propto \Sigma_{fn}^{(\sigma-1)/\theta} \cdot [\text{terms involving } P_n(c_{nf})].$$

The direct effect $\Sigma_{fn}^{(\sigma-1)/\theta}$ is strictly convex in Σ_{fn} when $(\sigma - 1)/\theta > 1$. This is the [Antràs et al. \(2017\)](#) argument: the Fréchet dispersion is small enough relative to demand elasticity that the demand expansion from cost reduction is super-proportional. The price-index feedback through P_n works against convexity at high market shares (Step 4 below derives the exact threshold), but the direct convexity dominates when shares are sufficiently small.

Step 3: Extension to Atkeson–Burstein markups. Under AB pricing, firm f sets $p_{nf} = \mu(s_{nf}) \cdot c_{nf}$ with:

$$\mu(s) = \frac{1}{1 - 1/\epsilon(s)}, \quad \frac{1}{\epsilon(s)} = \frac{1-s}{\sigma} + \frac{s}{\eta}, \quad (\text{A.4})$$

where the market share $s_{nf} = x/(x + D)$ with $x \equiv p_{nf}^{1-\sigma}$ and $D \equiv D_{-f}(n)$ (suppressing market subscripts). Variable profit is:

$$VP_{nf} = \frac{1}{\epsilon(s)} \cdot x \cdot \Omega_n \cdot (x + D)^{(\sigma-\eta)/(1-\sigma)}. \quad (\text{A.5})$$

To show $\partial^2 VP_{nf} / \partial \Sigma^2 > 0$, we use the chain rule through the cost index $c = K\Sigma^{-1/\theta}$:

$$\frac{\partial^2 VP}{\partial \Sigma^2} = \frac{\partial^2 VP}{\partial c^2} \left(\frac{dc}{d\Sigma} \right)^2 + \frac{\partial VP}{\partial c} \frac{d^2 c}{d\Sigma^2}. \quad (\text{A.6})$$

Term 2 (cost curvature). Since $c = K\Sigma^{-1/\theta}$:

$$\frac{dc}{d\Sigma} = -\frac{K}{\theta} \Sigma^{-1/\theta-1} < 0, \quad \frac{d^2 c}{d\Sigma^2} = \frac{K(1+\theta)}{\theta^2} \Sigma^{-1/\theta-2} > 0.$$

Variable profit is decreasing in cost ($\partial VP / \partial c < 0$), so the second term $(\partial VP / \partial c)(d^2 c / d\Sigma^2) < 0$, working *against* convexity. Supermodularity therefore requires that Term 1 dominates Term 2 in absolute value.

Term 1 (profit curvature in cost). The proof requires $\partial^2 VP / \partial c^2 > 0$ and sufficiently large. Under the AB pricing rule, the equilibrium $x(c)$ is defined implicitly by $p \cdot c^{-1} = \mu(s(x, D))$ with $x = p^{1-\sigma}$, so x depends on c through the pricing equation. Implicit differentiation yields:

$$\frac{dx}{dc} = \frac{(1-\sigma)x/c}{1 + (\sigma-1)\frac{\sigma-\eta}{\sigma\eta}s(1-s)\mu}. \quad (\text{A.7})$$

The denominator equals $1 + s(1-s)\beta_{nf}$ where $\beta_{nf} \equiv (\sigma-1)(\sigma-\eta)\mu/(\sigma\eta) > 0$ is the feedback parameter from (3.2), so $|dx/dc| < (\sigma-1)x/c$. The key bound is:

$$\left| \frac{c}{x} \frac{dx}{dc} \right| \leq (\sigma-1) \cdot \frac{1}{1 + \frac{\sigma-\eta}{\sigma\eta}s(1-s)\mu_{\min}} \leq (\sigma-1) \cdot C(\sigma, \eta), \quad (\text{A.8})$$

where $\mu_{\min} \equiv \sigma/(\sigma-1)$ is the CES benchmark markup (the lower bound on μ) and $C(\sigma, \eta) \leq 1$ is a finite constant (bounded because $s \in [0, 1]$, $\sigma > \eta > 1$, and $\mu \geq \mu_{\min}$). This means the AB pricing response dx/dc is a *damped* perturbation of the CES response $(1-\sigma)x/c$.

Working with the market share s as the state variable and $t = \ln \Sigma$, the exact AB share

law of motion is $ds/dt = g(s) \equiv \alpha s(1-s)\delta(s)$. Define

$$B \equiv \sigma - \eta > 0, \quad M(s) \equiv \eta(\sigma - 1) - Bs + B(\sigma - 1)s(1 - s),$$

so that $\delta(s) = [\eta(\sigma - 1) - Bs]/M(s)$. Since $VP^{AB}(s) = R_n(as + bs^2)$ with $a = 1/\sigma$ and $b = B/(\sigma\eta)$, the chain rule gives

$$\frac{d^2VP}{d\Sigma^2} = \frac{R_n g(s)}{\Sigma^2} H_{AB}(s), \quad H_{AB}(s) \equiv (a + 2bs)(g'(s) - 1) + 2b g(s).$$

Per-market convexity holds if and only if $H_{AB}(s) \geq 0$.

Proposition 36 (Unique AB per-market convexity threshold). *Suppose $\sigma > \eta > 1$ and $(\sigma - 1)/\theta > 1$. Then $H_{AB}(s)$ is strictly concave on $[0, 1]$:*

$$H''_{AB}(s) = -\frac{6(\sigma - \eta)(\sigma - 1)}{\eta\theta M(s)^4} P_{AB}(s),$$

where $P_{AB}(s) = \sum_{k=0}^4 \beta_k \binom{4}{k} s^k (1-s)^{4-k}$ is a degree-4 polynomial whose Bernstein coefficients $\beta_0, \dots, \beta_4$ are all strictly positive under $\sigma > \eta > 1$.⁸ Since

$$H_{AB}(0) = \frac{\sigma - \theta - 1}{\sigma\theta} > 0, \quad H_{AB}(1) = \frac{(\sigma + \theta - 1)(\eta - 2\sigma)}{\eta\sigma\theta} < 0,$$

there exists a unique $\bar{s}_{AB}^* \in (0, 1)$ such that $H_{AB}(\bar{s}_{AB}^*) = 0$. Per-market convexity holds if and only if $s < \bar{s}_{AB}^*$.

Proof. $M(s) \geq \sigma(\eta - 1) > 0$ on $[0, 1]$, so $M(s)^4 > 0$. Each Bernstein coefficient $\beta_k > 0$ under $\sigma > \eta > 1$, and the Bernstein basis polynomials are nonneg on $[0, 1]$, so $P_{AB}(s) > 0$. Hence $H''_{AB}(s) < 0$: strict concavity. The sign conditions at the endpoints follow by direct evaluation. A strictly concave continuous function positive at 0 and negative at 1 has a unique zero. $\square$

The AB threshold is parameter-dependent and need not uniformly exceed the CES benchmark: at the baseline $(\sigma, \eta, \theta) = (5, 2, 3)$, $\bar{s}_{AB}^* \approx 0.122$, slightly below $\bar{s}_{CES}^* = 0.125$.

⁸Explicitly: $\beta_0 = \eta^3(\sigma - \eta)(\sigma - 1)^3$, $\beta_1 = \eta^3(\eta - 1)(\sigma - 1)^3$, $\beta_2 = 2\eta^2\sigma(\eta - 1)^2(\sigma - 1)^2$, $\beta_3 = \eta\sigma(\eta - 1)^2(\sigma - 1)[2(\sigma - \eta)^2 + (5\eta - 4)(\sigma - \eta) + 3\eta(\eta - 1)]$, $\beta_4 = \sigma^2(\eta - 1)^2[2(\sigma - \eta)^3 + (9\eta - 8)(\sigma - \eta)^2 + (11\eta - 8)(\eta - 1)(\sigma - \eta) + 4\eta(\eta - 1)^2]$.

Since the maximum equilibrium share in the cloud application is 0.40, per-market convexity is violated under both CES and AB, and aggregate supermodularity must be verified computationally via part (ii).

Step 4: Direct characterization of per-market convexity. Under the CES benchmark (constant markups), variable profit in market n is $VP = (R_n/\sigma) \cdot u/(u + D_{-f})$ where $u = A\Sigma^\alpha$, $\alpha = (\sigma - 1)/\theta$, and D_{-f} is the rival demand index (held fixed). Writing $s = u/(u + D_{-f})$ for the market share, the chain rule gives:

$$\frac{d^2VP}{d\Sigma^2} = \frac{R_n \alpha s(1-s)}{\sigma(u + D_{-f}) \Sigma^2} \left[\underbrace{-2\alpha s}_{\text{demand curvature}} + \underbrace{(\alpha - 1)}_{\text{Fréchet convexity}} \right],$$

where the prefactor is strictly positive. Per-market convexity ($d^2VP/d\Sigma^2 > 0$) therefore holds if and only if:

$$s < \bar{s}^* \equiv \frac{\alpha - 1}{2\alpha} = \frac{\sigma - 1 - \theta}{2(\sigma - 1)}. \quad (\text{A.9})$$

The threshold $\bar{s}^* > 0$ whenever $(\sigma - 1)/\theta > 1$ (the maintained assumption) and $\bar{s}^* \rightarrow 1/2$ as $\alpha \rightarrow \infty$. At the baseline $(\sigma, \eta, \theta) = (5, 2, 3)$: $\bar{s}^* = 0.125$. Since the maximum equilibrium share is $0.39 \gg 0.125$, the per-market condition is violated, and aggregate supermodularity must be verified computationally via Step 5.

Step 5: Aggregate supermodularity. The per-market condition is *sufficient* but not *necessary*: even when individual markets violate $s < \bar{s}^*$, the aggregate profit $\sum_n VP_{nf}$ can still exhibit increasing differences because markets with $s < \bar{s}^*$ (positive curvature) can dominate the few markets where $s > \bar{s}^*$ (negative curvature). The overall increasing difference for sites i, i' is:

$$\Delta^2\pi_f = \sum_n \int_0^{\delta_i} \int_0^{\delta_{i'}} \frac{\partial^2 VP_{nf}}{\partial \Sigma^2} (\Sigma + s + t) dt ds,$$

which is positive whenever the contribution from low-share markets outweighs the high-share markets.

We verify computationally that $\Delta^2\pi_f > 0$ for all pairs of sites and all firms, across the full range of parameterizations (baseline, extension, and robustness sweeps), by checking the Jia–Tarski gap condition: a zero gap provides strong computational evidence that the

CDCP solution is a lattice optimum at the tested parameterization, which requires aggregate supermodularity in the sense of Remark 7. In all tested configurations—including those where $\max_{n,f} s_{nf}$ exceeds $\bar{s}^*$ —the gap is zero (see Section 5.5 and Appendix G).

Under AB markups, the pricing response dx/dc is damped by the factor $1/(1 + s(1 - s)\beta_{nf})$ (equation A.8). The exact per-market convexity condition under AB is characterized by $H_{AB}(s) \geq 0$; the corresponding threshold is parameter-dependent and need not be uniformly above the CES benchmark $\bar{s}^*$.

By the equivalence (A.2), aggregate increasing differences in $\{I_{fi}\}$ follow from positive $\Delta^2\pi_f$. Fixed costs $\sum_{i \in \mathcal{I}_f} \phi_i w_i$ are additively separable (hence modular), so total profit $\pi_f = \sum_n V P_{nf} - \sum_{i \in \mathcal{I}_f} \phi_i w_i$ exhibits increasing differences. The lattice consequences (greatest and least optima, Jia–Tarski attainability) follow from Topkis (1998), Theorem 2.8.1. $\square$

B. PROOF OF PROPOSITION 8

We prove that the extremal fixed points of Φ^2 bound all Nash equilibria, and establish convergence of iterated best response. The argument handles the key difficulty that the joint best-response map is *anti-monotone* (order-reversing) due to strategic substitutes, so Tarski’s theorem cannot be applied directly to Φ .

Setup. Let $L = \prod_{f=1}^F 2^{\mathcal{A}_f}$ be the joint activation space, ordered by component-wise set inclusion: $\mathbf{I} \leq \mathbf{I}'$ iff $\mathcal{I}_f \subseteq \mathcal{I}'_f$ for all f . The lattice L is finite and complete, with supremum $\hat{\mathbf{I}} = (\mathcal{A}_1, \dots, \mathcal{A}_F)$ and infimum $\check{\mathbf{I}} = (\emptyset, \dots, \emptyset)$.

By Proposition 6, each firm’s single-agent CDCP has a well-defined greatest optimum (and least optimum) for any fixed rival demand index. Define the joint best-response map $\Phi : L \rightarrow L$ by taking each firm’s greatest optimal activation set:

$$\Phi(\mathbf{I}) = (\mathcal{I}_1^*(\mathbf{I}_{-1}), \dots, \mathcal{I}_F^*(\mathbf{I}_{-F})).$$

Step 1: Anti-monotonicity of Φ . When rivals expand their activation sets ($\mathbf{I}_{-f} \leq \mathbf{I}'_{-f}$), the rival demand index $D_{-f}(n) = \sum_{g \neq f} p_{ng}^{1-\sigma}$ weakly increases (rivals have lower costs, hence lower prices, hence larger $p_g^{1-\sigma}$). Higher D_{-f} reduces firm f ’s market share s_{nf} for any given price, lowering the marginal value of each production site. Since the single-agent

problem is supermodular (Proposition 6), the optimal activation set shrinks by monotone comparative statics (Topkis, 1998): $I_f^*(\mathbf{I}'_{-f}) \subseteq I_f^*(\mathbf{I}_{-f})$.

Therefore $\mathbf{I} \leq \mathbf{I}'$ implies $\Phi(\mathbf{I}) \geq \Phi(\mathbf{I}')$: the map Φ is anti-monotone.

Step 2: Monotonicity of Φ^2 and Tarski's theorem. The squared map $\Phi^2 = \Phi \circ \Phi$ is monotone (increasing): if $\mathbf{I} \leq \mathbf{I}'$, then $\Phi(\mathbf{I}) \geq \Phi(\mathbf{I}')$, so $\Phi^2(\mathbf{I}) = \Phi(\Phi(\mathbf{I})) \leq \Phi(\Phi(\mathbf{I}')) = \Phi^2(\mathbf{I}')$. By Tarski's fixed-point theorem, the monotone map Φ^2 on the finite complete lattice L has a greatest fixed point $\bar{\mathbf{I}}$ and a least fixed point $\underline{\mathbf{I}}$.

Step 3: Bounding the Nash equilibrium set. Any Nash equilibrium $\mathbf{I}^*$ satisfies $\Phi(\mathbf{I}^*) = \mathbf{I}^*$, hence $\Phi^2(\mathbf{I}^*) = \mathbf{I}^*$: every Nash equilibrium is a fixed point of Φ^2 . By Tarski's theorem, $\underline{\mathbf{I}} \leq \mathbf{I}^* \leq \bar{\mathbf{I}}$.

If $\bar{\mathbf{I}}$ is itself a Nash equilibrium (i.e., $\Phi(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$), then it is the greatest Nash equilibrium by the bound above. In general, however, $\bar{\mathbf{I}}$ may be a two-cycle of Φ rather than a fixed point: $\Phi(\bar{\mathbf{I}}) \neq \bar{\mathbf{I}}$ while $\Phi^2(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$. Whether the extremal fixed points of Φ^2 are Nash equilibria depends on the specific game structure and must be verified. $\square$

Convergence (Proposition 9). Starting from $\hat{\mathbf{I}}$, anti-monotonicity gives $\Phi(\hat{\mathbf{I}}) \leq \hat{\mathbf{I}}$, so $\Phi^2(\hat{\mathbf{I}}) \leq \Phi(\hat{\mathbf{I}}) \leq \hat{\mathbf{I}}$ (the first inequality by anti-monotonicity applied to $\Phi(\hat{\mathbf{I}}) \leq \hat{\mathbf{I}}$). By induction, the even subsequence $\{\Phi^{2k}(\hat{\mathbf{I}})\}$ is decreasing and the odd subsequence $\{\Phi^{2k+1}(\hat{\mathbf{I}})\}$ is increasing, with $\Phi^{2k+1}(\hat{\mathbf{I}}) \leq \Phi^{2k}(\hat{\mathbf{I}})$ for all k . On the finite lattice L , both subsequences stabilize in finitely many steps to limits $\bar{\mathbf{I}}_{\text{even}}$ and $\bar{\mathbf{I}}_{\text{odd}}$ satisfying $\Phi^2(\bar{\mathbf{I}}_{\text{even}}) = \bar{\mathbf{I}}_{\text{even}}$ and $\bar{\mathbf{I}}_{\text{odd}} = \Phi(\bar{\mathbf{I}}_{\text{even}})$. Since $\{\Phi^{2k}(\hat{\mathbf{I}})\}$ is the largest decreasing sequence of iterates of Φ^2 , its limit $\bar{\mathbf{I}}_{\text{even}} = \bar{\mathbf{I}}$, the greatest fixed point of Φ^2 . If $\Phi(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$, the full sequence converges to $\bar{\mathbf{I}}$, the greatest Nash equilibrium.

For Gauss–Seidel iteration, define the composite update map $\Psi : L \rightarrow L$ that updates firms sequentially $(f_1, \dots, f_F)$, each using the latest rival profile. Unlike the Jacobi map Φ , the GS map Ψ is not obviously anti-monotone: firm f_j 's rival profile mixes updated information from earlier firms with stale information from later firms, so the comparability of rival profiles under perturbation is not preserved. We verify GS convergence computationally at all tested parameterizations (Section 5 and Appendix G). $\square$

C. JACOBIAN OF THE THRESHOLD MAP

We derive the factorization (3.11) and bound $\|J_v\|_\infty$ using CES primitives.

Step 1: Factorization. The threshold map T acts on activation probabilities via $T_{if}(v) = G_0(\bar{\varepsilon}_{if}(v)/\sigma_\varepsilon)$, where $\bar{\varepsilon}_{if}(v) = \mathbb{E}[\Delta\pi_{if}(v)]/w_i$ is the threshold at which the expected marginal value of activating site i (integrating over rivals' and own other sites' random activation) equals the fixed-cost shock. By the chain rule:

$$\frac{\partial T_{if}}{\partial v_{jg}} = \frac{g_0(\bar{\varepsilon}_{if}/\sigma_\varepsilon)}{\sigma_\varepsilon} \cdot \frac{1}{w_i} \cdot \frac{\partial \mathbb{E}[\Delta\pi_{if}]}{\partial v_{jg}}, \quad (\text{C.1})$$

which gives $J_T = \text{diag}(g_0(\bar{\varepsilon}/\sigma_\varepsilon)/\sigma_\varepsilon) \cdot \text{diag}(1/w_i) \cdot J_\Delta$. Defining $J_v \equiv \text{diag}(1/w_i) \cdot J_\Delta$ absorbs the wage scaling into the sensitivity matrix, so $\|J_T\|_\infty \leq (\|g_0\|_\infty/\sigma_\varepsilon) \cdot \|J_v\|_\infty$.

Step 2: Sign structure of J_v . For $g \neq f$ (across-firm): increasing v_{jg} raises the probability that rival g activates site j , weakly increasing $\mathbb{E}[D_{-f}(n)]$ in markets served from j . Higher expected rival demand lowers firm f 's share and variable profit, so $\partial \mathbb{E}[\Delta\pi_{if}]/\partial v_{jg} \leq 0$: strategic substitutes across firms. For $j \neq i$, $g = f$ (within-firm): increasing v_{jf} lowers firm f 's expected cost index (Fréchet complementarity (2.7)), raising the marginal value of site i by supermodularity (Proposition 6), so $\partial \mathbb{E}[\Delta\pi_{if}]/\partial v_{jf} \geq 0$: complementarity within firm.

Step 3: Bounding $\|J_v\|_\infty$ via the Perron–Frobenius weighted norm. Before bounding the row sums, we record the weighted-norm tool that replaces the Neumann-series (infinity-norm) bound used in earlier drafts. The substitution is essential because $\|\Lambda\|_\infty$ exceeds $1/(F-1)$ on roughly 12% of the per-market subgames enumerated in Proposition 28, so the Neumann bound $1/(1 - \|\Lambda\|_\infty)$ is not available throughout. The spectral radius $\rho(\Lambda) \leq 0.316 < 1$ is uniform, and Perron–Frobenius converts this into an operator-norm contraction in a weighted sup-norm.

Proposition 37 (Spectral-norm Jacobian bound). *Let $\Lambda \geq 0$ entrywise with $\rho(\Lambda) < 1$, and let $w \gg 0$ be the Perron–Frobenius eigenvector with $\Lambda w \leq \lambda w$ for some $\lambda \in [\rho(\Lambda), 1)$ (when Λ is irreducible, $\lambda = \rho(\Lambda)$; reducibility is handled by taking $\lambda = \rho(\Lambda) + \varepsilon$ for any $\varepsilon > 0$ and using a*

standard perturbation argument). Define the weighted sup-norm $\|x\|_w = \max_i |x_i|/w_i$ on $\mathbb{R}^{NF}$, with induced operator norm $\|A\|_w = \sup_{x \neq 0} \|Ax\|_w / \|x\|_w$. Then:

- (i) $\|\Lambda\|_w \leq \lambda$, so $(I - \Lambda)$ is invertible with $\|(I - \Lambda)^{-1}\|_w \leq 1/(1 - \lambda)$, which for irreducible Λ reduces to $1/(1 - \rho(\Lambda))$ and for reducible Λ holds with $\lambda = \rho(\Lambda) + \varepsilon$ for arbitrary $\varepsilon > 0$;
- (ii) applied to the downstream AB pricing Jacobian, a demand perturbation δD_{-f} propagates with amplification at most $1/(1 - \rho(\Lambda))$ in $\|\cdot\|_w$, so

$$\|J_v\|_w \leq \frac{L_D \cdot L_\mu}{1 - \rho(\Lambda)} + L_C,$$

with L_D, L_μ, L_C defined below.

The weighted norm $\|\cdot\|_w$ and the standard infinity norm $\|\cdot\|_\infty$ are equivalent: $\|x\|_w \leq \|x\|_\infty / \min_i w_i$ and $\|x\|_\infty \leq (\max_i w_i) \|x\|_w$. Hence a contraction in $\|\cdot\|_w$ yields a contraction in $\|\cdot\|_\infty$ with an at-worst constant-factor inflation $C_w = \max_i w_i / \min_i w_i$.

Proof. For (i): $\|\Lambda x\|_w = \max_i (\Lambda x)_i / w_i \leq \max_i (\Lambda |x|)_i / w_i$ (where $|x|$ denotes entrywise absolute value), and using $\Lambda w \leq \lambda w$ with $|x|_i \leq \|x\|_w \cdot w_i$:

$$(\Lambda |x|)_i \leq \sum_j \Lambda_{ij} \|x\|_w w_j = \|x\|_w (\Lambda w)_i \leq \|x\|_w \lambda w_i,$$

so $\|\Lambda\|_w \leq \lambda$. The Neumann series $(I - \Lambda)^{-1} = \sum_{k \geq 0} \Lambda^k$ converges in $\|\cdot\|_w$ with sum bounded by $1/(1 - \lambda)$. For (ii): the pricing-equilibrium map near a fixed point has Jacobian bounded entrywise by Λ (Proposition 11), so the linear response $\delta D' = (I - \Lambda)^{-1} \delta D$ obeys $\|\delta D'\|_w \leq \|\delta D\|_w / (1 - \rho(\Lambda))$. Composing with the demand- and cost-channel Lipschitz bounds L_D, L_μ, L_C yields the stated inequality. Norm-equivalence is a standard consequence of $w \gg 0$ with $\max_i w_i, \min_i w_i \in (0, \infty)$. $\square$

With Proposition 37 in hand, we decompose each row sum of $|J_v|$ into across-firm and within-firm contributions:

$$\sum_{(j,g)} |J_v|_{(i,f),(j,g)} = \underbrace{\sum_{g \neq f} \sum_j \frac{1}{w_i} \left| \frac{\partial \mathbb{E}[\Delta \pi_{if}]}{\partial v_{jg}} \right|}_{\text{across-firm (demand channel)}} + \underbrace{\sum_{j \neq i} \frac{1}{w_i} \left| \frac{\partial \mathbb{E}[\Delta \pi_{if}]}{\partial v_{jf}} \right|}_{\text{within-firm (cost channel)}}.$$

Across-firm terms: demand sensitivity L_D . Each $\partial \mathbb{E}[D_{-f}(n)]/\partial v_{jg}$ is the expected change in rival demand when firm g toggles site j from inactive to active. Under Fréchet costs (2.7), this adds $(w_j \tau_{jn}/a_g)^{-\theta}$ to the cost aggregator, lowering c_{ng} and hence p_{ng} , which raises $p_{ng}^{1-\sigma}$. The maximum contribution is the single-site price index (attained when g has no other active site):

$$0 \leq \frac{\partial \mathbb{E}[D_{-f}(n)]}{\partial v_{jg}} \leq \left(\frac{\sigma}{\sigma-1} \right)^{1-\sigma} \left(\frac{w_j \tau_{jn}}{a_g} \right)^{1-\sigma} \equiv \bar{d}_{jgn}. \quad (\text{C.2})$$

Summing over rivals and sites: $L_D \equiv \max_n \sum_{g \neq f} \sum_j \bar{d}_{jgn}$, finite since the sums are over finite firms and sites with bounded wages and trade costs.

Across-firm terms: marginal-value sensitivity L_μ . A perturbation $\delta D_{-f}(n)$ propagates through the AB pricing equilibrium with spectral amplification $\|(I - \Lambda)^{-1}\|_w \leq 1/(1 - \rho(\Lambda))$ in the Perron–Frobenius weighted sup-norm (Proposition 37 below). Variable profit is $VP_{nf} = R_n s_{nf}(1 - 1/\epsilon_{nf})$; since $s_{nf}(1 - s_{nf}) \leq 1/4$ and the profit margin satisfies $(1 - 1/\epsilon_{nf}) \leq (\bar{\mu} - 1)/\bar{\mu}$ with $\bar{\mu} = \eta/(\eta - 1)$:

$$L_\mu \equiv \max_{(i,f)} \frac{1}{w_i} \sum_n \left| \frac{\partial(\Delta VP_{nf})}{\partial D_{-f}(n)} \right| \leq \frac{1}{w_{\min}} \sum_n \frac{R_n (\bar{\mu} - 1)}{2 \bar{\mu} P_n^{1-\sigma}}, \quad (\text{C.3})$$

where $w_{\min} = \min_i w_i$ and the factor 2 accounts for evaluating ΔVP at configurations both with and without site i .

Within-firm terms: cost-channel sensitivity L_C . Increasing v_{jf} adds $(w_j \tau_{jn}/a_f)^{-\theta}$ to firm f 's Fréchet aggregator (2.7). The cost-index response satisfies $\partial c_{nf}^{1-\sigma}/\partial v_{jf} = \frac{\sigma-1}{\theta} (c_{nf}^{-\theta})^{(\sigma-1)/\theta-1} (w_j \tau_{jn})^{-\theta}$ which is finite and positive for any configuration with at least one active site ($c_{nf}^{-\theta} > 0$). The downstream effect on $\Delta \pi_{if}$ propagates through the same bounded-markup and share channels as L_μ , with an additional factor of $(\sigma - 1)/\theta$ from the cost complementarity. Summing over $|\mathcal{A}_f| - 1$ own sites and N markets, L_C is a finite product of CES primitives and equilibrium cost indices. Together:

$$\|J_v\|_w \leq \frac{L_D \cdot L_\mu}{1 - \rho(\Lambda)} + L_C, \quad (\text{C.4})$$

where $\|\cdot\|_w$ is the Perron–Frobenius weighted sup-norm from Proposition 37 below. The relationship $\|J_v\|_\infty \leq C_w \|J_v\|_w$ for the norm-equivalence constant $C_w = \max_i w_i / \min_i w_i$

(with w the Perron eigenvector) then yields the unweighted bound used in (3.12). All three bounds are finite under $\sigma > \eta > 1$ and $\theta > 0$: L_D depends only on primitives (wages, trade costs, TFP, and σ); L_μ on revenue shares and the markup range; L_C on the Fréchet aggregator and the supermodularity index $(\sigma - 1)/\theta$. At the baseline parameterization, $\rho(\Lambda) = 0.103$ (the tightest available rate; the conservative infinity-norm is $\|\Lambda\|_\infty = 0.193$), so the spectral amplification $1/(1 - \rho(\Lambda)) \approx 1.11$: the amplification is near unity, and the outer activation problem is nearly a constant-markup entry game. Using $\rho(\Lambda)$ rather than $\|\Lambda\|_\infty$ is essential because the latter exceeds $1/(F - 1)$ on roughly 12% of enumerated per-market subgames (Table 3), while the spectral radius satisfies $\rho(\Lambda) \leq 0.316 < 1$ uniformly. The script `main_bathtub.m` evaluates $\|J_T\|_\infty$ numerically at each σ_ε and confirms $\gamma_{\max} = 0.951 < 1$ (Figure 2).

Step 4: Tail behavior and convergence rate. *Large σ_ε .* From (3.12): $\gamma(\sigma_\varepsilon) \leq (\|g_0\|_\infty/\sigma_\varepsilon) \cdot \|J_v\|_\infty \rightarrow 0$ as $\sigma_\varepsilon \rightarrow \infty$, since $\|J_v\|_\infty$ is bounded on the compact set $[0, 1]^M$ (Step 3). For the normal density, $\|g_0\|_\infty = 1/\sqrt{2\pi}$.

Small σ_ε . By Part (iii), the BNE converges to a Nash equilibrium of the complete-information game as $\sigma_\varepsilon \rightarrow 0$. When the CI game has a unique equilibrium with gap $\delta_v = \min_{i,f} |\bar{\varepsilon}_{if}^{CI}|/w_i > 0$, continuity of the threshold map implies that for σ_ε small enough, every threshold satisfies $|\bar{\varepsilon}_{if}| \geq \delta_v w_i/2$. The standardized threshold $|z_{if}| = |\bar{\varepsilon}_{if}|/\sigma_\varepsilon \geq \delta_v w_{\min}/(2\sigma_\varepsilon) \rightarrow \infty$. For log-concave densities with tail bound $g_0(z) \leq C_g \exp(-c|z|)$ (e.g., normal: $C_g = e^{c^2/2}/\sqrt{2\pi}$, $c > 0$ arbitrary; logistic: $C_g = 1/4$, $c = 1$), each density evaluation satisfies $g_0(z_{if})/\sigma_\varepsilon \leq (C_g/\sigma_\varepsilon) \exp(-c \delta_v w_{\min}/(2\sigma_\varepsilon))$. Since the row sums of $|J_v|$ remain bounded (the equilibrium converges to the CI equilibrium with finite marginal values), the contraction modulus satisfies the explicit rate:

$$\gamma(\sigma_\varepsilon) \leq \frac{C_g}{\sigma_\varepsilon} \exp\left(-\frac{c \delta_v w_{\min}}{2\sigma_\varepsilon}\right) \|J_v\|_\infty \xrightarrow{\sigma_\varepsilon \rightarrow 0} 0, \quad (\text{C.5})$$

because the exponential tail decay dominates the polynomial $1/\sigma_\varepsilon$.

Uniqueness of the BNE branch. Define $F(v, \sigma_\varepsilon) = T_{\sigma_\varepsilon}(v) - v$. At any σ_ε^* where $\gamma(\sigma_\varepsilon^*) < 1$, the Jacobian $D_v F = J_T - I$ is nonsingular (since $\|J_T\|_\infty < 1$ implies that $I - J_T$ is invertible). By the implicit function theorem, the BNE extends as a unique C^1 branch $v^*(\sigma_\varepsilon)$ in a neighborhood of σ_ε^* . Since $\gamma \rightarrow 0$ at both extremes, the set $\{\sigma_\varepsilon > 0 : \gamma(\sigma_\varepsilon) < 1\}$ includes

neighborhoods of 0^+ and ∞ . If $\gamma_{\max} < 1$, this set is all of $(0, \infty)$, and $v^*(\sigma_\varepsilon)$ is the unique BNE for every $\sigma_\varepsilon > 0$. $\square$

D. NESTED-LOGIT VERIFICATION: TECHNICAL DETAILS

This appendix supplies the full algebra supporting Proposition 31 and Lemma 32 of Section 3.4.

Share derivatives. Differentiating (3.14) with respect to firm f 's own price $p_{n,f}$ gives

$$\begin{aligned}\frac{\partial s_{n,f}}{\partial p_{n,f}} &= -\frac{\gamma s_{n,f}}{\lambda_{n,k(f)}} \left[(1 - s_{n,f|k(f)}) + \lambda_{n,k(f)} s_{n,f|k(f)} (1 - s_{n,k(f)}) \right] \\ &= -\gamma s_{n,f} \left[\lambda_{n,k(f)}^{-1} (1 - s_{n,f|k(f)}) + s_{n,f|k(f)} - s_{n,f} \right],\end{aligned}$$

and with respect to a rival price $p_{n,g}$ ($g \neq f$) in the same nest $k(f) = k(g)$:

$$\frac{\partial s_{n,f}}{\partial p_{n,g}} = \gamma s_{n,f} s_{n,g|k(f)} \left[\lambda_{n,k(f)}^{-1} - s_{n,f}/s_{n,f|k(f)} \right],$$

and across nests ($k(f) \neq k(g)$):

$$\frac{\partial s_{n,f}}{\partial p_{n,g}} = \gamma s_{n,f} s_{n,g}.$$

All share derivatives are bounded in modulus by $\gamma \lambda_{\min}^{-1}$, since shares lie in $[0, 1]$ and within-nest conditional shares lie in $[0, 1]$.

Bertrand FOC uniqueness (proof of Proposition 31). The Bertrand FOC is $s_{n,f} + (p_{n,f} - c_{n,f}) \partial s_{n,f} / \partial p_{n,f} = 0$; substituting the derivative expression and rearranging yields (3.15). Define the log-share aggregator

$$\Psi_n(\mathbf{p}_n) = \sum_{k \in \mathcal{K}_n} D_{n,k}^{\lambda_{n,k}} = \frac{1 - s_{n,0}(\mathbf{p}_n)}{s_{n,0}(\mathbf{p}_n)}.$$

Nocke and Schutz (2018) show (their Theorem 1, Lemma 2) that a multiproduct Bertrand game with generalised extreme value demand is an *aggregative* game in Ψ_n with a strictly

quasi-concave universal payoff; for single-product firms and nested logit this yields a unique fixed point in prices. The argument extends the earlier nested-logit treatments of [Anderson et al. \(1992\)](#) (Chapter 7) and [Konovalov and Sándor \(2010\)](#). Strict concavity of the firm's log-share objective combined with finite-valued derivatives delivers C^1 dependence of the fixed point on c via the implicit function theorem.

Uniform bound on the share Jacobian (proof of Lemma 32). Fix a rival firm $g \neq f$ and a market n . Perturbing the presence indicator $\mathbf{1}_{g \in S_n}$ from 0 to 1 changes the equilibrium share vector by at most

$$\left| \frac{\partial s_{n,r}^*}{\partial v_{n,g}} \right|_{\infty} \leq \underbrace{\gamma \lambda_{\min}^{-1}}_{\text{share-price slope}} \cdot \underbrace{\|J_{p^*}\|_{\infty}}_{\text{equilibrium price response}} \leq L_s \equiv \frac{2\gamma}{\lambda_{\min}^2} \cdot \max_n (p_{n,f}^{\max} - c_{n,f}^{\min}),$$

where the price-response bound follows from the implicit function theorem applied to the FOC system at the unique root, and the factor $\lambda_{\min}^{-2}$ reflects the two rounds of λ^{-1} amplification (own-price and cross-price channels). Composing with the profit-Lipschitz constant $L_{\pi} = M_{\max}/(\gamma \lambda_{\min})$ (the largest market weight times the maximum unit margin) and the scalar $1/w_{\min}$ yields (3.16).

Closed-form $\gamma_{\max}^{\text{NL}}$ criterion. The large- σ tail of Lemma 15 combined with (3.16) gives

$$\gamma^{\text{NL}}(\sigma) \leq \frac{\|g_0\|_{\infty}}{\sigma} \cdot \frac{L_{\pi} L_s}{w_{\min}} = \frac{\|g_0\|_{\infty} M_{\max} (p^{\max} - c^{\min})}{w_{\min} \lambda_{\min}^3 \sigma}.$$

For small σ , the localisation argument of Lemma 16 yields $\gamma^{\text{NL}}(\sigma) \leq (K_S/\delta_v^{\text{NL}}) \sup_{|z| \geq \delta_v^{\text{NL}}/(2\sigma)} |z| g_0(z)$, which vanishes as $\sigma \rightarrow 0$ under the tail-regularity condition on g_0 . The single verifiable diagnostic $\gamma_{\max}^{\text{NL}} < 1$ is obtained by maximising over σ numerically on a log-spaced grid, identically to the CES-AB case.

E. CALIBRATION DETAILS AND ADDITIONAL COUNTERFACTUALS

This appendix provides the full calibration construction, specification comparison, robustness exercises, the data-localization counterfactual, and the interval-certification audit summarized in Section 5.

Computational verification summary. Our computational verification proceeds in four layers. First, Appendix A establishes analytically that the AB per-market convexity condition is governed by a strictly concave scalar function $H_{AB}(s)$, yielding a unique implicit threshold $\bar{s}_{AB}^*$ (Proposition 36). Second, pricing uniqueness is constructively witnessed by a Perron–Frobenius weighted-norm contraction with $\Lambda w \leq \lambda w$ and $\lambda < 1$ (Proposition 11). Third, the Jia–Tarski squeezing procedure is interval-audited conditional on the converged rival-demand vectors: all 129 decisive inclusion/exclusion inequalities are certified with zero inconclusive cases. Fourth, full-game uniqueness is verified numerically by coincidence of the supremum and infimum solutions.

Demand. Market demand is parameterized as $\Omega_n = (\text{GDP}_n/\text{GDP}_{\text{US}})^\beta$ with $\beta = 0.7$. The value $\beta = 0.7$ is chosen from a parsimonious calibration comparison: at common bounds and wage specification, $\beta = 0.7$ achieves 209/210 agreement versus 207/210 at $\beta = 1.0$, with a more plausible ϕ_i distribution. The counterfactual ranking is preserved at both values (AB-versus-CES profit decline: 30% at $\beta = 0.7$, 31% at $\beta = 1.0$).

Local costs. We use PWT 10.01 compensation per worker, adjusted for human capital: $w_i = \tilde{w}_i/(\text{hc}_i/\text{hc}_{\text{US}})$, where $\tilde{w}_i$ is the raw relative wage ($\tilde{w}_{\text{US}} = 1.0$) and hc_i is the PWT human capital index. The resulting w_i ranges from 0.16 (Philippines) to 1.94 (UAE). We also test TFP-adjusted wages ($w_i^{\text{tfp}} = \tilde{w}_i/\text{ctfp}_i$), which achieves 208/210 with virtually identical mechanism results (AB-versus-CES profit decline: 30.0% vs. 30.1%).

Trade costs. We construct a discrete three-tier iceberg matrix: $\tau_{in} = 1.0$ (same country), 1.15 (same continent or $< 5,000$ km), 1.5 (cross-continent), using capital-city coordinates. The tier values are motivated by the continent/cross-continent τ ratio implied by gravity estimates in the international trade literature at a CES elasticity in the range of our baseline; the discretization is adopted because our application has $N = 42$ markets and a full 42×42 gravity matrix is neither identified from entry-count moments alone nor central to the paper’s mechanism. This is a coarse approximation; cloud-specific frictions (latency, data-sovereignty regulations) would require a richer specification beyond the scope of this application. A sensitivity sweep over the cross-continent tier (e.g., $\tau \in \{1.3, 1.7\}$) is deferred to the revision, pending a clean rerun of the full counterfactual stack.

Specification comparison. Three specifications differ in which firm-level parameters are set externally: Spec A (preferred) fixes a_f from revenue, calibrates (g_f, ϕ_i) ; Spec B calibrates (a_f, ϕ_i) with $g_f = 1$; Spec C calibrates all three jointly. Spec A achieves 209/210 (99.5%), Spec B 201/210 (95.7%), Spec C 197/210 (93.8%). Spec A dominates despite having the same number of free firm parameters as Spec B (4 each plus 42 ϕ_i), because fixing a_f externally separates firm scale from deployment efficiency.

Fringe-firm TFP sensitivity. Rescaling Oracle’s a_f by +25% and Alibaba’s by +50% produces muted extensive-margin effects (≤ 1 additional market) but meaningful intensive-margin compression of Big 3 markups.

Profit-gap distribution. Figure 8 plots the distribution of per-site profit gaps ($VP_i - FC_i$) across all 128 active sites in the preferred specification. Of the 128 active sites, 66 have $|\text{profit gap}| < 0.005$ (close to the entry-indifference margin) and 33 have gaps exceeding 0.01 (deeply inframarginal). The 209/210 in-sample agreement rate therefore reflects genuine discrimination at the entry margin, not merely the fitting of inframarginal cells.

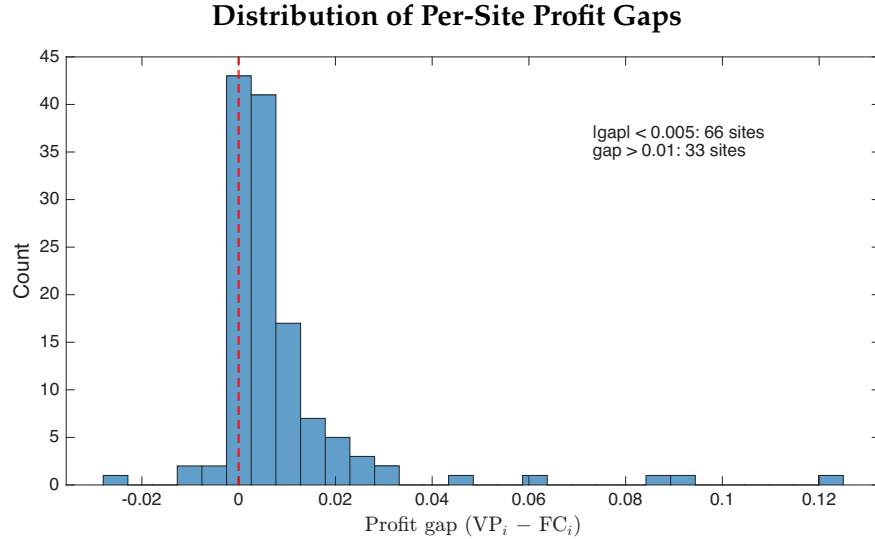


Figure 8: Distribution of per-site profit gaps ($VP_i - FC_i$) across 128 active firm-market pairs. Over half of active sites are close to the entry-indifference margin ($|\text{gap}| < 0.005$), indicating that the calibration is discriminating at the margin rather than fitting inframarginal cells.

Holdout cross-validation. The in-sample agreement figures (209/210, 99.5%) are informative but do not by themselves rule out interpolation of the observed activation matrix.

To assess predictive content, we report a 5-fold masked-cell holdout exercise within the same 210-cell cross section. The 210 firm–market cells are randomly partitioned into five disjoint blocks of 42 cells each; in each fold, ϕ_i and g_f are recalibrated using only the 168 training cells, and the fitted model is then solved on the full lattice and evaluated on the 42 held-out cells. Table 11 reports per-fold training agreement, holdout agreement, and the holdout true-positive and true-negative rates. Relative to the majority-class baseline of 0.614, the calibrated model achieves mean holdout agreement of 0.824 (SD 0.055), indicating substantial predictive content beyond exact in-sample matching. This exercise does not separately identify the model, and we note that because the equilibrium is solved on the full 5×42 lattice at the training-calibrated (ϕ_i, g_f) , general-equilibrium price and share feedback from the held-out cells enters the prediction—the exercise therefore tests the restriction *given the equilibrium structure*, not an out-of-sample prediction in the econometric sense. What it does show is that the calibrated fixed-cost structure retains predictive discipline on held-out cells within the observed cross section.⁹

Table 11: 5-Fold Holdout Cross-Validation

Fold	n_{train}	n_{hold}	Train agr.	Holdout agr.	Holdout TPR / TNR
1	168	42	0.940	0.786	0.679 / 1.000
2	168	42	0.946	0.762	0.724 / 0.846
3	168	42	0.952	0.833	0.739 / 0.947
4	168	42	0.952	0.905	0.917 / 0.889
5	168	42	0.946	0.833	0.720 / 1.000
Mean	—	—	0.948	0.824	0.756 / 0.936
SD	—	—	0.005	0.055	— / —
Majority-class baseline				0.614	—

Notes: Each fold holds out 42 of the 210 firm–market cells (20%) chosen uniformly at random; ϕ_i and g_f are recalibrated using per-market and per-firm moment counts restricted to the 168 training cells, and the fitted model is then solved on the full lattice. Agreement = $\frac{1}{|S|} \sum_{(f,i) \in S} \mathbf{1}[I_{fi}^{\text{obs}} = I_{fi}^{\text{model}}]$ for S = training or holdout. TPR and TNR are computed on the holdout cells only. The majority-class baseline is $\max(\bar{I}, 1 - \bar{I}) = 0.614$.

⁹The masked calibration is numerically ill-conditioned under the main-specification parameter bounds ($\phi_i \in [10^{-4}, 2]$, $g_f \in [0.1, 5]$): the outer inversion fails to converge within 5,000 iterations. We therefore run the holdout exercise on tighter bounds ($\phi_i \in [0.02, 0.30]$, $g_f \in [0.1, 3]$) that restore convergence. The exercise thus disciplines the entry/CDCP structure under a restricted parameter space; a full-bound holdout study would require a more robust masked solver and is left for future work.

EU compliance-cost ladder. Table 12 reports the extensive-margin response across a grid of additive compliance shocks $\Delta\phi$. The response saturates rapidly: by $\Delta\phi = 0.005$ (roughly twice the median baseline $\phi_i = 0.002$), nearly all EU firm-market pairs have exited. This indicates that most EU sites are near the entry margin at baseline, so even modest compliance wedges trigger substantial exit. The qualitative regressivity finding—lower-ranked firms lose proportionally more—is robust across the entire grid.

Table 12: EU Compliance-Cost Ladder

$\Delta\phi$	Active	Δ	AWS	MSFT	GOOG	ORCL	BABA
0 (baseline)	128	—	6	11	9	6	1
0.001	111	−17	−2	−5	−4	−6	−1
0.005	98	−30	−6	−10	−9	−6	−1
0.010	97	−31	−6	−11	−9	−6	−1
0.030	97	−31	−6	−11	−9	−6	−1

Notes: Columns AWS–BABA report EU market exits (negative) relative to baseline EU presence. The first row shows baseline EU market counts. The response saturates between $\Delta\phi = 0.005$ and 0.01.

Interval certification of Jia–Tarski sign checks. The JT gap = 0 diagnostic is strengthened by a post-hoc interval audit of the squeezing procedure. The auditor re-evaluates every decisive site-level marginal-value comparison used by the Jia–Tarski solver, conditional on the converged rival-demand vector D_{-f} . For each candidate site, market shares are enclosed by monotone scalar root-bracketing on the strictly increasing AB pricing equation (bisection to 10^{-12} tolerance), these enclosures are mapped into variable-profit intervals via the monotonicity of $VP(s) = R_n(s/\sigma + bs^2)$, and the resulting site-level marginal values are aggregated across markets with conservative interval arithmetic. A site is certified for inclusion when the lower bound on its marginal value is strictly positive, and for exclusion when the upper bound is strictly negative. In the baseline cloud illustration, all 129 decisive sign checks are certified with zero inconclusive cases; the remaining five AWS exclusions arise from the exogenous feasible-set restriction $\mathcal{A}_f$ and do not require sign certification. The resulting JT characterization is a finite interval certificate for the single-agent activation subproblems, conditional on the equilibrium rival-demand vectors.

Data-localization counterfactual. Removing China from all non-Alibaba feasible sets leaves the total at 128 active pairs (AWS gains two markets while losing China). The reverse scenario—China opens and Alibaba’s feasible set expands to all 42 markets—increases total pairs from 128 to 155, with Alibaba entering 28 new countries. The binding constraint is market access ($\mathcal{A}_f$), not entry costs (ϕ_i): reducing ϕ_{China} by 90% without feasible-set expansion produces zero additional entry. This counterfactual is sensitive to the deployment-efficiency specification and is reported as illustrative of the mechanism.

F. ALGORITHM PERFORMANCE

F.1 Numerical Algorithm Comparison

Table 13 reports the results for all four algorithm variants in the baseline economy ($N = 20, F = 8$). All converge, and the Jia–Tarski gap is zero for all eight firms in every iteration—providing strong computational evidence for single-agent supermodularity (Proposition 6).

Table 13: Algorithm Comparison ($N = 20, F = 8$)

Variant	Iters	Time (s)	CDCP Time (s)	JT Gap
Jia–Tarski + Jacobi	120	2.38	2.19	0
Jia–Tarski + Gauss–Seidel	115	1.92	1.86	0
AES + Jacobi	120	1.16	1.12	—
AES + Gauss–Seidel	115	1.12	1.07	—

Gauss–Seidel converges 4% faster than Jacobi (115 vs. 120 iterations). The AES solver is approximately 40% faster than Jia–Tarski per call: under supermodularity, the squeezing phase prunes most candidates, reducing the branching tree. The value of Jia–Tarski is not speed but its *diagnostic*: a gap of zero is strong computational evidence for the supermodular structure (necessary condition), whereas AES provides no such diagnostic.

Inner-solver consistency. To verify that the two inner solvers agree, we run a pairwise comparison: JT+supremum versus AES+supremum, and JT+infimum versus AES+infimum. In both cases, the solvers produce identical activation sets at every outer iteration, confirming that both algorithms find the same optimum of the single-agent

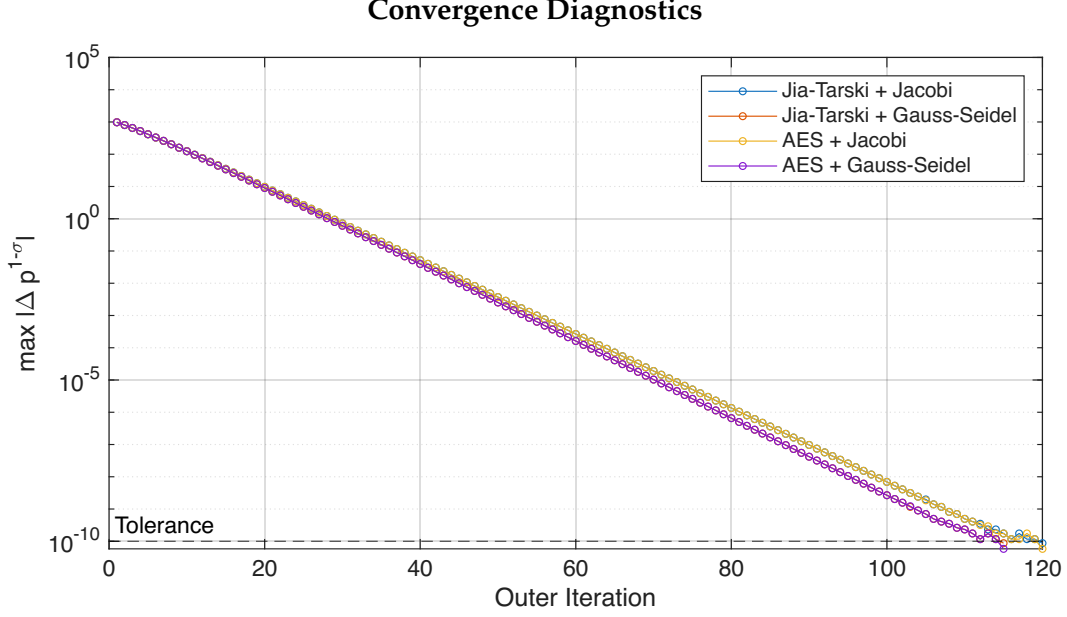


Figure 9: Convergence of Outer Iterations ($N = 20, F = 8$)

CDCP. We further test each firm in isolation by calling both solvers with identical rival demand indices D_{-f} ; all firms return the same optimal activation set from both algorithms. These diagnostics confirm that the numerical results are solver-invariant.

Figure 9 plots the convergence metric $\max_{n,f} |\Delta p_{nf}^{1-\sigma}|$. Convergence is monotone and geometric, consistent with Proposition 9.

F.2 Extension: $N = 40, F = 10$

To demonstrate scalability, we double the economy to $N = 40$ countries (eight clusters of five, forming two “hemispheres”) and $F = 10$ firms. Firms 1 and 2 now face $2^{40} \approx 1.1 \times 10^{12}$ candidate subsets each—exhaustive enumeration is impossible.

Table 14: Algorithm Comparison ($N = 40, F = 10$)

Variant	Iters	Time (s)	CDCP Time (s)	JT Gap
Jia-Tarski + Jacobi	114	17.44	17.04	0
Jia-Tarski + Gauss-Seidel	110	14.48	14.27	0
AES + Jacobi	114	7.29	7.11	—
AES + Gauss-Seidel	110	6.84	6.66	—

Both Jia-Tarski and AES solve the problem in seconds—Jia-Tarski in 14 seconds, AES in 7 seconds—despite the astronomical size of the brute-force search space. The Jia-Tarski

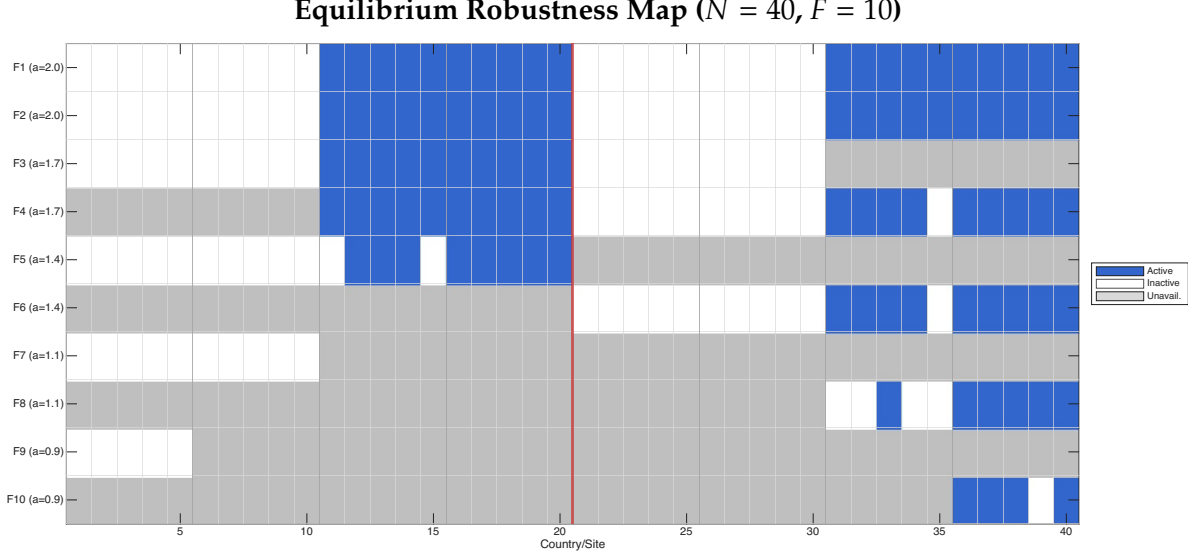


Figure 10: Site Activation Map ($N = 40, F = 10$, unique equilibrium). Blue: active sites; white: available but inactive; gray: unavailable.

gap remains zero for all ten firms, confirming that supermodularity extends to this scale. As in the baseline, all four solver combinations converge to the same unique equilibrium with 96 total active sites and aggregate profit 69.19.

G. SYNTHETIC STRESS TESTS

The cloud application in Section 5 demonstrates that the theoretical results hold at empirical parameter values ($F = 5, N = 42$). This appendix provides complementary stress tests using a synthetic economy ($N = 20, F = 8$) that permits systematic sweeps over the parameter space, including regions where multiplicity emerges.

G.1 Parameterization

The synthetic economy uses $\sigma = 5, \eta = 2, \theta = 3$, matching the cloud application. Twenty countries are organized in four income clusters with dispersed wages, expenditures, and trade costs. Eight firms span a TFP hierarchy from global players ($a_f = 2.0, |\mathcal{A}_f| = 20$) to regional players ($a_f = 1.0, |\mathcal{A}_f| = 5$). Wages are drawn on a log-linear grid $w_i \in [0.15, 1.90]$, expenditures Ω_n on a log-linear grid $[0.05, 1.00]$; trade costs follow the three-tier iceberg structure of Section 5 (1.00 same country, 1.15 same cluster, 1.50 cross-cluster).

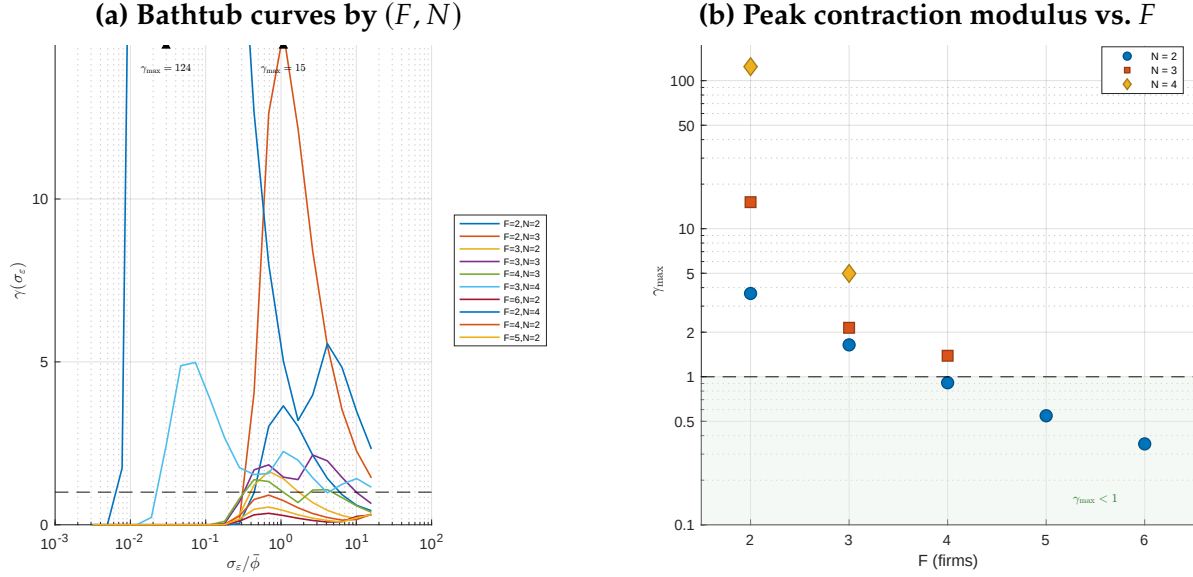


Figure 11: Scaling of the contraction modulus with (F, N) . (a) Bathtub curves for economies of different sizes. (b) Peak $\gamma_{\max}$ vs. F .

G.2 Contraction Modulus: The Bathtub

As shown in Figure 2 (Section 3.6), the contraction modulus has a bathtub shape with peak $\gamma_{\max} = 0.951 < 1$ in the small economy. The bathtub shape is preserved across economies of different sizes (Figure 11). Two opposing forces determine $\gamma_{\max}$: more firms means smaller market shares and weaker marginal-value sensitivity ($\gamma_{\max}$ decreases with F), while more sites create additional near-marginal activation decisions ($\gamma_{\max}$ increases with N). The contraction condition is most easily satisfied in the empirically relevant regime of many firms with moderate market shares.

G.3 Uniqueness Boundary: AB versus CES

To identify where multiplicity emerges and how AB markups shift the boundary relative to CES, we conduct a joint sweep over TFP dispersion $\delta = a_{\max}/a_{\min}$ and fixed-cost scale α in the synthetic economy.

Figure 12 maps the uniqueness region. The equilibrium is unique across most of the parameter space; multiplicity emerges only at extreme TFP dispersion ($\delta \geq 7$) combined with high fixed costs ($\alpha \geq 1.5$), where a single marginal site produces two equilibria.

Figure 13 compares the AB and CES uniqueness regions. On the 11×11 grid, AB

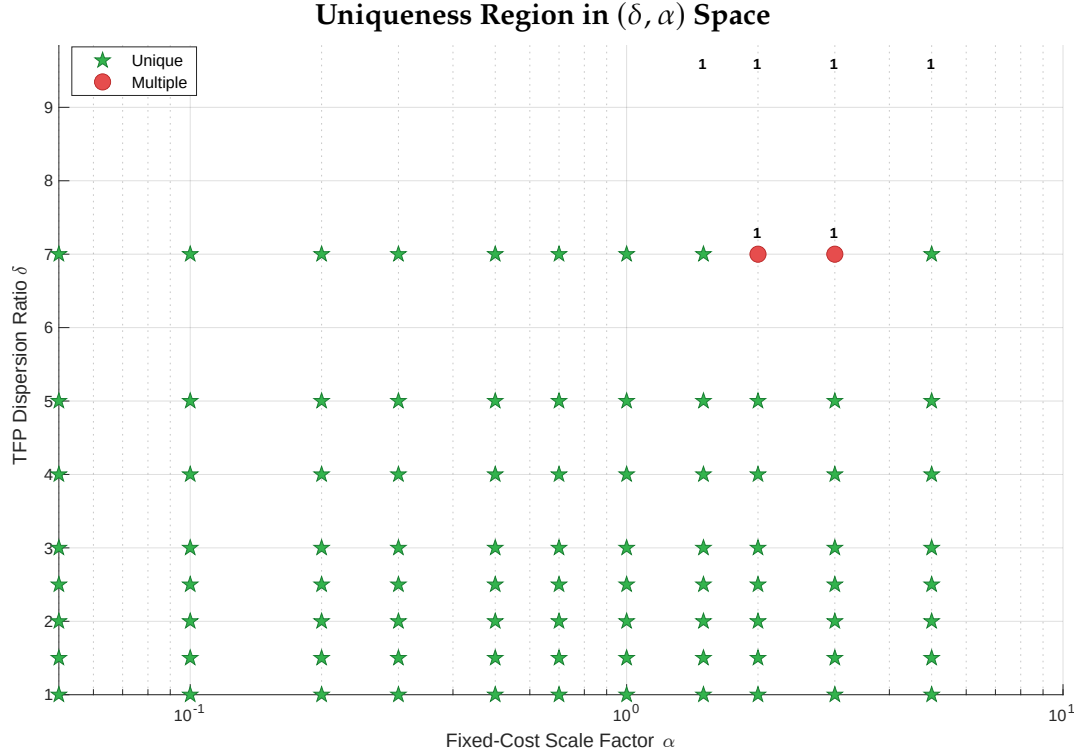


Figure 12: Uniqueness region in (δ, α) space ($N = 20, F = 8$). Green: unique equilibrium; red: multiple equilibria. Multiplicity requires extreme TFP dispersion and high fixed costs.

markups *expand* the uniqueness region: 23 cells have a unique equilibrium under AB but multiple equilibria under CES, whereas only 1 cell shows the reverse. The mechanism is that AB markup compression dampens the dominant firm's effective demand response, keeping marginal sites' profit gaps bounded away from zero.

G.4 Broader Parameter Sensitivity

Table 15 extends the analysis to a $3 \times 3 \times 4$ grid over (σ, η, θ) . In 18 of 28 configurations the analytic supermodularity condition fails $((\sigma - 1)/\theta \leq 1)$. Nevertheless, the Jia–Tarski gap equals zero and the equilibrium is unique at all 28 configurations. The spectral radius $\rho(\Lambda)$ reaches 0.225 at $(\sigma, \eta) = (8, 1.5)$ but remains well below one throughout.

G.5 Welfare Decomposition: Variety versus Markup Channel

The AB and CES pricing rules have offsetting welfare effects. Under AB, the 16% activation expansion (37 vs. 32 active sites in the synthetic baseline) generates additional product

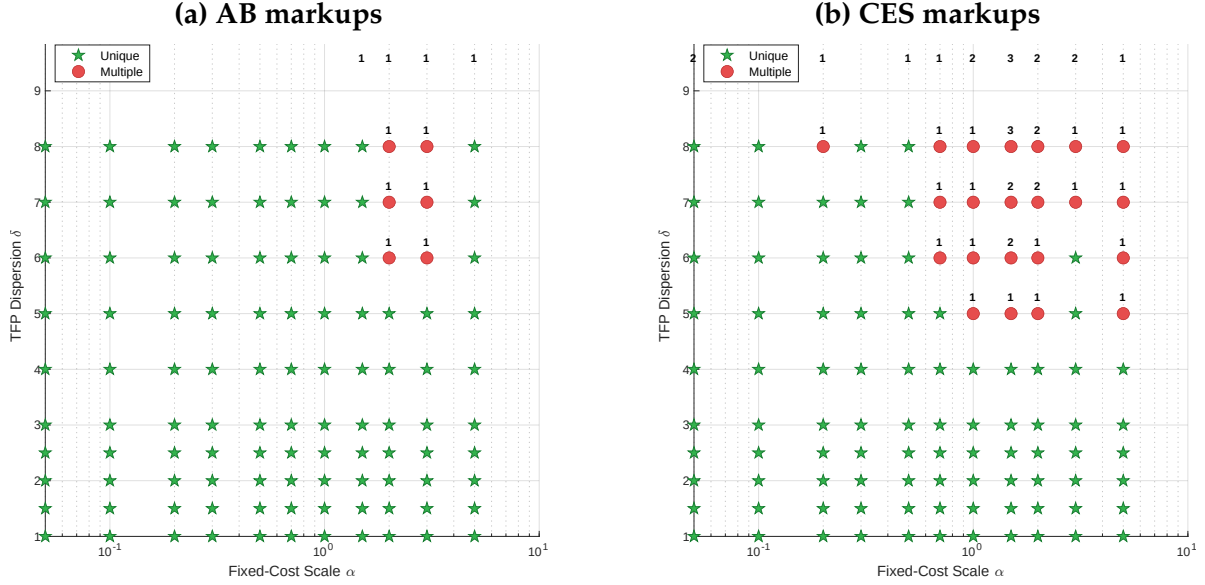


Figure 13: Uniqueness region: AB (left) vs. CES (right). AB markups expand the uniqueness region—23 of 121 cells switch from multiple to unique equilibria.

variety, lowering the sectoral price index P_n . But the higher variable markups ($\bar{\mu}^{AB} = 1.36$ vs. $\mu^{CES} = 1.25$) raise prices. Computing the CES real-income index $W_n = R_n/P_n$ at each market (holding R_n fixed at its initial value), AB welfare is on average 11% *lower* than CES welfare in the synthetic economy. The markup channel dominates the variety channel at all 20 markets, with the welfare gap largest in advanced markets (where concentration is highest) and smallest in frontier markets.

This decomposition underscores that the activation decision—the dimension on which this article’s uniqueness result operates—has quantitatively significant welfare consequences that constant-markup models misattribute to other channels. However, the welfare comparison is partial equilibrium: R_n is held fixed, and general-equilibrium feedbacks through wages or demand could alter the ranking. The result should be interpreted as a diagnostic, not as a normative recommendation.

G.6 Sufficient Conditions: Infinity-Norm versus Spectral Radius

At the synthetic baseline, the maximum feedback coefficient $\max \kappa_{nf} = 0.193$ exceeds the infinity-norm threshold $1/(F - 1) = 0.143$ from Proposition 11, because the dominant firms’ high market shares (up to $s = 0.39$, above the per-market threshold $\bar{s} = 0.339$ from Corollary 26) generate large pricing feedback. However, the tighter spectral condition

Table 15: Uniqueness Diagnostics across (σ, η, θ) : Synthetic Economy

σ	θ	$\eta = 1.5$				$\eta = 2$				$\eta = 3$			
		$\frac{\sigma-1}{\theta}$	ρ	JT	sites	$\frac{\sigma-1}{\theta}$	ρ	JT	sites	$\frac{\sigma-1}{\theta}$	ρ	JT	sites
3	3	0.67	.047	0	39	0.67	.025	0	38				
3	5	0.40	.047	0	19	0.40	.026	0	18				
3	8	0.25	.048	0	16	0.25	.026	0	16				
5	3	1.33	.125	0	38	1.33	.096	0	37	1.33	.053	0	36
5	5	0.80	.123	0	17	0.80	.095	0	16	0.80	.052	0	16
5	8	0.50	.123	0	15	0.50	.095	0	15	0.50	.052	0	15
8	3	2.33	.225	0	39	2.33	.208	0	36	2.33	.160	0	34
8	5	1.40	.222	0	18	1.40	.204	0	18	1.40	.157	0	18
8	8	0.88	.222	0	15	0.88	.204	0	14	0.88	.157	0	13

Notes: Synthetic economy ($N = 20, F = 8$). $\sigma > \eta$ required; $(\sigma, \eta) = (3, 3)$ excluded. JT: Jia–Tarski gap (0 at all configurations). All 28 configurations yield a unique equilibrium.

$\rho(\Lambda) = 0.096 < 1$ confirms pricing uniqueness. The same pattern holds in the cloud application ($\rho(\Lambda) = 0.103$): the infinity-norm condition is conservative, while the spectral condition resolves the intensive margin comfortably. The two numbers relate as $\|\Lambda\|_\infty = \max_{n,f} \kappa_{nf}$ (the row-sum norm equals the largest feedback coefficient because Λ has a single nonzero diagonal entry per row in our parameterization), so $\rho(\Lambda) \leq \|\Lambda\|_\infty$ is simply the standard eigenvalue–row-sum inequality.

This gap between the two sufficient conditions—the infinity norm violated but the spectral radius well below one—is a generic feature of economies with firm-level TFP heterogeneity. Heterogeneous TFP concentrates market shares in a few dominant firms, inflating $\max \kappa$ and violating the per-market condition, but the spectral radius accounts for the fact that high- κ pairs are few and their feedback does not compound across the full matrix.

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